TODOs list

- **p. 4** Cite/mention algebra structure.
- **p. 4** Introduce module structure to axis action maps. And be careful with left/right action conventions.
- **p. 5** Review the factoring out radical portion.
- **p. 5** Justify the map is well-defined.
- **p. 6** Justify equal dimension of the tensor product vector space and the bimap space so Ξ is an isomorphism
- **p. 7** Review and rewrite the prior work section
- **p. 17** "regularity condition" for quicksylver needs more precision.
- **p. 20** "regularity condition" for derivations needs more precision.
- **p. 36** Add some performance numbers/charts from Quicksylver/derivation calculations from the algorithm implementation in Julia
- **p. 40** cite non-associative algebra book for definitions
- **p. 42** Shorten and sharpen this direction.
- **p. 44** Shorten or make clearer this concrete nucleus deecomposition example.
- **p. 45** Shorten or get rid of this explicit derivation of a tensor product bimap example!

Chris Liu, Date: March 26, 2026

1. Introduction

A scalar records one measurement. A vector records a magntiude and direction, so it has one axis, where an axis means an independent mode of access. A matrix has two axes, and many scientific and engineering settings require tensors with three or more axes. Tensors encapsulate all of these abstractions in one unifying concept. As as result, they are used in various disciplines to record information for some fixed reference frame. Some complementary perspectives studying tensors include [Bro97] [KB09] [Lan12] [RS18] [DLDMV00] [Tuc66].

Below we give two examples.

The first is the stress tensor in mechanics. For an object such as a steel beam supporting traffic, we can track how the internal force depends on the orientation of a tiny surface element inside the material. At each point this gives a 3×3 Cauchy stress tensor: diagonal entries capture normal compression/tension, while off-diagonal entries capture shear [GFA10]. To model deformation laws, one often maps one stress state to another via a four axes tensor, since each output component can depend on each input component across two input and two output axes.

The second comes from recommender systems, of which the Netflix tensor is the most famous. The starting point is to store a user $\times$ movie matrix of ratings. The factorization of this matrix was central to recommenders competing for the Netflix Prize [KBV09]. But recommendation systems may need more axes, for example, by considering preferences over time, and considering multiple devices. This necessitates higher valence tensors, for instance with axes user $\times$ movie $\times$ time. Tensor factorization methods explicitly model and work with these higher valence tensors [Kar+10].

1.1. Main Results.

We briefly describe the two main problems tackled by my dissertation.

The first is about efficient computing of algebraic invariants of tensors. They have been key to discovering basis independent cluster pattern in tensors [BKW24], decomposing p-groups [Wil09], computing endomorphisms of modules [BL08], and advances in isomorphism testing [IQ19], [BMW17] [BW12] [BMW22]

Throughout, $\mathbb{F}$ will denote a field. Suppose we are given a 3-tensor $T \in \mathbb{F}^{m \times n \times r}$ which represents trilinear measurements in a fixed basis. Let the (i, j, k)th component be denoted $T[i, j, k]$, and let $T_1, \ldots, T_r \in \mathbb{M}_{m \times n}(\mathbb{F})$ be slices using the Matlab/Julia inspired slice notation, where $T_i \coloneqq T[:, :, i]$. For $Z = (z_{ji}) \in \mathbb{M}_r(\mathbb{F})$, define T^Z as the tensor with slices

$$(T^Z)_i \coloneqq \sum_{j=1}^{r} z_{ji} T_j.$$

Equations for three algebraic invariants (adjoint, centroid, and derivation) we study are:

$$\begin{aligned}
&\mathrm{Adj}(T) \coloneqq \{(X,Y) \in \mathbb{M}_m(\mathbb{F}) \times \mathbb{M}_n(\mathbb{F}) : (\forall i),\ XT_i = T_iY\ \},\\
&\mathrm{Cen}(T) \coloneqq \{(X,Y,Z) \in \mathbb{M}_m(\mathbb{F}) \times \mathbb{M}_n(\mathbb{F}) \times \mathbb{M}_r(\mathbb{F}) : (\forall i),\ XT_i = T_iY = (T^Z)_i\},\\
&\mathrm{Der}(T) \coloneqq \{(X,Y,Z) \in \mathbb{M}_m(\mathbb{F}) \times \mathbb{M}_n(\mathbb{F}) \times \mathbb{M}_r(\mathbb{F}) : (\forall i),\ XT_i + T_iY = (T^Z)_i\}.
\end{aligned} \tag{1.1}$$

Each corresponds to linear system. Our first main result, Theorem 3.33, proves given $T \in \mathbb{F}^{a\times b\times c}$, there is an algorithm for computing the derivation algebra of a tensor in an asymptotically faster fashion than solving the full linear system in certain scenarios.

The second result is about understanding structure when we combine two tensors into one via multiplication. Multiplication is a natural way to inductively build up data from smaller pieces. For instance, if A and B are $\mathbb{F}$-algebras, then $A \otimes_{\mathbb{F}} B$ is again an algebra. An example of this is the extensions of scalars construction $A \otimes_{\mathbb{F}} L$ for a field extension $L/\mathbb{F}$. We define a product of tensors that serves the same role for tensors by combining interactions while preserving multilinearity. We are given tensors as as bilinear maps

$$s : \mathbb{F}^m \times \mathbb{F}^n \rightarrowtail \mathbb{F}^l, \qquad t : \mathbb{F}^p \times \mathbb{F}^q \rightarrowtail \mathbb{F}^r.$$

A convention in this dissertation is to give bilinear maps with lower case letters, and multiway arrays as upper case letters. Define the product tensor

$$s \otimes t : (\mathbb{F}^m \otimes \mathbb{F}^p) \times (\mathbb{F}^n \otimes \mathbb{F}^q) \rightarrowtail \mathbb{F}^l \otimes \mathbb{F}^r$$

on coordinate vectors by

$$(s \otimes t)(e_i \otimes f_a,\ e_j \otimes f_b) \coloneqq s(e_i, e_j) \otimes t(f_a, f_b),$$

and extend bilinearly.

Example 1.1. Consider the multiplication tensors of $\mathbb{F}[x]/(x^2+1)$ and the multiplication tensor of $\mathbb{F}[y]/(y^2+5y+10)$. Each algebra is given a $2\times 2\times 2$ multiplication table, and their combined multiplication is as shown below.

	1	x
1	1	x
x	x	-1

$\otimes$

	1	y
1	1	y
y	y	$-5y-10$

$=$

	1	x	y	xy
1	1	x	y	xy
x	x	-1	xy	$-y$
y	y	xy	$-5y-10$	$-5xy-10x$
xy	xy	$-y$	$-5xy-10x$	$5y+10$

Our second result, Theorem 5.1, is a theorem that states given 3-tensors s and t, we can understand $\mathrm{Der}(s \otimes t)$ by individually understanding the adjoint, centroid, and derivation algebras of s and t.

2. Preliminaries and Prior Work

Given $\mathbb{F}$-vector spaces U, V, and W, a function $f : U \times V \to W$ is $\mathbb{F}$-bilinear if $f(ku, v) = f(u, kv) = kf(u, v)$ for all k in $\mathbb{F}$, u in U, and v in V. We write $f : U \times V \rightarrowtail W$ ($\rightarrowtail$ for bilinear). When context is clear, we avoid the prefix $\mathbb{F}$.

Example 2.1. Examples of bilinear maps include:

(1) If A is an $\mathbb{F}$-algebra, then multiplication map $\mu : A \times A \rightarrowtail A$ is bilinear.
(2) For positive integers m, n, p, the rectangular matrix multiplication map

$$\mathbb{M}_{m\times n}(\mathbb{F}) \times \mathbb{M}_{n\times p}(\mathbb{F}) \rightarrowtail \mathbb{M}_{m\times p}(\mathbb{F}), \qquad (X, Y) \mapsto XY$$

is bilinear.
(3) For $U = \mathbb{F}^2$ and $V = \mathbb{F}^3$, any bilinear form $t : U \times V \rightarrowtail \mathbb{F}$ can be represented by a matrix $M \in \mathbb{M}_{2\times 3}(\mathbb{F})$ via $t(u, v) = u^T Mv$.

The above extends to explain $\mathbb{F}$-trilinear and $\mathbb{F}$-multilinear in general. Throughout this dissertation, $\mathbb{F}$-bilinear, trilinear, and multilinear maps are also called *tensors*.

For vector spaces U and V, the *tensor product* of U and V is the vector space $U \otimes V$ with the canonical map $\varphi : U \times V \rightarrowtail U \otimes V$, such that for every bilinear map f with domain $U \times V$, there is a unique induced linear map $\hat{f}$ satisfying $f = \hat{f} \circ \varphi$. For vectors $s \in U$ and $t \in V$, the notation $s \otimes t$ is denotes the image of (s, t) under φ. The elements $s \otimes t$ are called pure tensors. Next we describe subspaces of a tensor product space.

Lemma 2.2. Let U, V be vector spaces, and $A \leq U$ and $X \leq V$ be subspaces, with inclusion maps $\iota_A : A \hookrightarrow U$ and $\iota_X : X \hookrightarrow V$. Then inside $U \otimes V$,

$$A \otimes X = (U \otimes X) \cap (A \otimes V).$$

Proof. The intersection of $U \otimes X$ and $A \otimes V$ in $U \otimes V$ is the pullback of the two inclusion maps. Thus it suffices to show the following square is also a pullback:

$$\begin{tikzcd}
A \otimes X \arrow[r, "\iota_A \otimes \mathrm{id}_X"] \arrow[d, "\mathrm{id}_A \otimes \iota_X"'] & U \otimes X \arrow[d, "\mathrm{id}_U \otimes \iota_X"] \\
A \otimes V \arrow[r, "\iota_A \otimes \mathrm{id}_V"'] & U \otimes V
\end{tikzcd}$$

Let T be a vector space with maps $q_1 : T \to A \otimes V$ and $q_2 : T \to U \otimes X$ such that $(\iota_A \otimes \mathrm{id}_V) \circ q_1 = (\mathrm{id}_U \otimes \iota_X) \circ q_2$. Then we must construct the unique u making both triangles commute in the below diagram

$$\begin{tikzcd}
T \arrow[dr, dashed, "u"] \arrow[drr, bend left, "q_2"] \arrow[ddr, bend right, "q_1"'] & & \\
& A \otimes X \arrow[r, "\iota_A \otimes \mathrm{id}_X"] \arrow[d, "\mathrm{id}_A \otimes \iota_X"'] & U \otimes X \arrow[d, "\mathrm{id}_U \otimes \iota_X"] \\
& A \otimes V \arrow[r, "\iota_A \otimes \mathrm{id}_V"'] & U \otimes V
\end{tikzcd}$$

Let $q : V \to V/X$ be the quotient map. Since $q \circ \iota_X = 0$, we know $(\mathrm{id}_U \otimes q) \circ (\mathrm{id}_U \otimes \iota_X) = 0$.

Also, as elements in $\mathrm{Hom}(A \otimes V, U \otimes (V/X))$,

$$(\mathrm{id}_U \otimes q) \circ (\iota_A \otimes \mathrm{id}_V) = (\iota_A \otimes \mathrm{id}_{V/X}) \circ (\mathrm{id}_A \otimes q), \tag{2.1}$$

because both sides send the pure tensor $a \otimes v$ to $\iota_A(a) \otimes q(v)$. Therefore

$$\begin{aligned} 0 = (\mathrm{id}_U \otimes q) \circ (\mathrm{id}_U \otimes \iota_X) \circ q_2 &= (\mathrm{id}_U \otimes q) \circ (\iota_A \otimes \mathrm{id}_V) \circ q_1 \\ &= (\iota_A \otimes \mathrm{id}_{V/X}) \circ (\mathrm{id}_A \otimes q) \circ q_1. \end{aligned}$$

Where the last equality is by (2.1). Since $\iota_A \otimes \mathrm{id}_{V/X}$ is injective, $(\mathrm{id}_A \otimes q) \circ q_1 = 0$. This means $\mathrm{Im}(q_1) \leq \mathrm{Ker}(\mathrm{id}_A \otimes q)$.

The following is always a short exact sequence of vector spaces:

$$0 \to X \xrightarrow{\iota_X} V \xrightarrow{q} V/X \to 0.$$

Since tensoring over a field is exact, tensoring with A gives

$$0 \to A \otimes X \xrightarrow{\mathrm{id}_A \otimes \iota_X} A \otimes V \xrightarrow{\mathrm{id}_A \otimes q} A \otimes (V/X) \to 0.$$

Hence $\mathrm{Ker}(\mathrm{id}_A \otimes q) = \mathrm{Im}(\mathrm{id}_A \otimes \iota_X)$.

As $\mathrm{Im}(q_1) \leq \mathrm{Ker}(\mathrm{id}_A \otimes q) = \mathrm{Im}(\mathrm{id}_A \otimes \iota_X)$, there is a unique $u : T \to A \otimes X$ with $q_1 = (\mathrm{id}_A \otimes \iota_X) \circ u$. Finally, we need to show $q_2 = (\iota_A \otimes \mathrm{id}_X) \circ u$. We have

$$\begin{aligned} (\mathrm{id}_U \otimes \iota_X) \circ (\iota_A \otimes \mathrm{id}_X) \circ u &= (\iota_A \otimes \mathrm{id}_V) \circ (\mathrm{id}_A \otimes \iota_X) \circ u \\ &= (\iota_A \otimes \mathrm{id}_V) \circ q_1 \end{aligned}$$

$$= (\mathrm{id}_U \otimes \iota_X) \circ q_2,$$

Therefore $(\mathrm{id}_U \otimes \iota_X) \circ \big((\iota_A \otimes \mathrm{id}_X) \circ u - q_2\big) = 0$. As $\mathrm{id}_U \otimes \iota_X$ is injective, we get $q_2 = (\iota_A \otimes \mathrm{id}_X) \circ u$ as needed. □

2.1. **Algebras and Actions on 3-tensors.** Existing work such as [Jac10], [Mya90], [BW14], and [Wil16] highlights the role of the three algebras we saw in Equation (1.1). In this subsection, we expand on that and describe some operations and properties that we need for the future sections.

As before U, V, and W are finite-dimensional $\mathbb{F}$-vector spaces. Let $\mathrm{End}(U) = \mathrm{Hom}(U, U)$ be the endomorphisms of a vector space U, and throughout this subsection, let $b : U \times V \rightarrowtail W$ be a bilinear map. For $\alpha \in \mathrm{End}(U)$, define $b \circ_2 \alpha : U \times V \rightarrowtail W$ by $(u, v) \mapsto b(\alpha u, v)$. For $\beta \in \mathrm{End}(V)$, define $b \circ_1 \beta$ by $(u, v) \mapsto b(u, \beta v)$. For $\gamma \in \mathrm{End}(W)$, define $b \circ_0 \gamma$ by $(u, v) \mapsto \gamma(b(u, v))$.

Definition 2.3. The nuclei of b are

$$\begin{aligned}
\mathcal{L}(b) &\coloneqq \{(\alpha, \gamma) \in \mathrm{End}(U) \times \mathrm{End}(W) : b \circ_2 \alpha = b \circ_0 \gamma\},\\
\mathcal{M}(b) &\coloneqq \{(\alpha, \beta) \in \mathrm{End}(U) \times \mathrm{End}(V) : b \circ_2 \alpha = b \circ_1 \beta\},\\
\mathcal{R}(b) &\coloneqq \{(\beta, \gamma) \in \mathrm{End}(V) \times \mathrm{End}(W) : b \circ_1 \beta = b \circ_0 \gamma\}.
\end{aligned}$$

Notice $\mathcal{M}(b)$ is the coordinate free description of the adjoint algebra in Equation (1.1).

The centroid is

$$\mathcal{C}(b) \coloneqq \{(\alpha, \beta, \gamma) \in \mathrm{End}(U) \times \mathrm{End}(V) \times \mathrm{End}(W) : b \circ_2 \alpha = b \circ_1 \beta = b \circ_0 \gamma\},$$

and the derivation algebra is

$$\mathrm{Der}(b) \coloneqq \{(\alpha, \beta, \gamma) \in \mathrm{End}(U) \times \mathrm{End}(V) \times \mathrm{End}(W) : b \circ_2 \alpha + b \circ_1 \beta = b \circ_0 \gamma\}.$$

Remark 2.4. Each of

$$\mathcal{L}(b),\ \mathcal{M}(b),\ \mathcal{R}(b),\ \mathcal{C}(b),\ \mathrm{Der}(b)$$

is an $\mathbb{F}$-vector subspace of the corresponding product of endomorphism spaces. **TODO: Cite/mention algebra structure.**

2.2. **Radicals and non-degeneracy of tensors.** Let $b : U \times V \rightarrowtail W$ be a bimap. For $u \in U$ and $v \in V$, let

$$\begin{aligned}
b(u, V) &\coloneqq \{b(u, v') : v' \in V\},\\
b(U, v) &\coloneqq \{b(u', v) : u' \in U\},\\
b(U, V) &\coloneqq \mathrm{span}\{b(u', v') : u' \in U,\ v' \in V\}.
\end{aligned}$$

The *left radical*, *right radical*, and *coradical* of a bimap b are respectively

$$\begin{aligned}
\mathrm{Rad}_L(b) &\coloneqq \{u \in U : b(u, V) = 0\},\\
\mathrm{Rad}_R(b) &\coloneqq \{v \in V : b(U, v) = 0\},\\
\mathrm{Corad}(b) &\coloneqq W/b(U, V).
\end{aligned}$$

We call b *fully nondegenerate* if $\mathrm{Rad}_L(b) = \mathrm{Rad}_R(b) = 0$ and $\mathrm{Corad}(b) = 0$. We also define the axis action maps

$$\begin{aligned}
\Lambda_2^b &: \mathrm{End}(U) \to \mathrm{Bil}(U, V; W), \quad \alpha \mapsto b \circ_2 \alpha,\\
\Lambda_1^b &: \mathrm{End}(V) \to \mathrm{Bil}(U, V; W), \quad \beta \mapsto b \circ_1 \beta,\\
\Lambda_0^b &: \mathrm{End}(W)^{\mathrm{op}} \to \mathrm{Bil}(U, V; W), \quad \gamma \mapsto b \circ_0 \gamma.
\end{aligned}$$

The superscript is suppressed when context is clear. **TODO: Introduce module structure to axis action maps. And be careful with left/right action conventions.**

Lemma 2.5. If $b : U \times V \rightarrowtail W$ is fully nondegenerate, then Λ_2, Λ_1, and Λ_0 are injective.

Proof. If $\Lambda_2(\alpha) = 0$, then $b(\alpha u, v) = 0$ for all $u \in U$ and $v \in V$, so $\alpha u \in \mathrm{Rad}_L(b)$ for every u. Since $\mathrm{Rad}_L(b) = 0$, we get $\alpha = 0$. The proof for Λ_1 is the same.

If $\Lambda_0(\gamma) = 0$, then $\gamma(b(u, v)) = 0$ for all u, v, so γ vanishes on $\mathrm{span}(b(U, V)) = W$. Hence $\gamma = 0$. □

Fact 2.6. For any bimap $b : U \times V \rightarrowtail W$, there is a canonical induced bimap

$$b/\sqrt{b} : \big(U/\mathrm{Rad}_L(b)\big) \times \big(V/\mathrm{Rad}_R(b)\big) \rightarrowtail b(U, V),$$

defined by

$$\big(b/\sqrt{b}\big)\big(u + \mathrm{Rad}_L(b), v + \mathrm{Rad}_R(b)\big) := b(u, v).$$

This map is fully nondegenerate.

material between these lines is not trusted, and to be reviewed

TODO: Review the factoring out radical portion.

Understanding the algebraic invariants of $b/\sqrt{b}$ gives insight into the algebraic invariants of b. In fact, every $(\alpha, \gamma) \in \mathcal{L}(b)$ satisfies

$$\alpha(\mathrm{Rad}_L(b)) \leq \mathrm{Rad}_L(b), \qquad \gamma\big(b(U, V)\big) \leq b(U, V).$$

The proof is as follows. Let $u \in \mathrm{Rad}_L(b)$. Then $b(u, v) = 0$ for all v, so $\gamma(b(u, v)) = \gamma(0) = 0$. But $\gamma(b(u, v)) = b(\alpha u, v)$ as well, so $\alpha u \in \mathrm{Rad}_L(b)$. The proof for $\gamma(b(U, V)) \leq b(U, V)$ is analogous. Therefore there is a canonical surjection

$$\rho_L : \mathcal{L}(b) \to \mathcal{L}(b/\sqrt{b}), \qquad (\alpha, \gamma) \mapsto (\bar{\alpha}, \gamma|_{b(U,V)}),$$

where $\bar{\alpha}$ is induced by α on $U/\mathrm{Rad}_L(b)$. **TODO: Justify the map is well-defined.** Its kernel is

$$\mathrm{Ker}(\rho_L) = \mathrm{Hom}(U, \mathrm{Rad}_L(b)) \times \{f \in \mathrm{End}(W) : f|_{b(U,V)} = 0\} \cong \mathrm{Hom}(U, \mathrm{Rad}_L(b)) \times \mathrm{Hom}(\mathrm{Corad}(b), W).$$

Hence we have the below exact sequence

$$0 \to \mathrm{Hom}(U, \mathrm{Rad}_L(b)) \times \mathrm{Hom}(\mathrm{Corad}(b), W) \to \mathcal{L}(b) \xrightarrow{\rho_L} \mathcal{L}(b/\sqrt{b}) \to 0.$$

material between these lines is not trusted, and to be reviewed

2.3. **Product of tuples of endomorphisms.** For $i \in \{0, 1, 2\}$, let U_i and V_i be finite-dimensional $\mathbb{F}$-vector spaces. We use the Kronecker isomorphisms

$$\kappa_i : \mathrm{End}(U_i) \otimes \mathrm{End}(V_i) \xrightarrow{\sim} \mathrm{End}(U_i \otimes V_i),$$

defined by $\kappa_i(\alpha \otimes f)(u_i \otimes v_i) := \alpha u_i \otimes f v_i$. We use the following notation for combining tuples of endomorphisms of multiple tensors.

Definition 2.7. Let I be finite. For each $i \in I$, let U_i, V_i be finite-dimensional $\mathbb{F}$-vector spaces. Define

$$\boxtimes : \Big(\prod_{i \in I} \mathrm{End}(U_i)\Big) \times \Big(\prod_{i \in I} \mathrm{End}(V_i)\Big) \to \prod_{i \in I} \mathrm{End}(U_i \otimes V_i)$$

by

$$((\alpha_i)_{i \in I}, (f_i)_{i \in I}) \longmapsto (\kappa_i(\alpha_i \otimes f_i))_{i \in I} =: (\alpha_i)_{i \in I} \boxtimes (f_i)_{i \in I}.$$

For subspaces $A \leq \prod_{i \in I} \mathrm{End}(U_i)$ and $B \leq \prod_{i \in I} \mathrm{End}(V_i)$, define

$$A \boxtimes B := \mathrm{span}\{a \boxtimes b : a \in A,\ b \in B\} \leq \prod_{i \in I} \mathrm{End}(U_i \otimes V_i).$$

2.4. **Product of tensors.** As described in Section 1, there is a need to combine two tensors into one using a product.

Definition 2.8. Fix two bimaps

$$s : U_2 \times U_1 \rightarrowtail U_0, \qquad t : V_2 \times V_1 \rightarrowtail V_0.$$

The *tensor product bimap*

$$s \otimes t : (U_2 \otimes V_2) \times (U_1 \otimes V_1) \rightarrowtail U_0 \otimes V_0$$

is the unique bilinear map satisfying $(s \otimes t)((u_2 \otimes v_2), (u_1 \otimes v_1)) = s(u_2, u_1) \otimes t(v_2, v_1)$.

Lemma 2.9. If s and t are fully nondegenerate bimaps, then $s \otimes t$ is fully nondegenerate.

Proof. We prove nondegeneracy for $U_2 \otimes V_2$. Write $x \in U_2 \otimes V_2$ as $\sum_j u_{2,j} \otimes v_{2,j}$ and assume

$$(s \otimes t)(x, (u_1 \otimes v_1)) = 0 \quad \text{for all } u_1 \in U_1,\ v_1 \in V_1.$$

Expanding gives

$$(\forall u_1 \in U_1, v_1 \in V_1) \qquad \sum_j s(u_{2,j}, u_1) \otimes t(v_{2,j}, v_1) = 0.$$

Since s and t are fully nondegenerate, this forces $x = 0$. The argument for $U_1 \otimes V_1$ is analogous.

For the coradical, write $u_0 = \sum_a s(u_{2,a}, u_{1,a})$ and $v_0 = \sum_b t(v_{2,b}, v_{1,b})$. Then

$$\begin{aligned} u_0 \otimes v_0 &= \left(\sum_a s(u_{2,a}, u_{1,a})\right) \otimes \left(\sum_b t(v_{2,b}, v_{1,b})\right) \\ &= \sum_{a,b} (s \otimes t)((u_{2,a} \otimes v_{2,b}), (u_{1,a} \otimes v_{1,b})), \end{aligned}$$

so $\mathrm{Corad}(s \otimes t) = 0$. □

Remark 2.10. If s and t are fully nondegenerate, then for each $i \in \{0, 1, 2\}$ the map

$$\Lambda_i^{s \otimes t} : \mathrm{End}(U_i \otimes V_i) \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0)$$

is injective by Lemmas 2.5 and 2.9.

Next we come to a key correspondence that allows us to analyze actions on tensor product bimaps by *linear algebra* techniques. Recall for U, V vector spaces, and $(U \otimes V, \varphi : U \times V \to U \otimes V)$ the tensor product space of U and V, the element $\varphi(u, v) \in U \otimes V$ is the pure tensor often written as $u \otimes v$. To be clear for the rest of this subsection, we write $\varphi(u, v)$.

Let $U := \mathrm{Bil}(U_2, U_1; U_0)$, $V := \mathrm{Bil}(V_2, V_1; V_0)$, and define a linear map.

$$\Theta : U \times V \to \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0), \quad \Theta(s, t) := s \otimes t \text{as the tensor product bimap.}$$

The universal property of the tensor product space $U \otimes V$ gives a unique induced linear map

$$\widetilde{\Theta} : U \otimes V \xrightarrow{\sim} \mathrm{Bil}(U_2 \otimes V_2, U_1 \otimes V_1; U_0 \otimes V_0). \tag{2.2}$$

The two spaces have equal dimension, so $\widetilde{\Theta}$ is an isomorphism. **TODO: Justify equal dimension of the tensor product vector space and the bimap space so Ξ is an isomorphism** Set $\Xi := \widetilde{\Theta}^{-1}$.

Lemma 2.11. For all $i \in \{0, 1, 2\}$, $\alpha \in \mathrm{End}(U_i)$, and $f \in \mathrm{End}(V_i)$,

$$\Xi\big((s \otimes t) \circ_i \kappa_i(\alpha \otimes f)\big) = \varphi(s \circ_i \alpha, t \circ_i f).$$

Proof. By definition, $\widetilde{\Theta}(s \circ_i \alpha \otimes t \circ_i f) = \varphi(s \circ_i \alpha, t \circ_i f)$. Evaluating on pure tensors gives the stated identity for each i. □

As a consequence, for $i \in \{0, 1, 2\}$ and $x \in \mathrm{End}(U_i \otimes V_i)$ with decomposition $x = \sum_j \kappa_i(\alpha_j \otimes f_j)$, linearity gives

$$\Xi\big((s \otimes t) \circ_i x\big) = \sum_j \Xi\big((s \otimes t) \circ_i \kappa_i(\alpha_j \otimes f_j)\big)$$

$$= \sum_j \varphi(s \circ_i \alpha_j, t \circ_i f_j).$$

material between these lines is not trusted, and to be reviewed

2.5. **Prior work.** **TODO: Review and rewrite the prior work section**

We build on a number of methods which we briefly summarize below. Myasnikov, Wilson, and others in [Mya90], [Wil16], [Wil12], [MM10], [Wil09] proves for the bimap t the algebras $\mathrm{Adj}(t)$ and $\mathrm{Cen}(t)$ control direct sum decompositions and automorphisms of t, using them to prove properties for the originating algebraic structures. In recent work [BKW24], it's shown that derivations also controls the original tensor's null patterns, limiting the subspaces on which the tensor is nonzero.

There has also been prior work on derivations of tensors coming from the multiplication tensors of algebras. Leger and Luks study generalized derivations, quasiderivations, and quasicentroids as extensions of algebra derivations [LL00], and Novotný and Hřivnák study (α, β, γ)-derivations together with associated invariant functions [NH08].

Recent work generalizing from bimaps finds a long exact sequence linking the various generalized adjoints, centroids, and derivations of a higher valence tensor [BMW20]. Further work by First, Maglione, and Wilson [FMW20] defines a ternary Galois connection between tensors, operators, and polynomial ideals. Alongside it, they define a generalized (P, Ω)-tensor product of vector spaces $U_1, \ldots, U_n$, for any $\Omega \subset \prod_i \mathrm{End}(U_i)$ and polynomials $P \subset \mathbb{F}[x_1, \ldots, x_n]$.

2.5.1. *Related Works.* For the first main result there are is existing literature in solving the adjoint equation, which is a type of Simultaneous Sylvester Equation. Lee and Vu study simultaneous matrix Sylvester systems together with spectral separation criteria and idempotent separators [LV10]. In the operator setting, Lee and Vu prove existence and uniqueness for simultaneous operator Sylvester equations under disjoint joint spectra [LV01].

For the second main result, prior work includes derivations of tensor products of nonassociative algebras [Bre17], derivations and automorphisms of nonassociative matrix algebras [BO81], and derivations of tensor products of algebras [Aza08].

material between these lines is not trusted, and to be reviewed

In the next three sections, we describe in detail the problems we investigated and contributions we made as part of this dissertation.

3. Quick Derivation Solutions

The descriptions of the adjoint, centroid, and derivation algebras in Equation (1.1) are linear in the unknown matrix entries. So each admit a naive solution as a large system of linear equations. In collaboration with Joshua Maglione and James B. Wilson, we have results for an asymptotically lower complexity algorithm for computing the adjoint algebra, inspired by analogous results for matrices known as the Bartels-Stewart algorithm [BS72]. In this chapter, I state and prove results from this joint work using a novel approach. I also state and prove my own original result, which concerns the derivation algebra, following a similar approach.

Theorem 3.21. For cubic $(n \times n \times n)$ tensors, under the regularity conditions (established in Definition 3.32), SimultaneousSylvesterSystem is solved in the same cost as the full deterministic verification. This is $O(n^4)$ cost, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

Theorem 3.33. For cubic $(n \times n \times n)$ tensors, under the regularity conditions (established in Definition 3.38), DerivationSystem is solved in cost $O(n^{4.5})$. This is an additional $n^{1/2}$ overhead compared to the deterministic verification cost of $O(n^4)$, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

Our strategy in this section is to reduce each large problem to a set of smaller linear algebra problems. Their solutions are affine subspaces, which we define below.

3.1. **Preliminaries.** An *affine space* is a pair (A, U) where A is a non-empty set and U is a vector space acting on A by a fixed-point free and transitive action. Write the action as $+ : A \times U \to A$, $(a, u) \mapsto a + u$. We call U the *direction space* of A. When U is clear from context, we also say "A is an affine space".

Lemma 3.1. Let (A, U) be an affine space. For any $a, b \in A$, there is a unique $u \in U$ such that $a + u = b$. We denote this vector by $b - a$.

Proof. Existence follows from transitivity. If $a + u = b = a + v$, then $a + (u - v) = a$. Fixed-point freeness gives $u - v = 0$, so $u = v$. □

Let (A, U) be a d-dimensional affine space. An *affine frame* is a tuple $(a_0, \ldots, a_d)$ in A such that $\{a_1 - a_0, \ldots, a_d - a_0\}$ is a basis of U. Every $a \in A$ has unique coordinates $\lambda = (\lambda_1, \ldots, \lambda_d) \in \mathbb{F}^d$ with $a = a_0 + \sum_{i=1}^{d} \lambda_i (a_i - a_0)$.

Convention 3.2. For computation, we encode an affine space by an affine frame (one basepoint plus a basis of the direction space).

Let (A, U) and (B, V) be affine spaces. A map $f : A \to B$ is an *affine map* if there exists a linear map $L : U \to V$ such that $f(a + u) = f(a) + L(u)$ for all $a \in A$ and $u \in U$. We call L the underlying linear map of f. Below we list some properties of affine space and maps that we will need in following sections.

(1) Every vector space V is also an affine space with direction space V.
(2) If A, B are vector spaces and $f : A \to B$ is a linear map, then f is also an affine map.
(3) If (A, U) and (B, V) are affine spaces, then $(A \times B, U \times V)$ is an affine space with action $(a, b) + (u, v) := (a + u, b + v)$.
(4) An affine map $f : A \times B \to C$ has an underlying linear map $L : U \times V \to W$ with $f\big((a, b) + (u, v)\big) = f(a, b) + L(u, v)$. This is in contrast to when f is a bilinear map in two variables.
(5) Let U and V be vector spaces, and let $L : U \to V$ be linear. If (A, W) is an affine subspace of U, then $L|_A : A \to V$ is affine, with underlying linear map $L|_W$.

Example 3.3. Let $A = \mathbb{R}^3$, $a_0 = (1, 2, 3)$, and $W = \{(x, y, 0) : x, y \in \mathbb{R}\} \leq \mathbb{R}^3$. Then $a_0 + W$ is the plane $z = 3$, an affine subspace of $(\mathbb{R}^3, \mathbb{R}^3)$.

Finally, we describe how we analyze algorithms and our notation for tensors.

Convention 3.4. For functions $f, g : \mathbb{N} \to \mathbb{N}$, we write $f \in O(g)$ if there exist constants $C, n_0 \in \mathbb{N}$ such that $f(n) \leq C\, g(n)$ for all $n \geq n_0$.

Convention 3.5. We always use the standard basis for coordinate vector spaces $\mathbb{F}^n$. For $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$, we use the bilinear pairing $\langle u, v \rangle := \sum_{i=1}^{n} u_i v_i$.

Convention 3.6. We analyze algorithms by their arithmetic complexity, meaning the number of field operations over $\mathbb{F}$ (addition, subtraction, multiplication, and division).

Definition 3.7. Let $T \in \mathbb{F}^{a \times b \times c}$ be a tensor, let $I \subseteq [a]$, $J \subseteq [b]$, and let $k \in [c]$. We use the slice notation

$$T[I, :, k] \in \mathbb{F}^{|I| \times b}, \qquad T[:, J, k] \in \mathbb{F}^{a \times |J|}, \qquad T[I, J, k] \in \mathbb{F}^{|I| \times |J|}.$$

For $M \in \mathbb{F}^{a \times b}$ a matrix, we write

$$M[I, :] \in \mathbb{F}^{|I| \times b}, \qquad M[:, J] \in \mathbb{F}^{a \times |J|}.$$

The indexing of sliced tensors and matrices is by the fixed ordering of the slicing sets. For instance, for $I = \{i_1 < \cdots < i_t\} \subset [a]$, the restricted matrix $X[I, :] \in \mathbb{F}^{t \times r}$ has $X[I, :][i_j] = X[j, :]$. We use this convention for all restricted tensor and matrix indexing.

Example 3.8. Let

$$T_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad T_2 = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix},$$

then define $T \in \mathbb{F}^{2\times 2\times 2}$ by $T[:,:,1] = T_1$, $T[:,:,2] = T_2$. For $I = \{1\}$, $J = \{2\}$, we have

$$T[I,:,1] = \begin{bmatrix} 1 & 2 \end{bmatrix}, \qquad T[:,J,1] = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

For $J = \{2,3\}$, we have

$$M = \begin{bmatrix} 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix} \in \mathbb{F}^{2\times 3}, \qquad M[:,J] = \begin{bmatrix} 10 & 11 \\ 13 & 14 \end{bmatrix} \in \mathbb{F}^{2\times 2}.$$

Definition 3.9. We define vectorization for matrices and tensors. For $M = (m_{ij}) \in \mathbb{F}^{a\times b}$, $\mathrm{vec}(M) \in \mathbb{F}^{ab}$ for $(i,j) \in [a] \times [b]$ by

$$\mathrm{vec}(M)_{\,i+(j-1)a} = m_{ij}.$$

For $T = (t_{ijk}) \in \mathbb{F}^{a\times b\times c}$, define $\mathrm{vec}(T) \in \mathbb{F}^{abc}$ for $(i,j,k) \in [a] \times [b] \times [c]$ by

$$\mathrm{vec}(T)_{\,i+(j-1)a+(k-1)ab} = t_{ijk}.$$

Example 3.10. For

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{F}^{2\times 3}, \qquad T \in \mathbb{F}^{2\times 2\times 2}, \quad T[:,:,1] = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad T[:,:,2] = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix},$$

we have $\mathrm{vec}(M) = (1,4,2,5,3,6)$, and $\mathrm{vec}(T) = (1,3,2,4,5,7,6,8)$.

3.2. Linear Algebra subroutines.

Lemma 3.11. Let (A,U) and (B,V) be affine spaces, let $f : A \to B$ be affine with underlying linear map $L : U \to V$, and let (S,W) be an affine subspace of B. Then $f^{-1}(S) \subseteq A$ is either empty or an affine subspace of A.

Proof. If $f^{-1}(S) = \varnothing$, there is nothing to prove. Assume $f^{-1}(S) \neq \varnothing$, choose $a_0 \in f^{-1}(S)$, and set $U_0 := L^{-1}(W) \leq U$. For $u \in U$,

$$f(a_0 + u) = f(a_0) + L(u).$$

Since $f(a_0) \in S$ and S has direction space W,

$$f(a_0 + u) \in S \iff L(u) \in W \iff u \in U_0.$$

Hence $f^{-1}(S) = a_0 + U_0$, so $f^{-1}(S)$ is an affine subspace. □

Problem LinSolve($\mathbb{F}$).

Given: $\mathbb{F}$-vector spaces U, V, a linear map $L : U \to V$, and $v \in V$.
Return: A computational encoding of the solution set $\{u \in U : L(u) = v\}$.

Proposition 3.12. LinSolve($\mathbb{F}$) returns either an empty set or an affine subspace of U.

Proof. The map L is affine and $\{v\}$ is an affine subspace of V, so the solution set is $L^{-1}(\{v\})$. Applying Lemma 3.11 concludes the proposition. □

Proposition 3.13. A coordinatized instance of LinSolve($\mathbb{F}$), with

$$U = \mathbb{F}^{n\times \ell}, \qquad V = \mathbb{F}^{m\times \ell}, \qquad L(X) = MX, \qquad M \in \mathbb{F}^{m\times n},$$

is solvable in cost $\mathtt{T}_{p1}(m,n,\ell) \in O(\min\{m,n\}^2 \max\{m,n\} + mn\ell)$.

This is the standard dense Gaussian-elimination bound for one shared coefficient matrix and ℓ right-hand sides. Next is the workhorse for our solution strategy, allowing us to lift subproblem solutions to solutions of the whole problem.

Problem LinearEqualsAffine($\mathbb{F}$).

Given: $\mathbb{F}$-vector spaces U, W, a linear map $L : U \to W$, an affine space (Θ, V), and an affine map $N : \Theta \to W$.
Return: A computational encoding of the solution set $\{(u, \theta) \in U \times \Theta : L(u) = N(\theta)\}$.

Proposition 3.14. The solution set of LinearEqualsAffine($\mathbb{F}$) is either empty or an affine subspace of $U \times \Theta$.

Proof. Define $F : U \times \Theta \to W$ by $F(u, \theta) := L(u) - N(\theta)$. Then F is affine and the solution set is $F^{-1}(\{0\})$. Apply Lemma 3.11. □

Proposition 3.15. Let $\rho : W \to W/\operatorname{im}(L)$ be the quotient map and define

$$\Theta_{\text{feas}} := \{\theta \in \Theta : \rho(N(\theta)) = 0\}.$$

Then (u, θ) solves $L(u) = N(\theta)$ if and only if $\theta \in \Theta_{\text{feas}}$ and $u \in u_0(\theta) + \operatorname{Ker}(L)$ for some fixed solution $u_0(\theta)$ of $L(u) = N(\theta)$.

Proof. For fixed θ, solvability is $N(\theta) \in \operatorname{im}(L)$, which is exactly $\rho(N(\theta)) = 0$. If $u_0(\theta)$ is one solution, then $L(u) = N(\theta)$ if and only if $L(u - u_0(\theta)) = 0$ if and only if $u - u_0(\theta) \in \operatorname{Ker}(L)$. □

Proposition 3.16. Given a coordinatized instance of LinearEqualsAffine($\mathbb{F}$), with

$$U = \mathbb{F}^{n\times\ell}, \qquad W = \mathbb{F}^{m\times\ell}, \qquad L(X) = MX, \qquad M \in \mathbb{F}^{m\times n},$$

and with affine-frame coordinates $\theta \in \mathbb{F}^p$ for Θ and $N(\theta) = N_0 + \sum_{i=1}^{p} \theta_i N_i$, where

$$\kappa := \dim \operatorname{Ker}(M^T), \qquad q := \dim \Theta_{\text{feas}} \text{ if } \Theta_{\text{feas}} \neq \varnothing, \text{ and } 0 \text{ otherwise.}$$

The problem is solvable in cost

$$\mathtt{T}_{p2}(m, n, \ell, p, q, \kappa)$$

with

$$\begin{aligned}\mathtt{T}_{p2}(m, n, \ell, p, q, \kappa) \in O\Big(&mn^2 + p\kappa m\ell + (\kappa\ell)p^2 \\ &+ mn^2 + n\ell \\ &+ (q+1)p\,m\ell + mn^2 + (q+1)mn\ell\Big).\end{aligned}$$

Proof. The first line is the cost to solve for feasible parameters. Compute a basis $w_1, \ldots, w_\kappa$ of $\operatorname{Ker}(M^T)$ in $O(mn^2)$. By the standard pairing,

$$x \in \operatorname{im}(M) \iff \langle w_i, x\rangle = 0 \text{ for all } i \in [\kappa].$$

Let C be the matrix with rows $w_1, \ldots, w_\kappa$. Applying columnwise to $N(\theta)$ means $\rho(N(\theta)) = 0$ exactly when $CN(\theta) = 0$. Recall $N(\theta)$ is coordinatized as $N(\theta) = N_0 + \sum_{i=1}^{p} \theta_i N_i$. Thus we solve the inhomogeneous system $A_0 + \sum_{i=1}^{p} \theta_i A_i = 0$ with $A_i := CN_i$. Hence the three costs in line 1 are $O(mn^2)$ to compute w_i, $O(p\kappa m\ell)$ to form A_i, and $O((\kappa\ell)p^2)$ to solve the $(\kappa\ell) \times p$ system.

The second line is to compute a basis for the kernel of L by computing a basis of $\operatorname{Ker}(M)$ in $O(mn^2)$ and using the columnwise description $L(X) = MX$.

The third line is to solve on an affine frame of the feasible parameters. We evaluate N on this affine frame of $(q+1)$ points in $O((q+1)p\,m\ell)$, row-reduce M once in $O(mn^2)$, and solve the $(q+1)$ right-hand sides with that reduction in $O((q+1)mn\ell)$. □

Example 3.17. In the following example, we have $m = 3, n = 2, p = 2$, and $\ell = 2$. And set

$$M = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \in \mathbb{F}^{3\times 2}, \qquad N : \mathbb{F}^2 \to \mathbb{F}^{3\times 2}, \qquad N(\theta_1, \theta_2) = N_0 + \theta_1 N_1 + \theta_2 N_2,$$

$$N_0 = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ -1 & -2 \end{bmatrix}, \qquad N_1 = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 1 & 2 \end{bmatrix}, \qquad N_2 = \begin{bmatrix} 0 & -1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}.$$

In the three-part process

(1) **Feasible parameters.** Here $\mathrm{im}(M) = \{(u, v, 0)^T : u, v \in \mathbb{F}\}$, so feasibility is exactly vanishing of the third row of $N(\theta_1, \theta_2)$. This is accomplished by the condition

$$\theta_1 + \theta_2 = 1.$$

Hence $\Theta_{\text{feas}} = \{(\theta_1, \theta_2) \in \mathbb{F}^2 : \theta_1 + \theta_2 = 1\}$.

(2) **Kernel.** $\mathrm{Ker}(M) = \{0\}$, so $\mathrm{Ker}(L) = \{0\}$.

(3) **Solve on an affine frame of Θ_{feas}.** Take frame points $\theta^{(0)} = (1, 0)$ and $\theta^{(1)} = (0, 1)$. Solving $MX = N(\theta^{(0)})$ and $MX = N(\theta^{(1)})$ gives

$$X^{(0)} = \begin{bmatrix} 3 & -5 \\ 0 & 4 \end{bmatrix}, \qquad X^{(1)} = \begin{bmatrix} 0 & -10 \\ 1 & 6 \end{bmatrix}.$$

Therefore parametrizing $\theta \in \Theta_{\text{feas}}$ as $\theta = (1 - \lambda, \lambda)$,

$$X(\theta) = X^{(0)} + \lambda\big(X^{(1)} - X^{(0)}\big).$$

Remark 3.18. If L has a left inverse G, then for each $\theta \in \Theta_{\text{feas}}$, $L(u) = N(\theta)$ has exactly one solution, $u = G(N(\theta))$.

The compatability criterion has to be met in the lifting process from subproblem solutions are instances of the following problem.

Problem AffinePullback($\mathbb{F}$).

Given: $\mathbb{F}$-vector spaces U, V, W and affine maps $f : U \to W$, $g : V \to W$.
Return: A computational encoding of $P := \{(u, v) \in U \times V : f(u) = g(v)\}$.

The set P is either empty or an affine subspace of $U \times V$, as $P = h^{-1}(\{0\})$ for $h(u, v) := f(u) - g(v)$.

Proposition 3.19. Given a coordinate instance of AffinePullback($\mathbb{F}$) with

$$U = \mathbb{F}^d, \quad V = \mathbb{F}^e, \quad W = \mathbb{F}^m, \quad f(u) = Au + \alpha, \quad g(v) = Bv + \beta,$$

and $A \in \mathbb{F}^{m \times d}$ and $B \in \mathbb{F}^{m \times e}$, it is solvable in cost

$$\mathtt{T}_{p1}(m, d + e, 1).$$

Proof. The condition $f(u) = g(v)$ is the block linear system

$$\begin{bmatrix} A & -B \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \beta - \alpha,$$

which is one instance of LinSolve($\mathbb{F}$). □

3.3. **Simultaneous Sylvester Equations for $\mathrm{Adj}(T)$.** We work towards Theorem 3.21 in this section. Throughout, a, b, c, r, s will be natural numbers, and we use the notation

$$\mathcal{X} := \mathbb{F}^{a \times r}, \qquad \mathcal{Y} := \mathbb{F}^{s \times b},$$

to denote vector spaces. This notation may have subscripts to indicate restricted solution spaces. As an example, for $I \subseteq [a]$, $\mathcal{X}_I$ is the space $\mathbb{F}^{|I| \times r}$.

Problem SimultaneousSylvesterSystem.

Given: Tensors $R \in \mathbb{F}^{r \times b \times c}$, $S \in \mathbb{F}^{a \times s \times c}$, and $T \in \mathbb{F}^{a \times b \times c}$.
Return: Either return $\varnothing$, or the affine subspace

$$\{(X, Y) \in \mathcal{X} \times \mathcal{Y} : (\forall k \in [c])\ X\, R[:, :, k] + S[:, :, k]\, Y = T[:, :, k]\}.$$

This problem has a naive solution using LinSolve($\mathbb{F}$) with $m = abc$, $n = ar + bs$, and $\ell = 1$. When $R = S$ and $T = 0$, solution to the above problem computes the adjoint of the tensor R.

Example 3.20. Below is an example of laying out the equations for SimultaneousSylvesterSystem as a naive matrix vector system. Take $a = 2$, $b = 2$, $c = 2$, $r = 3$, and $s = 4$, with slices

$$R[:,:,1] = \begin{bmatrix} 1 & 5 \\ 0 & -1 \\ 2 & 11 \end{bmatrix}, \qquad R[:,:,2] = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ -1 & 1 \end{bmatrix},$$

$$S[:,:,1] = \begin{bmatrix} 1 & 0 & 3 & 0 \\ 1 & 1 & 10 & -1 \end{bmatrix}, \qquad S[:,:,2] = \begin{bmatrix} 0 & 1 & 7 & -1 \\ 1 & -1 & -4 & 1 \end{bmatrix},$$

$$T[:,:,1] = \begin{bmatrix} 1 & 19 \\ -11 & 4 \end{bmatrix}, \qquad T[:,:,2] = \begin{bmatrix} -10 & 4 \\ 8 & 2 \end{bmatrix}.$$

The corresponding augmented linear system is shown in Figure 1.

ijk	x_{00}	x_{01}	x_{02}	x_{10}	x_{11}	x_{12}	y_{00}	y_{10}	y_{20}	y_{30}	y_{01}	y_{11}	y_{21}	y_{31}	t_{ijk}
000	1	0	2				1	0	3	0					1
001	0	1	−1				0	1	7	−1					−10
010	5	−1	11								1	0	3	0	19
011	1	1	1								0	1	7	−1	4
100				1	0	2	1	1	10	−1					−11
101				0	1	−1	1	−1	−4	1					8
110				5	−1	11					1	1	10	−1	4
111				1	1	1					1	−1	−4	1	2

FIGURE 1. An augmented matrix for the Simultaneous Sylvester System in Example 3.20.

We describe an alternative approach, culminating in the following.

Theorem 3.21. For cubic ($n \times n \times n$) tensors, under the regularity conditions (established in Definition 3.32), SimultaneousSylvesterSystem is solved in the same cost as the full deterministic verification. This is $O(n^4)$ cost, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

Example 3.22. We use the following running example throughout Section 3.3. Work over $\mathbb{F} = \mathbb{Q}$, with $a = b = r = s = 2$ and $c = 2$. Define

$$R[:,:,1] = S[:,:,1] = T[:,:,1] = \operatorname{diag}(1,0),$$
$$R[:,:,2] = S[:,:,2] = \operatorname{diag}(0,1), \quad T[:,:,2] = \operatorname{diag}(0,2).$$

The full solution family is

$$X = \operatorname{diag}(\alpha, \beta), \qquad Y = \operatorname{diag}(1-\alpha, 2-\beta), \qquad \alpha, \beta \in \mathbb{F}.$$

Our strategy for SimultaneousSylvesterSystem is to solve smaller systems with a subset of the constraints, and then lift them. We start with the following.

Problem DoubleRestricted.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times c}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$ with $|I| = a'$ and $|J| = b'$.

Return: Return $\varnothing$, or the affine subspace

$$\mathcal{S}_{\mathrm{DR}} := \{(X_I, Y_J) \in \mathcal{X}_I \times \mathcal{Y}_J : (\forall k \in [c])\ X_I R[:,J,k] + S[I,:,k] Y_J = T[I,J,k]\}.$$

To solve, vectorizing yields an instance of LinSolve($\mathbb{F}$) which costs

$$\mathtt{T}_{p1}(m = a'b'c,\ n = a'r + b's,\ \ell = 1) \in O\big((a'b'c)(a'r + b's)^2\big).$$

We refer to this cost as $\mathtt{T}_{\mathrm{DR}}(a,b,c,r,s,a',b')$. When context is clear, we drop the parameters and the cost is denoted $\mathtt{T}_{\mathrm{DR}}$. By Proposition 3.12, the returned solution set is affine.

Example 3.23. In the running example, take $I = J = \{1\}$. Then $a' = b' = 1$, and

$$X_I = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \in \mathbb{F}^{1\times 2}, \qquad Y_J = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{F}^{2\times 1}.$$

Here DoubleRestricted $x_1 + y_1 = 1$, with no other constraints on (x_2, y_2). Hence

$$\mathcal{S}_{\text{DR}} = \left\{ \left(\begin{bmatrix} \alpha & \beta \end{bmatrix}, \begin{bmatrix} 1-\alpha \\ \gamma \end{bmatrix} \right) : \alpha, \beta, \gamma \in \mathbb{F} \right\}.$$

Problem RowRestricted.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times c}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$ with $|I| = a'$ and $|J| = b'$.

Return: Return $\varnothing$, or the affine subspace

$$\mathcal{S}_{\text{R}} := \{(X_I, Y) \in \mathcal{X}_I \times \mathcal{Y} : (\forall k \in [c])\ X_I R[:,:,k] + S[I,:,k]Y = T[I,:,k]\}.$$

Proposition 3.24. For fixed $I \subseteq [a]$ and $J \subseteq [b]$, RowRestricted is solved by solving DoubleRestricted and then LinearEqualsAffine($\mathbb{F}$), with cost

$$\begin{aligned} O\big(&\mathtt{T}_{p1}(m = a'b'c,\ n = a'r + b's,\ \ell = 1) \\ &+ d\,c\,a'r(b-b') \\ &+ \mathtt{T}_{p2}(m = ca',\ n = s,\ \ell = b-b',\ p = d,\ q = q_R,\ \kappa = \kappa_R)\big). \end{aligned}$$

where $a' = |I|$, $b' = |J|$, d is the dimension of the solution set of DoubleRestricted, q_R is the feasible parameter dimension, and κ_R is the left nullspace dimension parameter in the LinearEqualsAffine($\mathbb{F}$) instance (these three parameters are all 0 if $\varnothing$ is returned)

Proof. **Algorithm. Step 1.** Solve DoubleRestricted. If it returns $\varnothing$, return $\varnothing$ and stop.

Step 2. Set $\widehat{J} := [b] \setminus J$ and define an affine map $N_R : \mathcal{S}_{\text{DR}} \to \mathbb{F}^{(ca')\times(b-b')}$ as well as an element $M_R \in \mathbb{F}^{(ca')\times s}$ by

$$N_R(X_I, Y_J) := \begin{bmatrix} T[I,\widehat{J},1] - X_I R[:,\widehat{J},1] \\ \vdots \\ T[I,\widehat{J},c] - X_I R[:,\widehat{J},c] \end{bmatrix} \in \mathbb{F}^{(ca')\times(b-b')}, \qquad M_R := \begin{bmatrix} S[I,:,1] \\ \vdots \\ S[I,:,c] \end{bmatrix} \in \mathbb{F}^{(ca')\times s} \tag{3.1}$$

Apply LinearEqualsAffine($\mathbb{F}$) to

$$\begin{aligned} &U = \mathcal{Y}_{\widehat{J}}, \quad W = \mathbb{F}^{(ca')\times(b-b')}, \\ &L_R : U \to W, \quad \widehat{Y} \mapsto M_R\widehat{Y}, \quad \Theta = \mathcal{S}_{\text{DR}}, \quad N = N_R. \end{aligned}$$

This returns either $\varnothing$, in which case our algorithm stops, or an affine subspace

$$\mathcal{A}_R \subseteq \mathcal{S}_{\text{DR}} \times \mathcal{Y}_{\widehat{J}}$$

of pairs $((X_I, Y_J), \widehat{Y})$ satisfying $M_R\widehat{Y} = N_R(X_I, Y_J)$.

Step 3. For each $((X_I, Y_J), \widehat{Y}) \in \mathcal{A}_R$, form the unique $Y \in \mathcal{Y}$ such that $Y[:,J] = Y_J$ and $Y[:,\widehat{J}] = \widehat{Y}$. Return the affine subspace of $\mathcal{X}_I \times \mathcal{Y}$ obtained by sending each $((X_I, Y_J), \widehat{Y}) \in \mathcal{A}_R$ to (X_I, Y).

Timing. Step 1 is one call to DoubleRestricted, costing $\mathtt{T}_{\text{DR}}(a,b,c,r,s,a',b')$. Step 2 has two contributions. Recall $\dim(\mathcal{S}_{\text{DR}}) = d$, and we are given an affine frame $(X_{I,i}, Y_{J,i})_{i=0}^{d}$. Building the resulting coordinatized affine map N_R from Equation (3.1) costs

$$O\big(d\,c\,a'r(b-b')\big),$$

because each of the d matrices has size $a' \times r$, each $R[:,\widehat{J},k]$ has size $r \times (b-b')$, and we compute c multiplications between them. Invoking Proposition 3.16 on (M_R, N_R) adds $\mathtt{T}_{p2}(m = ca',\ n = s,\ \ell = b-b',\ p = d,\ q = q_R,\ \kappa = \kappa_R)$. So the total cost is

$$O\big(\mathtt{T}_{\text{DR}} + d\,c\,a'r(b-b') + \mathtt{T}_{p2}\big).$$

Correctness. Step 1 gives exactly $\mathcal{S}_{\text{DR}}$, i.e. all (X_I, Y_J) that satisfy the (I, J)-restricted equations. Step 2 gives exactly the elements $\widehat{Y}$ for which the combined Y and $\widehat{Y}$ satisfies the row restricted equation. Step 3 combines $(Y_J, \widehat{Y})$ into Y, so $\mathcal{S}_{\text{R}}$ is exactly the set of row-restricted solutions. □

Example 3.25. Continue the running example with $I = J = \{1\}$, so $\widehat{J} = \{2\}$. From Example 3.23, write

$$X_I = \begin{bmatrix} \alpha & \beta \end{bmatrix}, \qquad Y_J = \begin{bmatrix} 1-\alpha \\ \gamma \end{bmatrix}.$$

Step 2 of Proposition 3.24 uses

$$M_R = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad N_R(X_I, Y_J) = \begin{bmatrix} 0 \\ -\beta \end{bmatrix}.$$

Solving $M_R \widehat{Y} = N_R(X_I, Y_J)$ forces $\widehat{Y} = \begin{bmatrix} 0 \\ \delta \end{bmatrix}$ and $\beta = 0$. Therefore

$$\mathcal{S}_{\text{R}} = \left\{ \left(\begin{bmatrix} \alpha & 0 \end{bmatrix}, \begin{bmatrix} 1-\alpha & 0 \\ \gamma & \delta \end{bmatrix} \right) : \alpha, \gamma, \delta \in \mathbb{F} \right\}.$$

Problem ColumnRestricted.

Given: Tensors $R \in \mathbb{F}^{r \times b \times c}$, $S \in \mathbb{F}^{a \times s \times c}$, $T \in \mathbb{F}^{a \times b \times c}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$ with $|I| = a'$ and $|J| = b'$.

Return: Return $\varnothing$, or the affine subspace

$$\mathcal{S}_{\text{C}} := \{(X, Y_J) \in \mathcal{X} \times \mathcal{Y}_J : (\forall k \in [c])\ XR[:, J, k] + S[:, :, k]Y_J = T[:, J, k]\}.$$

Proposition 3.26. For fixed $I \subseteq [a]$ and $J \subseteq [b]$, ColumnRestricted is solved by solving DoubleRestricted and then LinearEqualsAffine($\mathbb{F}$), in cost

$$\begin{aligned} O\big(&\mathtt{T}_{p1}(m = a'b'c,\ n = a'r + b's,\ \ell = 1) \\ &+ d\,c\,(a - a')sb' \\ &+ \mathtt{T}_{p2}(m = cb',\ n = r,\ \ell = a - a',\ p = d,\ q = q_C,\ \kappa = \kappa_C)\big). \end{aligned}$$

where $a' = |I|$, $b' = |J|$, d is the dimension of the solution set of DoubleRestricted, q_C is the feasible parameter dimension, and κ_C is the left nullspace dimension parameter in the LinearEqualsAffine($\mathbb{F}$) instance (these three parameters are all 0 if $\varnothing$ is returned)

Proof. **Algorithm. Step 1.** Solve DoubleRestricted. If it returns $\varnothing$, return $\varnothing$ and stop.

Step 2. Set $\widehat{I} := [a] \setminus I$ and define an affine map $N_C : \mathcal{S}_{\text{DR}} \to \mathbb{F}^{(cb') \times (a-a')}$ as well as an element $M_C \in \mathbb{F}^{(cb') \times r}$ by

$$N_C(X_I, Y_J) := \begin{bmatrix} (T[\widehat{I}, J, 1] - S[\widehat{I}, :, 1]Y_J)^T \\ \vdots \\ (T[\widehat{I}, J, c] - S[\widehat{I}, :, c]Y_J)^T \end{bmatrix} \in \mathbb{F}^{(cb') \times (a-a')}, \qquad M_C := \begin{bmatrix} (R[:, J, 1])^T \\ \vdots \\ (R[:, J, c])^T \end{bmatrix} \in \mathbb{F}^{(cb') \times r}.$$

Apply LinearEqualsAffine($\mathbb{F}$) to

$$\begin{aligned} &U = \mathcal{X}_{\widehat{I}}, \quad W = \mathbb{F}^{(cb') \times (a-a')}, \\ &L_C : U \to W, \quad \widehat{X} \mapsto M_C \widehat{X}, \quad \Theta = \mathcal{S}_{\text{DR}}, \quad N = N_C. \end{aligned}$$

This returns either $\varnothing$, or an affine subspace

$$\mathcal{A}_C \subseteq \mathcal{S}_{\text{DR}} \times \mathcal{X}_{\widehat{I}}$$

of pairs $((X_I, Y_J), \widehat{X})$ satisfying $M_C \widehat{X} = N_C(X_I, Y_J)$. If it returns $\varnothing$, return $\varnothing$ for ColumnRestricted and stop.

Step 3. For each $((X_I, Y_J), \widehat{X}) \in \mathcal{A}_C$, form the unique $X \in \mathcal{X}$ such that $X[I, :] = X_I$ and $X[\widehat{I}, :] = \widehat{X}^T$. Return $\mathcal{S}_{\text{C}}$, the affine subspace of $\mathcal{X} \times \mathcal{Y}_J$ obtained by sending each $((X_I, Y_J), \widehat{X}) \in \mathcal{A}_C$ to (X, Y_J).

Timing. The same cost accounting as Proposition 3.24. Step 1 costs $\mathtt{T}_{\mathrm{DR}}(a,b,c,r,s,a',b')$, Step 2 has cost in $O\big(d\,c\,(a-a')sb' + \mathtt{T}_{p2}(m = cb',\ n = r,\ \ell = a-a',\ p = d,\ q = q_C,\ \kappa = \kappa_C)\big)$. So the total cost is

$$O\big(\mathtt{T}_{\mathrm{DR}} + d\,c\,(a-a')sb' + \mathtt{T}_{p2}\big).$$

Correctness. Same as Proposition 3.24. □

Example 3.27. Continue the running example with $I = J = \{1\}$, so $\widehat{I} = \{2\}$. From Example 3.23, write

$$X_I = \begin{bmatrix}\alpha & \beta\end{bmatrix}, \qquad Y_J = \begin{bmatrix}1-\alpha \\ \gamma\end{bmatrix}.$$

Step 2 of Proposition 3.26 gives

$$M_C = \begin{bmatrix}1 & 0 \\ 0 & 0\end{bmatrix}, \qquad M_C\widehat{X} = N_C(X_I, Y_J) = \begin{bmatrix}0 \\ -\gamma\end{bmatrix},$$

with $\widehat{X}$ as the unknown. Solving gives $\widehat{X} = \begin{bmatrix}0 \\ \epsilon\end{bmatrix}$ and $\gamma = 0$. Therefore

$$\mathcal{S}_{\mathrm{C}} = \left\{\left(\begin{bmatrix}\alpha & \beta \\ 0 & \epsilon\end{bmatrix}, \begin{bmatrix}1-\alpha \\ 0\end{bmatrix}\right) : \alpha, \beta, \epsilon \in \mathbb{F}\right\}.$$

Problem LiftRestricted.

Given: Solutions to RowRestricted and ColumnRestricted.
Return: Solution to SimultaneousSylvesterSystem.

Solving RowRestricted and ColumnRestricted gives $\mathcal{S}_{\mathrm{R}} \subseteq \mathcal{X}_I \times \mathcal{Y}$ and $\mathcal{S}_{\mathrm{C}} \subseteq \mathcal{X} \times \mathcal{Y}_J$. For $z_R = (X_I, Y) \in \mathcal{S}_{\mathrm{R}}$ and $z_C = (X, Y_J) \in \mathcal{S}_{\mathrm{C}}$, define

$$\pi_J : \mathcal{X}_I \times \mathcal{Y} \to \mathcal{X}_I \times \mathcal{Y}_J, \qquad \pi_J(z_R) := (X_I, Y[:, J]),$$
$$\pi_I : \mathcal{X} \times \mathcal{Y}_J \to \mathcal{X}_I \times \mathcal{Y}_J, \qquad \pi_I(z_C) := (X[I, :], Y_J).$$

Proposition 3.28. Let $\mathcal{S}_{\mathrm{R}} \subseteq \mathcal{X}_I \times \mathcal{Y}$, and $\mathcal{S}_{\mathrm{C}} \subseteq \mathcal{X} \times \mathcal{Y}_J$ be the solutions of RowRestricted and ColumnRestricted. Let the pullback of the projection maps be defined as

$$P := \{(z_R, z_C) \in \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}} : \pi_J(z_R) = \pi_I(z_C)\},$$

the output map be defined as

$$\Psi : P \to \mathcal{X} \times \mathcal{Y}, \qquad \Psi\big((X_I, Y), (X, Y_J)\big) := (X, Y),$$

and the error map be defined as

$$\mathrm{Err} : \mathcal{X} \times \mathcal{Y} \to \mathbb{F}^{a\times b\times c}, \qquad \mathrm{Err}(X, Y) := \big(XR[:,:,k] + S[:,:,k]Y - T[:,:,k]\big)_{k\in[c]}.$$

Then $\{\Psi(z) : z \in P \text{ and } \mathrm{Err}(\Psi(z)) = 0\}$ is the solution set to LiftRestricted.

Proof. The restricted pairs agreeing from $\mathcal{S}_{\mathrm{R}}$ and $\mathcal{S}_{\mathrm{C}}$ are exactly P. For each $z \in P$, $\Psi(z)$ is a candidate solution. This candidate is in the solution set of SimultaneousSylvesterSystem exactly when its residual is zero. □

Proposition 3.29. Let $\mathcal{S}_{\mathrm{R}}, \mathcal{S}_{\mathrm{C}}$ be solutions to RowRestricted and ColumnRestricted, and let

$$d := \max\{\dim \mathcal{S}_{\mathrm{R}}, \dim \mathcal{S}_{\mathrm{C}}\}, \qquad P := \{(z_R, z_C) \in \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}} : \pi_J(z_R) = \pi_I(z_C)\},$$
$$\tau := \dim P \text{ if } P \neq \varnothing, \text{ and } 0 \text{ otherwise.}$$

Then LiftRestricted is solvable in cost

$$O\big((a'r + b's)d^2 + \tau\,c\,a\,b\,(r+s) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1)\big).$$

Proof. **Algorithm.**

If either RowRestricted or ColumnRestricted returns $\varnothing$, then we return $\varnothing$ and stop.

Step 1 (compute the candidates). From the outputs of RowRestricted and ColumnRestricted, we get the affine frames

$$z_{R,0}, \dots, z_{R,d_R} \in \mathcal{S}_{\mathrm{R}}, \qquad z_{C,0}, \dots, z_{C,d_C} \in \mathcal{S}_{\mathrm{C}},$$

and thereby a coordinate realization of the pullback P, which is solved for via an instance of AffinePullback($\mathbb{F}$) with

$$\begin{aligned} &U = \mathbb{F}^{d_R}, \quad V = \mathbb{F}^{d_C}, \quad W = \mathcal{X}_I \times \mathcal{Y}_J, \\ &f : U \to W,\ \lambda \mapsto \pi_J(\chi_R(\lambda)), \quad g : V \to W,\ \mu \mapsto \pi_I(\chi_C(\mu)), \end{aligned}$$

where χ_R, χ_C are the coordinate maps to $\mathcal{S}_{\mathrm{R}}$ and $\mathcal{S}_{\mathrm{C}}$.

If $P = \varnothing$, return $\varnothing$. Otherwise output the frame $(p_0, \dots, p_\tau) \subseteq \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}}$ from the coordinate pullback solutions.

Step 2 (compute the error map on a frame). For each $t \in \{0, \dots, \tau\}$, compute

$$(X_t, Y_t) := \Psi(p_t), \qquad E_t := \mathrm{Err}(X_t, Y_t) \in \mathbb{F}^{a \times b \times c},$$

and vectorize $e_t := \mathrm{vec}(E_t) \in \mathbb{F}^{abc}$.

If $\tau = 0$, this means at most one solution exists. We check if $e_0 = 0$. In the case $e_0 \neq 0$, return $\varnothing$. When $e_0 = 0$, the solution is the unique point $\Psi(p_0)$.

Else, $\tau > 0$, so we solve the inhomogeneous linear system

$$Mx = -e_0, \qquad M := \begin{bmatrix} e_1 - e_0 & \cdots & e_\tau - e_0 \end{bmatrix} \in \mathbb{F}^{abc \times \tau}$$

in the frame coordinates via LinSolve($\mathbb{F}$). If it returns $\varnothing$, return $\varnothing$. Otherwise output the frame of solutions in $\mathcal{X} \times \mathcal{Y}$ by applying Ψ to the coordinate solutions of the linear system.

Timing. Step 1 is one instance of AffinePullback($\mathbb{F}$), with $d = d_R$, $e = d_C$, and $m = a'r + sb'$. Under the assumptions that d_R, d_C are $O(d)$, Proposition 3.19 takes $O(d^2(a'r + sb'))$ operations. Evaluating Err at one point costs $O(c\,a\,b\,(r + s))$, so all $\tau + 1$ evaluations cost $O(\tau\, c\,a\,b\,(r + s))$. Solving $Mx = -e_0$ is one call to LinSolve($\mathbb{F}$) with $m = abc$, $n = \tau$, and $\ell = 1$, so it costs $\mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1)$. □

Example 3.30. From Examples 3.25 and 3.27, a compatible pair in the pullback has the form

$$X = \begin{bmatrix} \alpha & 0 \\ 0 & \epsilon \end{bmatrix}, \qquad Y = \begin{bmatrix} 1 - \alpha & 0 \\ 0 & \delta \end{bmatrix}.$$

for $\alpha, \epsilon, \delta \in \mathbb{F}$. Step 2 checks

$$\mathrm{Err}(X, Y) = \big(XR[:,:,k] + S[:,:,k]Y - T[:,:,k]\big)_{k \in [2]},$$

and the remaining condition is $\epsilon + \delta = 2$. Therefore LiftRestricted returns the same as the naive solution after renaming ϵ to β, namely $X = \mathrm{diag}(\alpha, \beta)$, and $Y = \mathrm{diag}(1 - \alpha, 2 - \beta)$ for $\alpha, \beta \in \mathbb{F}$.

Now we are ready to solve SimultaneousSylvesterSystem.

Theorem 3.31. Let $I \subseteq [a]$ and $J \subseteq [b]$. Then SimultaneousSylvesterSystem is solvable in cost

$$\mathtt{T}_{\mathrm{DR}} + \mathtt{T}_{\mathrm{RR}} + \mathtt{T}_{\mathrm{CR}} + \mathtt{T}_{\mathrm{lift}},$$

where $\mathtt{T}_{\mathrm{DR}}$ is the cost for DoubleRestricted, $\mathtt{T}_{\mathrm{RR}}, \mathtt{T}_{\mathrm{CR}}$ are the costs from RowRestricted and ColumnRestricted, not including the double restricted solve, and $\mathtt{T}_{\mathrm{lift}}$ the cost for LiftRestricted.

Proof. By correctness of DoubleRestricted, we obtain the doubly-restricted solution family for (I, J). By Propositions 3.24 and 3.26, the row- and column-restricted families computed from that data are exactly the solutions of RowRestricted and ColumnRestricted. Then LiftRestricted is applied to these two families. Its correctness means we solve SimultaneousSylvesterSystem. □

An end-to-end example can be found in Example B.1.

Finally we describe a class of problems that this solve and lift algorithm performs best on. **TODO: "regularity condition" for quicksylver needs more precision.**

Definition 3.32. The *regularity* condition that is the hypothesis of Theorem 3.21 is when

$$a, b, c, r, s \in O(n),$$

$I \subseteq [a], J \subseteq [b]$ satisfies

$$a' = |I| \in O(1), \qquad b' = |J| \in O(1),$$

and $\dim \mathcal{S}_{\mathrm{DR}},\ q_R,\ q_C,\ \tau \in O(1)$. Furthermore, the row/column restricted systems have the property that

$$N_R(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_R), \qquad N_C(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_C), \qquad \mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \{0\}.$$

The above assumptions are plausible in the cubic case ($n \times n \times n$ tensors, with $r = s = n$ as well). When $a', b' \in O(1)$, the matrices M_R and M_C have dimensions

$$M_R \in \mathbb{F}^{(a'c)\times s}, \qquad M_C \in \mathbb{F}^{(b'c)\times r}$$

respectively. In the cubic regime, both the row counts and column counts of these systems are in $O(n)$, and for generic dense slice data one expects full column rank, so $\mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \{0\}$ is likely.

The inclusions $N_R(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_R)$ and $N_C(\mathcal{S}_{\mathrm{DR}}) \subseteq \mathrm{im}(M_C)$ assert that all double restricted solutions can be lifted. Whether this condition is satisfied is particular to the specific instance of the question.

Proof of Theorem 3.21. Assume the conditions in Definition 3.32. For the doubly-restricted step,

$$\mathtt{T}_{\mathrm{DR}} \in O\big((a'b'c)(a'r + b's)^2\big),$$

which is $O(n^3)$ since $a', b' \in O(1)$ and $c, r, s \in O(n)$.

For the row-restricted step, Proposition 3.24 gives

$$\mathtt{T}_{\mathrm{RR}} \in O\big(\mathtt{T}_{\mathrm{DR}} + d_{\mathrm{DR}}\, c\, a' r\, (b - b') + \mathtt{T}_{p2}(m = ca',\ n = s,\ \ell = (b - b'),\ p = d_{\mathrm{DR}},\ q = q_R,\ \kappa = \kappa_R)\big).$$

Here $ca', s, (b - b') \in O(n)$ and $d_{\mathrm{DR}}, q_R \in O(1)$. Also $\kappa_R \le ca'$, so in this regime $\kappa_R \in O(n)$, and the $\mathtt{T}_{p2}$ term is still $O(n^3)$. Therefore $\mathtt{T}_{\mathrm{RR}} \in O(n^3)$. The same argument gives $\mathtt{T}_{\mathrm{CR}} \in O(n^3)$.

For the lift step, Proposition 3.29 gives

$$\mathtt{T}_{\mathrm{lift}} \in O\big((a'r + b's)d^2 + \tau\, c\, a\, b\, (r + s) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1)\big),$$

with $d = \max\{\dim \mathcal{S}_{\mathrm{R}}, \dim \mathcal{S}_{\mathrm{C}}\}$. Under Definition 3.32, we have $\dim \mathcal{S}_{\mathrm{R}} = \dim \mathcal{S}_{\mathrm{C}} = \dim \mathcal{S}_{\mathrm{DR}} \in O(1)$, so the first term is $O(n)$, the last is $O(n^3)$, and

$$\tau\, c\, a\, b\, (r + s) \in O(n^4).$$

Hence $\mathtt{T}_{\mathrm{lift}} \in O(n^4)$, so the total solve-and-lift cost is $O(n^4)$.

Finally, full deterministic verification checks all abc constraints and has cost $O(cab(r + s)) = O(n^4)$. Therefore SimultaneousSylvesterSystem is solved in the same order of arithmetic complexity as full deterministic verification. □

3.4. **Derivation Equations for** $\mathrm{Der}(T)$**.** In this section, we work towards Theorem 3.33. Throughout, a, b, c, r, s, t will be natural numbers, and we denote the spaces

$$\mathcal{X} := \mathbb{F}^{a\times r}, \qquad \mathcal{Y} := \mathbb{F}^{s\times b}, \qquad \mathcal{Z} := \mathbb{F}^{t\times c}.$$

This notation may have subscripts to indicate restricted solution spaces. As an example, for $I \subseteq [a]$, $\mathcal{X}_I$ is the space $\mathbb{F}^{|I|\times r}$. For a matrix $Z \in \mathbb{F}^{t\times c}$, define T^Z, the outer action of Z on T, by specifying its slices for $k \in [c]$ as

$$(T^Z)[:, :, k] := \sum_{j\in[t]} T[:, :, j]\, Z[j, k]. \tag{3.2}$$

Problem DerivationSystem.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$.
Return: Either return $\varnothing$, or the affine subspace

$$\left\{(X,Y,Z) \in \mathcal{X}\times\mathcal{Y}\times\mathcal{Z} : (\forall k \in [c])\ X\,R[:,:,k] + S[:,:,k]\,Y = (T^Z)[:,:,k]\right\}.$$

This problem has a naive solution using LinSolve($\mathbb{F}$) with $m = abc$, $n = ar + bs + ct$, and $\ell = 1$. When $R = S = T$, solutions to the equation correspond to the derivation algebra of a tensor T, as seen in Equation (1.1). We describe an alternative approach, culminating in the following.

Theorem 3.33. For cubic ($n \times n \times n$) tensors, under the regularity conditions (established in Definition 3.38), DerivationSystem is solved in cost $O(n^{4.5})$. This is an additional $n^{1/2}$ overhead compared to the deterministic verification cost of $O(n^4)$, as compared to the naive $O(n^7)$ cost by solving as a linear system of equations.

As in the above section, we solve the derivation system by solving on a subset of constraints, and then lifting those solutions to the full system.

Problem TripleRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$ and index sets $I \subseteq [a], J \subseteq [b], K \subseteq [c]$, with $|I| = a'$, $|J| = b'$, and $|K| = c'$.
Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{I,J,K} := \left\{(X_I, Y_J, Z_K) \in \mathcal{X}_I\times\mathcal{Y}_J\times\mathcal{Z}_K : (\forall k \in K)\ X_I\,R[:,J,k] + S[I,:,k]\,Y_J = (T^{Z_K})[I,J,k]\right\}.$$

This is a dense linear system with $ra' + sb' + tc'$ unknowns and $a'b'c'$ equations. Hence it is solvable in cost

$$\mathtt{T}_{p1}(m = a'b'c',\ n = ra' + sb' + tc',\ \ell = 1). \tag{3.3}$$

We now solve three restricted systems which gives solution families X, Y, Z, respectively.

Problem ColDepthRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$, $K \subseteq [c]$ with $|I| = a'$, $|J| = b'$, and $|K| = c'$.
Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{[a],J,K} := \left\{(X, Y_J, Z_K) \in \mathcal{X}\times\mathcal{Y}_J\times\mathcal{Z}_K : (\forall k \in K)\ X\,R[:,J,k] + S[:,:,k]\,Y_J = (T^{Z_K})[:,J,k]\right\}.$$

Problem RowDepthRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$, $K \subseteq [c]$ with $|I| = a'$, $|J| = b'$, and $|K| = c'$.
Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{I,[b],K} := \left\{(X_I, Y, Z_K) \in \mathcal{X}_I\times\mathcal{Y}\times\mathcal{Z}_K : (\forall k \in K)\ X_I\,R[:,:,k] + S[I,:,k]\,Y = (T^{Z_K})[I,:,k]\right\}.$$

Problem RowColRestrictedDer.

Given: Tensors $R \in \mathbb{F}^{r\times b\times c}$, $S \in \mathbb{F}^{a\times s\times c}$, $T \in \mathbb{F}^{a\times b\times t}$, and index sets $I \subseteq [a]$, $J \subseteq [b]$, $K \subseteq [c]$ with $|I| = a'$, $|J| = b'$, and $|K| = c'$.
Return: Either return $\varnothing$, or the affine subspace

$$\mathcal{D}_{I,J,[c]} := \left\{(X_I, Y_J, Z) \in \mathcal{X}_I\times\mathcal{Y}_J\times\mathcal{Z} : (\forall k \in [c])\ X_I\,R[:,J,k] + S[I,:,k]\,Y_J = (T^{Z})[I,J,k]\right\}.$$

Proposition 3.34. Each of ColDepthRestrictedDer, RowDepthRestrictedDer, and RowColRestrictedDer are solved via an instance of TripleRestrictedDer followed by an instance of LinearEqualsAffine($\mathbb{F}$) instance.

Proof. All three restricted problems are solved via computing $\mathcal{D}_{I,J,K}$ using TripleRestrictedDer (this is only done once and the solution is shared), and then LinearEqualsAffine($\mathbb{F}$) to find the missing block.

We detail RowDepthRestrictedDer. Let $\widehat{J} := [b] \setminus J$, and write $K = \{k_1, \ldots, k_{c'}\}$. Define the affine map $N_R : \mathcal{D}_{I,J,K} \to \mathbb{F}^{(a'c') \times (b-b')}$ and $M_R \in \mathbb{F}^{a'c' \times s}$ by

(3.4)

$$N_R(X_I, Y_J, Z_K) := \begin{bmatrix} (T^{Z_K})[I, \widehat{J}, k_1] - X_I\, R[:, \widehat{J}, k_1] \\ \vdots \\ (T^{Z_K})[I, \widehat{J}, k_{c'}] - X_I\, R[:, \widehat{J}, k_{c'}] \end{bmatrix} \in \mathbb{F}^{(a'c') \times (b-b')}, \qquad M_R := \begin{bmatrix} S[I, :, k_1] \\ \vdots \\ S[I, :, k_{c'}] \end{bmatrix} \in \mathbb{F}^{(a'c') \times s}.$$

Then, for each $(X_I, Y_J, Z_K) \in \mathcal{D}_{I,J,K}$, solve

$$M_R \widehat{Y} = N_R(X_I, Y_J, Z_K)$$

as an instance of LinearEqualsAffine($\mathbb{F}$).

ColDepthRestrictedDer is solved similarly, while RowColRestrictedDer requires some new notation to set up the system. Let $\widehat{K} = [c] \setminus K$. Define a matrix $M_D \in \mathbb{F}^{a'b' \times t}$ by specifying its jth column $M_D[:, j] (j \in [t])$ as $\mathrm{vec}(T[I, J, j]) \in \mathbb{F}^{a'b'}$.

And define the affine map $N_D : \mathcal{D}_{I,J,K} \to \mathbb{F}^{a'b' \times (c-c')}$ by specifying the jth column of its output:

$$N_D(X_I, Y_J, Z_K)[:, j] := \mathrm{vec}\big(X_I\, R[:, J, k_j] + S[I, :, k_j]\, Y_J\big) \quad (j \in [c - c'], k_j \in \widehat{K})$$

Now solving for $\widehat{Z} \in \mathbb{F}^{t \times (c-c')}$ in $M_D \widehat{Z} = N_D(X_I, Y_J, Z_K)$ as an instance of LinearEqualsAffine($\mathbb{F}$) concludes the solution. □

Proposition 3.35. Let $d := \dim \mathcal{D}_{I,J,K}$ and let (q_R, κ_R), (q_C, κ_C), (q_Z, κ_Z) be the feasible parameter and left nullspace dimensions respectively of the three LinearEqualsAffine($\mathbb{F}$) instances solving Proposition 3.34. Then

RowDepthRestrictedDer is solvable in cost

$$O\big(\mathtt{T}_{p1}(m = a'b'c',\ n = a'r + sb' + tc',\ \ell = 1) + d\, c'\, a'(b - b')\,(r + t) \\ + \mathtt{T}_{p2}(m = a'c',\ n = s,\ \ell = b - b',\ p = d,\ q = q_R,\ \kappa = \kappa_R)\big),$$

ColDepthRestrictedDer is solvable in cost

$$O\big(\mathtt{T}_{p1}(m = a'b'c',\ n = a'r + sb' + tc',\ \ell = 1) + d\, c'\,(a - a')\, b'\,(s + t) \\ + \mathtt{T}_{p2}(m = b'c',\ n = r,\ \ell = a - a',\ p = d,\ q = q_C,\ \kappa = \kappa_C)\big),$$

RowColRestrictedDer is solvable in cost

$$O\big(\mathtt{T}_{p1}(m = a'b'c',\ n = a'r + sb' + tc',\ \ell = 1) + d\,(c - c')\, a'b'\,(r + s) \\ + \mathtt{T}_{p2}(m = a'b',\ n = t,\ \ell = c - c',\ p = d,\ q = q_Z,\ \kappa = \kappa_Z)\big).$$

Proof. Each cost is read directly from the data needed to build the corresponding LinearEqualsAffine($\mathbb{F}$) instance. For RowDepthRestrictedDer, use Equation (3.4): for each $k \in K$, $X_I \in \mathbb{F}^{a' \times r}$ multiplies $R[:, \widehat{J}, k] \in \mathbb{F}^{r \times (b-b')}$, and $T^{Z_K}[I, \widehat{J}, k]$ is an $a' \times (b - b')$ block formed from $Z_K \in \mathbb{F}^{t \times c'}$ and the t slices of T. Hence each block costs $a'(b - b')(r + t)$, with c' blocks per frame point and d frame points, giving $d\, c'\, a'(b - b')(r + t)$. The ColDepthRestrictedDer and RowColRestrictedDer bounds are obtained in the same way from their stacked block dimensions. □

Problem LiftRestrictedDer.

Given: Solutions to ColDepthRestrictedDer, RowDepthRestrictedDer, and RowColRestrictedDer.

Return: The solution to DerivationSystem.

The overall strategy is the same as the solution of Problem 8. Each lifted solution space has a restriction map to $\mathcal{D}_{I,J,K}$.

$$\begin{aligned}
\rho_X &: \mathcal{D}_{[a],J,K} \to \mathcal{D}_{I,J,K}, & (X, Y_J, Z_K) &\mapsto (X[I,:], Y_J, Z_K) \\
\rho_Y &: \mathcal{D}_{I,[b],K} \to \mathcal{D}_{I,J,K}, & (X_I, Y, Z_K) &\mapsto (X_I, Y[:,J], Z_K), \\
\rho_Z &: \mathcal{D}_{I,J,[c]} \to \mathcal{D}_{I,J,K}, & (X_I, Y_J, Z) &\mapsto (X_I, Y_J, Z[:,K]).
\end{aligned}$$

We first form the 2-way pullback as an instance of Problem 3. Solving it gives compatible partial solution triples (X, Y, Z_K).

$$Q_{XY} := \{(u,v) \in \mathcal{D}_{[a],J,K} \times \mathcal{D}_{I,[b],K} : \rho_X(u) = \rho_Y(v)\},$$

and then as a second pullback calculation, solve

$$Q := \{((u,v),w) \in Q_{XY} \times \mathcal{D}_{I,J,[c]} : \rho_X(u) = \rho_Y(v) = \rho_Z(w)\}.$$

Each point of Q determines a full candidate triple $(X, Y, Z) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$.

What remains is to enforce zero residual on Q. Define

$$\mathrm{Err} : \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \to \mathbb{F}^{a \times b \times c}, \qquad \mathrm{Err}(X,Y,Z)[:,:,k] := X\, R[:,:,k] + S[:,:,k]\, Y - (T^Z)[:,:,k],$$

and solve for the candidate solutions that have zero residual error in a linear system.

An end-to-end example can be found in Example B.2.

Proposition 3.36. Let $\mathcal{D}_{I,J,K}$, $\mathcal{D}_{[a],J,K}$, $\mathcal{D}_{I,[b],K}$, and $\mathcal{D}_{I,J,[c]}$ be the affine spaces returned by TripleRestrictedDer, ColDepthRestrictedDer, and RowDepthRestrictedDer, and RowColRestrictedDer respectively. Let Q_{XY} and Q be the two pullbacks built in LiftRestrictedDer. Write

$$d := \dim \mathcal{D}_{I,J,K}, \quad d_X := \dim \mathcal{D}_{[a],J,K}, \quad d_Y := \dim \mathcal{D}_{I,[b],K}, \quad d_Z := \dim \mathcal{D}_{I,J,[c]},$$

$$d_{XY} := \dim Q_{XY}, \qquad \tau := \dim Q \text{ if } Q \neq \varnothing, \text{ and } 0 \text{ otherwise.}$$

Then LiftRestrictedDer is solvable in cost

$$\begin{aligned}
&\mathtt{T}_{p1}(m = d,\ n = d_X + d_Y,\ \ell = 1) + \mathtt{T}_{p1}(m = d,\ n = d_{XY} + d_Z,\ \ell = 1) \\
&\qquad + O\big((\tau + 1)\, abc(r + s + t)\big) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1).
\end{aligned}$$

Proof. The first two summands are the costs of the two applications of AffinePullback($\mathbb{F}$). After the second pullback, we evaluate the Err map on an affine frame

$$q_0, \ldots, q_\tau \in Q.$$

Each evaluation costs $O(abc(r + s + t))$, since for each $k \in [c]$ we must compute

$$X\, R[:,:,k], \qquad S[:,:,k]\, Y, \qquad (T^Z)[:,:,k].$$

Finally, solving for the coordinates of Q that satisfies $\mathrm{Err}(q) = 0$ requires an application of LinSolve($\mathbb{F}$) with

$$m = abc, \qquad n = \tau, \qquad \ell = 1.$$ □

Theorem 3.37. Fix $I \subseteq [a]$, $J \subseteq [b]$, and $K \subseteq [c]$. Then DerivationSystem is solvable in cost

$$\mathtt{T}_{\mathrm{core}} + \mathtt{T}_{\mathrm{col}} + \mathtt{T}_{\mathrm{row}} + \mathtt{T}_{\mathrm{slice}} + \mathtt{T}_{\mathrm{lift}},$$

where $\mathtt{T}_{\mathrm{core}}$ is the TripleRestrictedDer cost from Equation (3.3), $\mathtt{T}_{\mathrm{col}}$, $\mathtt{T}_{\mathrm{row}}$, and $\mathtt{T}_{\mathrm{slice}}$ are the three restricted lift costs from Proposition 3.35, not counting solving the triple restricted system, and $\mathtt{T}_{\mathrm{lift}}$ is the cost of computing compatible solutions from Proposition 3.36.

Proof. Same reasoning as Theorem 3.31. □

Finally we describe a class of problems that this solve and lift algorithm performs best on. **TODO: "regularity condition" for derivations needs more precision.**

Definition 3.38. The *regularity* condition that is the hypothesis of Theorem 3.33 is when

$$a, b, c, r, s, t \in O(n),$$

and for $I \subseteq [a], J \subseteq [b], K \subseteq [c]$,

$$a' = |I|,\ b' = |J|,\ c' = |K| \in O(n^{1/2}).$$

Furthermore,

$$\dim \mathcal{D}_{I,J,K},\ q_R,\ q_C,\ q_Z,\ \tau \in O(1),$$

and the three lift systems satisfy

$$N_R(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_R), \qquad N_C(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_C), \qquad N_D(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_D),$$
$$\mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \mathrm{Ker}(M_D) = \{0\}.$$

The above assumptions are plausible in the cubic case ($n \times n \times n$ tensors). The matrices used for lifting solutions have dimensions

$$M_R \in \mathbb{F}^{(a'c') \times s}, \qquad M_C \in \mathbb{F}^{(b'c') \times r}, \qquad M_D \in \mathbb{F}^{(a'b') \times t},$$

so in the cubic regime when $a', b', c' \in O(n^{1/2})$ they are all of order $n \times n$. For generic dense slice data this makes injectivity plausible, so $\mathrm{Ker}(M_R) = \mathrm{Ker}(M_C) = \mathrm{Ker}(M_D) = \{0\}$ is a reasonable condition.

The inclusions $N_R(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_R)$, $N_C(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_C)$, $N_D(\mathcal{D}_{I,J,K}) \subseteq \mathrm{im}(M_D)$ are lift-feasibility conditions. Whether they are satisfied is particular to the problem instance.

Proof of Theorem 3.33. Equation (3.3) gives

$$\mathtt{T}_{p1}(m = a'b'c',\ n = ra' + sb' + tc',\ \ell = 1).$$

Assuming Definition 3.38, we have $a', b', c' \in O(n^{1/2})$ and $r, s, t \in O(n)$, so the two matrix dimensions in Equation (3.3) are both in $O(n^{3/2})$. Therefore this term is $O(n^{4.5})$.

For the three restricted lift stages, Proposition 3.35 gives $O(n^4)$ under the same assumptions: here $d \in O(1)$, $q_R, q_C, q_Z \in O(1)$, and $\kappa_R \le a'c'$, $\kappa_C \le b'c'$, $\kappa_Z \le a'b'$. For the final lifting stage, Proposition 3.36 gives

$$\mathtt{T}_{p1}(m = d,\ n = d_X + d_Y,\ \ell = 1) + \mathtt{T}_{p1}(m = d,\ n = d_{XY} + d_Z,\ \ell = 1)$$
$$+O\big((\tau + 1)abc(r + s + t)\big) + \mathtt{T}_{p1}(m = abc,\ n = \tau,\ \ell = 1).$$

Using $d = d_X = d_Y = d_Z = d_{XY} = \tau = \dim \mathcal{D}_{I,J,K} \in O(1)$, the two pullback solves are $O(1)$, the last term is $O(abc) = O(n^3)$, and the dominant term is $O(abc(r + s + t)) = O(n^4)$. Hence the total cost is $O(n^{4.5})$.

□

4. Nuclei of Product of Bimaps

In this section, we work towards the below theorem. As a reminder, for a bimap $b : U \times V \rightarrowtail W$, its left nucleus is $\mathcal{L}(b) = \{(\alpha, \gamma) \in \mathrm{End}(U) \times \mathrm{End}(W) : b \circ_2 \alpha = b \circ_0 \gamma\}$. $\mathcal{M}(b), \mathcal{R}(b)$ are defined in the same way by matching the corresponding axis actions.

Theorem 4.1. Suppose $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ are fully nondegenerate bimaps, then

$$\mathcal{L}(s \otimes t) = \mathcal{L}(s) \boxtimes \mathcal{L}(t), \quad \mathcal{M}(s \otimes t) = \mathcal{M}(s) \boxtimes \mathcal{M}(t), \quad \mathcal{R}(s \otimes t) = \mathcal{R}(s) \boxtimes \mathcal{R}(t).$$

As a result of this theorem, we also have the following.

Corollary 4.2. Suppose $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ are fully nondegenerate bimaps, then $\mathcal{C}(s \otimes t) = \mathcal{C}(s) \boxtimes \mathcal{C}(t)$.

4.1. **Previous results.** The following is an existing result when additional restrictions are placed.

Proposition 4.3 ([Wil09, Propositions 7.8 and 7.9])**.** Let $d : U \times U \rightarrowtail C$ be a nondegenerate Hermitian C-form and $b : V \times V \rightarrowtail W$ a k-bilinear map. Then $d \otimes b$ is Hermitian and

$$\mathrm{Adj}(d \otimes b) = \mathrm{Adj}(d) \otimes \mathrm{Adj}(b).$$

In our notation, $\mathrm{Adj}(b) = \mathcal{M}(b)$. In the Hermitian setting, the adjoint ring is faithfully represented on one axis, so the notation $\mathrm{Adj}(d) \otimes \mathrm{Adj}(b)$ is unambiguous. Theorem 4.1 recovers this case via $\mathcal{M}(d \otimes b) = \mathcal{M}(d) \boxtimes \mathcal{M}(b)$.

In the rest of this section, we'll use the key observation in Lemma 2.11 to convert an equation about axis actions on tensor product bimaps to a matrix factorization statement.

4.2. **Algebra preliminaries.**

Definition 4.4. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces, and $t \in U \otimes V$. Define

$$\begin{aligned} \Theta_L &: U \otimes V \to \mathrm{Hom}(V^*, U), & \Theta_L(u \otimes v)(\varphi) &= \varphi(v)u, \\ \Theta_R &: U \otimes V \to \mathrm{Hom}(U^*, V), & \Theta_R(u \otimes v)(\psi) &= \psi(u)v \\ \mathrm{Im}_L(t) &:= \mathrm{Im}(\Theta_L(t)) \leq U, & \mathrm{Im}_R(t) &:= \mathrm{Im}(\Theta_R(t)) \leq V. \end{aligned}$$

Proposition 4.5. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces and $t \in U \otimes V$. Then

$$t \in \mathrm{Im}_L(t) \otimes \mathrm{Im}_R(t) \leq U \otimes V.$$

Proof. Let $A := \mathrm{Im}_L(t) \leq U$, and $X := \mathrm{Im}_R(t) \leq V$. Choose a basis $v_1, \ldots, v_n$ of V with dual basis $\varphi_1, \ldots, \varphi_n$. Then

$$t = \sum_{j=1}^{n} \Theta_L(t)(\varphi_j) \otimes v_j \in A \otimes V,$$

since each $\Theta_L(t)(\varphi_j)$ lies in A. Choose a basis $u_1, \ldots, u_m$ of U with dual basis $\psi_1, \ldots, \psi_m$. Then

$$t = \sum_{i=1}^{m} u_i \otimes \Theta_R(t)(\psi_i) \in U \otimes X,$$

since each $\Theta_R(t)(\psi_i)$ lies in X. Therefore by Lemma 2.2, $t \in (A \otimes V) \cap (U \otimes X) = A \otimes X$. □

Remark 4.6. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces. Let $\{u_i\}$ be an ordered bases of U and $\{v_j\}$ be an ordered basis V. Then t can be uniquely written as

$$t = \sum_{i,j} T_{ij}\, u_i \otimes v_j \in U \otimes V.$$

This identifies t with the matrix $T = (T_{ij})$.

Example 4.7. Let $U = \mathbb{F}^2$ with standard bases e_1, e_2 and $V = \mathbb{F}^3$ with standard basis f_1, f_2, f_3. Set

$$t = e_1 \otimes f_1 + e_2 \otimes f_1 + e_2 \otimes f_2, \qquad T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Hence $\mathrm{Im}_L(t) = \mathrm{span}\{e_1, e_2\}$, and $\mathrm{Im}_R(t) = \mathrm{span}\{f_1, f_2\}$.

Definition 4.8. Let U, V be finite-dimensional $\mathbb{F}$-vector spaces and $t \in U \otimes V$. The *rank* of t is the minimum number r such that t can be written as a sum $t = \sum_{i=1}^{r} u_i \otimes v_i$, for $u_i \in U$, and $v_i \in V$.

An observation is for $t \in U \otimes V$, the rank of t is less than or equal to the minimum of $\dim U$, and $\dim V$, as we can always expand t either in a basis of U or a basis of V.

Lemma 4.9. Let $t = \sum_{i=1}^{r} u_i \otimes v_i$ in $U \otimes V$, where $r = \mathrm{rank}(t)$. Then $\{u_1, \ldots, u_r\}$ and $\{v_1, \ldots, v_r\}$ are linearly independent, and

$$\mathrm{Im}_L(t) = \mathrm{span}\{u_1, \ldots, u_r\}, \qquad \mathrm{Im}_R(t) = \mathrm{span}\{v_1, \ldots, v_r\}.$$

Proof. Set $W \coloneqq \operatorname{span}\{u_1, \dots, u_r\}$ and $X \coloneqq \operatorname{span}\{v_1, \dots, v_r\}$. Since $t \in W \otimes V$, the above observation gives $\operatorname{rank}(t) \leq \dim W$. Since W is spanned by r vectors, $\dim W \leq r$. Together with $\operatorname{rank}(t) = r$, this gives $\dim W = r$, so $\{u_1, \dots, u_r\}$ is linearly independent. An analogous argument gives $\dim X = r$, so $\{v_1, \dots, v_r\}$ is linearly independent.

Choose $\varphi_i \in V^*$ with $\varphi_i(v_j) = \delta_{ij}$ for $1 \leq i, j \leq r$. Then

$$\Theta_L(t)(\varphi_i) = u_i,$$

so $W \leq \operatorname{Im}_L(t)$. For the reverse inclusion, we see for all $\varphi \in V^*$,

$$\Theta_L(t)(\varphi) = \sum_{i=1}^{r} \varphi(v_i)\, u_i \in W,$$

so $\operatorname{Im}_L(t) \leq W$. Hence $\operatorname{Im}_L(t) = W$. An analogous argument concludes $\operatorname{Im}_R(t) = X$. □

Remark 4.10. Suppose $\sum_{i=1}^{r} a_i \otimes x_i = \sum_{i=1}^{r} b_i \otimes y_i$ are two minimal rank decompositions in $U \otimes V$. Fix ordered bases of U and V, and let A, B be coordinates of a_i, b_i as columns respectively, and X, Y be coordinates of x_i, y_i as rows respectively. In these fixed coordinates, we get the equality $AX = BY$. Because A and B are both rank r, there is a unique $M \in \mathrm{GL}_r(\mathbb{F})$ such that

$$B = AM, \qquad Y = M^{-1}X.$$

Example 4.11. Take $\mathbb{F} = \mathbb{F}_5$, $U = V = \mathbb{F}^3$, and $r = 2$. Define

$$A = \left[\begin{array}{c|c} a_1 & a_2 \end{array}\right] = \left[\begin{array}{c|c} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{array}\right], \quad X = \left[\begin{array}{c} b_1 \\ \hline b_2 \end{array}\right] = \left[\begin{array}{ccc} 1 & 2 & 0 \\ \hline 0 & 1 & 1 \end{array}\right]$$

$$B = \left[\begin{array}{c|c} c_1 & c_2 \end{array}\right] = \left[\begin{array}{c|c} 1 & 2 \\ 0 & 3 \\ 3 & 4 \end{array}\right], \quad Y = \left[\begin{array}{c} d_1 \\ \hline d_2 \end{array}\right] = \left[\begin{array}{ccc} 3 & 2 & 1 \\ \hline 4 & 0 & 2 \end{array}\right].$$

A direct multiplication verifies

$$AX = BY = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

meaning $a_1 \otimes x_1 + a_2 \otimes x_2 = b_1 \otimes y_1 + b_2 \otimes y_2$. As per Remark 4.10, we find

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad M^{-1} = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix},$$

satisfying

$$AM = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 3 & 4 \end{bmatrix} = B, \qquad M^{-1}X = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 0 & 2 \end{bmatrix} = Y.$$

Now we begin analyzing actions on the product of bimaps.

Lemma 4.12. Let $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ be fully nondegenerate bimaps. Let $I \subseteq \{2, 1, 0\}$ be nonempty, and $w \in \operatorname{Bil}(U_2, U_1; U_0) \otimes \operatorname{Bil}(V_2, V_1; V_0) =: U \otimes V$ be an element such that for each $k \in I$, there is some $x_k \in \operatorname{End}(U_k \otimes V_k)$ satisfying

$$w = \Xi\big((s \otimes t) \circ_k x_k\big) \in U \otimes V.$$

Fix a minimal rank decomposition $w = \sum_{i=1}^{r} u_i \otimes v_i$. Then for each $k \in I$, there exists $\{\alpha_{k,i} \in \operatorname{End}(U_k)\}_{i=1}^{r}$ and $\{f_{k,i} \in \operatorname{End}(V_k)\}_{i=1}^{r}$ such that $s \circ_k \alpha_{k,i} = u_i$, and $t \circ_k f_{k,i} = v_i$. And furthermore, $x_k = \sum_{i=1}^{r} \kappa_k(\alpha_{k,i} \otimes f_{k,i})$.

Proof. Fix $k \in I$. By Equation (2.2) and Lemma 2.11, we have

$$w \in \operatorname{Im}(\Lambda_k^s) \otimes \operatorname{Im}(\Lambda_k^t), \qquad \operatorname{Im}_L(w) \leq \operatorname{Im}(\Lambda_k^s), \qquad \operatorname{Im}_R(w) \leq \operatorname{Im}(\Lambda_k^t).$$

Since $w = \sum_i u_i \otimes v_i$ is a minimal rank decomposition, apply Lemma 4.9 to conclude for each i that the specific elements $u_i \in \operatorname{Im}_L(w) \leq \operatorname{Im}(\Lambda_k^s)$ and $v_i \in \operatorname{Im}_R(w) \leq \operatorname{Im}(\Lambda_k^t)$ can lift to endomorphisms.

Specifically, there exist $\alpha_{k,i} \in \mathrm{End}(U_k)$ and $f_{k,i} \in \mathrm{End}(V_k)$ such that

$$s \circ_k \alpha_{k,i} = u_i, \qquad t \circ_k f_{k,i} = v_i.$$

Finally, for each k, set

$$x'_k := \sum_{i=1}^{r} \kappa_k(\alpha_{k,i} \otimes f_{k,i}).$$

Since $(s \otimes t) \circ_k x_k = (s \otimes t) \circ_k x'_k$ and $\Lambda_k^{s \otimes t}$ is injective, we conclude $x_k = x'_k$. □

4.3. **Key reduction from being in the nuclei to a matrix equality.** Recall from Equation (2.2) and Lemma 2.11 that axis actions on the product bimap $s \otimes t$ can be analyzed in the tensor product space $\mathrm{Bil}(U_2, U_1; U_0) \otimes \mathrm{Bil}(V_2, V_1; V_0)$. The next lemma is the reduction used in the nuclei case.

Lemma 4.13. Let $s : U_2 \times U_1 \rightarrowtail U_0$ and $t : V_2 \times V_1 \rightarrowtail V_0$ be bimaps, let $k \in \{2, 1, 0\}$ and $x \in \mathrm{End}(U_k \otimes V_k)$ where

$$w := \Xi\big((s \otimes t) \circ_k x\big) \in (\mathrm{Bil}(U_2, U_1; U_0)) \otimes (\mathrm{Bil}(V_2, V_1; V_0)).$$

Then $w \in \mathrm{Im}_L(w) \otimes \mathrm{Im}_R(w) \leq \mathrm{Im}(\Lambda_k^s) \otimes \mathrm{Im}(\Lambda_k^t)$.

Proof. Expand $\kappa_k^{-1}(x) = \sum_j \alpha_j \otimes f_j$. By Lemma 2.11,

$$w = \sum_j (s \circ_k \alpha_j) \otimes (t \circ_k f_j) \in \mathrm{Im}(\Lambda_k^s) \otimes \mathrm{Im}(\Lambda_k^t).$$

By Proposition 4.5, $w \in \mathrm{Im}_L(w) \otimes \mathrm{Im}_R(w) \leq \mathrm{Im}(\Lambda_k^s) \otimes \mathrm{Im}(\Lambda_k^t)$. □

4.4. **Proof of Theorem 4.1.** In this subsection, we prove the middle nucleus case. The left and right nucleus cases are analogous.

Proof. ($\subseteq$) Let $(x, y) \in \mathcal{M}(s \otimes t)$. Then $(s \otimes t) \circ_2 x = (s \otimes t) \circ_1 y$. Set $w := \Xi\big((s \otimes t) \circ_2 x\big) = \Xi\big((s \otimes t) \circ_1 y\big) \in U \otimes V$. Choose a minimal-rank decomposition

$$w = \sum_{i=1}^{r} u_i \otimes v_i.$$

Apply Lemma 4.12 with $I = \{2, 1\}$. We obtain $\alpha_i \in \mathrm{End}(U_2)$, $\beta_i \in \mathrm{End}(U_1)$, $f_i \in \mathrm{End}(V_2)$, $g_i \in \mathrm{End}(V_1)$ such that

$$s \circ_2 \alpha_i = u_i = s \circ_1 \beta_i, \qquad t \circ_2 f_i = v_i = t \circ_1 g_i,$$

and

$$x = \sum_{i=1}^{r} \kappa_2(\alpha_i \otimes f_i), \qquad y = \sum_{i=1}^{r} \kappa_1(\beta_i \otimes g_i).$$

Hence $(\alpha_i, \beta_i) \in \mathcal{M}(s)$ and $(f_i, g_i) \in \mathcal{M}(t)$ for each i, so

$$(x, y) = \sum_{i=1}^{r} \big((\alpha_i, \beta_i) \boxtimes (f_i, g_i)\big) \in \mathcal{M}(s) \boxtimes \mathcal{M}(t).$$

($\supseteq$) Let $(\alpha, \beta) \in \mathcal{M}(s)$ and $(f, g) \in \mathcal{M}(t)$. Then

$$(s \otimes t) \circ_2 \kappa_2(\alpha \otimes f) = (s \circ_2 \alpha) \otimes (t \circ_2 f) = (s \circ_1 \beta) \otimes (t \circ_1 g) = (s \otimes t) \circ_1 \kappa_1(\beta \otimes g).$$

So $(\alpha, \beta) \boxtimes (f, g) = (\kappa_2(\alpha \otimes f), \kappa_1(\beta \otimes g)) \in \mathcal{M}(s \otimes t)$. Since $\mathcal{M}(s) \boxtimes \mathcal{M}(t)$ is the span of such pure terms, we get $\mathcal{M}(s) \boxtimes \mathcal{M}(t) \subseteq \mathcal{M}(s \otimes t)$. □

A worked out nucleus decomposition example is the Appendix (Example D.1).

Now we wrap up this chapter with the proof that $\mathcal{C}(s \otimes t) = \mathcal{C}(s) \boxtimes \mathcal{C}(t)$.

Proof of Corollary 4.2. We prove the direction $\mathcal{C}(s \otimes t) \subseteq \mathcal{C}(s) \boxtimes \mathcal{C}(t)$. Let $(x_2, x_1, x_0) \in \mathcal{C}(s \otimes t)$. Then

$$(s \otimes t) \circ_2 x_2 = (s \otimes t) \circ_1 x_1 = (s \otimes t) \circ_0 x_0.$$

Set $U := \mathrm{Bil}(U_2, U_1; U_0)$, and $V := \mathrm{Bil}(V_2, V_1; V_0)$. Let

$$w := \Xi\big((s \otimes t) \circ_2 x_2\big) \in U \otimes V.$$

Fix a minimal rank decomposition $w = \sum_{i=1}^{r} u_i \otimes v_i$, and apply Lemma 4.12 with $I = \{2, 1, 0\}$, then follow the same argument as the proof of Theorem 4.1.

The direction $\mathcal{C}(s \otimes t) \supseteq \mathcal{C}(s) \boxtimes \mathcal{C}(t)$ is analogous to the proof $\mathrm{Nuc}(s \otimes t) \supseteq \mathcal{M}(s) \boxtimes \mathcal{M}(t)$ in Theorem 4.1. □

5. Derivation of Product of Bimaps

The main result of this section is the following.

Theorem 5.1. If s and t are fully nondegenerate bimaps, then

$$\begin{aligned}\mathrm{Der}(s \otimes t) = {} & \mathrm{Der}(s) \boxtimes \mathcal{C}(t) + \mathcal{C}(s) \boxtimes \mathrm{Der}(t) \\ & + \iota_{20}(\mathcal{L}(s) \boxtimes \mathcal{L}(t)) + \iota_{10}(\mathcal{R}(s) \boxtimes \mathcal{R}(t)) + \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t)),\end{aligned} \tag{5.1}$$

where $\iota_{20}(\alpha, \gamma) = (\alpha, 0, \gamma)$, $\iota_{10}(\beta, \gamma) = (0, \beta, \gamma)$, and $\iota_{21}(\alpha, \beta) = (-\alpha, \beta, 0)$.

The strategy we take to prove it will be similar to the previous section where we analyze equalities in tensor product vector spaces, and relate them to equalities of axis actions on tensor prodcut bimaps.

5.1. Previous results.

In this subsection, algebras are finite-dimensional over $\mathbb{F}$, and not necessarily associative. For an algebra A, its derivation refers to

$$\mathcal{D}_{\mathrm{alg}}(A) := \{d \in \mathrm{End}(A) : \forall x, y \in A,\ d(xy) = d(x)y + xd(y)\}.$$

Brešar proved the following.

Theorem 5.2 (Brešar, Theorem 3.1 [Bre17])**.** Let R and S be nonassociative unital algebras. Suppose that either at least one of R and S is finite-dimensional or they are both finitely generated. Then every derivation d of $R \otimes S$ can be written as

$$d = \mathrm{ad}(u) + \sum_{j=1}^{p} \lambda_{z_j} \otimes f_j + \sum_{i=1}^{q} g_i \otimes \lambda_{w_i}, \tag{5.2}$$

where $u \in \mathrm{Nuc}(R) \otimes \mathrm{Nuc}(S)$, $z_j \in Z(R)$, $w_i \in Z(S)$, $f_j \in \mathcal{D}_{\mathrm{alg}}(S)$, and $g_i \in \mathcal{D}_{\mathrm{alg}}(R)$.

Proposition 5.3. Let R and S be unital algebras over $\mathbb{F}$, with multiplication maps μ_R, μ_S. Then Theorem 5.1 implies Theorem 5.2.

Proof. See Appendix C. □

Specialization to Matrix Algebras. Let A be a unital finite-dimensional $\mathbb{F}$-algebra. As a vector space, $\mathbb{M}_n(A) \cong \mathbb{M}_n(\mathbb{F}) \otimes A$, and the multiplication on $\mathbb{M}_n(A)$ is defined by $(X \otimes a)(Y \otimes b) = XY \otimes ab$.

Theorem 5.4 (Benkart–Osborn, Corollary 4.9 [BO81])**.** Assume $\mathrm{char}\,\mathbb{F} \neq 2, 3$ and $n > 3$. Then for any unital algebra A, $\mathcal{D}_{\mathrm{alg}}(\mathbb{M}_n(A)) = \mathcal{D}_{\mathrm{alg}}(L'_n(A)) = \mathcal{D}_{\mathrm{alg}}(K_n(A)) = (\mathcal{D}_{\mathrm{alg}}(A))^{\#} + \mathrm{ad}_{\mathbb{M}_n(N)}$.

Remark 5.5. In Theorem 5.4, N denotes the nucleus of A, $L_n(A)$ is $\mathbb{M}_n(A)$ under the Lie product, $L'_n(A) = [L_n(A), L_n(A)]$, and $K_n(A) = L'_n(A)/Z(L'_n(A))$. Also, $(\mathcal{D}_{\mathrm{alg}}(A))^{\#}$ denotes entrywise application of derivations of A. Thus in the notation of this chapter,

$$N = \mathrm{Nuc}(A), \qquad (\mathcal{D}_{\mathrm{alg}}(A))^{\#} = I_n \otimes \mathcal{D}_{\mathrm{alg}}(A), \qquad \mathrm{ad}_{\mathbb{M}_n(N)} = \mathrm{ad}(\mathbb{M}_n(\mathrm{Nuc}(A))).$$

So Theorem 5.4 proves $\mathcal{D}_{\mathrm{alg}}(\mathbb{M}_n(A)) = \big(I_n \otimes \mathcal{D}_{\mathrm{alg}}(A)\big) + \mathrm{ad}(\mathbb{M}_n(\mathrm{Nuc}(A)))$.

Benkart–Osborn note that specifically for Corollary 4.9, no restrictions on $\mathrm{char}\,\mathbb{F}$ or n are needed. As a result, we have the following.

Proposition 5.6. Theorem 5.1 implies Theorem 5.4.

Proof. Theorem 5.1 implies Theorem 5.2 by Proposition 5.3, and Theorem 5.2 implies Theorem 5.4 by [Bre17, Cor. 4.2]. □

5.2. **Preliminaries.** This decomposition features prominently in the proof of Theorem 5.1.

Definition 5.7. Let $p, q, r \in U \otimes V$ be nonzero tensors satisfying $p + q = r$. Define

$$A := \mathrm{Im}_L(p), \quad B := \mathrm{Im}_L(q), \quad C := \mathrm{Im}_L(r),$$
$$X := \mathrm{Im}_R(p), \quad Y := \mathrm{Im}_R(q), \quad Z := \mathrm{Im}_R(r).$$

By Proposition 4.5, $p \in A \otimes X$, $q \in B \otimes Y$, $r \in C \otimes Z$.

A (p, q, r) *locally independent unified decomposition* (LIU decomposition) consists of a natural number n, and vectors

$$\forall i \in [n], \quad a_i \in A,\ b_i \in B,\ c_i \in C, \qquad x_i \in X,\ y_i \in Y,\ z_i \in Z$$

satisfying the following:

- **Decomposition equality.**
$$p = \sum_{i=1}^{n} a_i \otimes x_i, \qquad q = \sum_{i=1}^{n} b_i \otimes y_i, \qquad r = \sum_{i=1}^{n} c_i \otimes z_i.$$
- **Local equality.** For all $i \in [n]$, $a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i$.

Example 5.8. Let $U = V = \mathbb{F}^2$ with bases u_1, u_2 and v_1, v_2. Define

$$p := u_1 \otimes v_1 + u_1 \otimes v_2, \qquad q := u_2 \otimes v_1 - u_1 \otimes v_2, \qquad r := u_1 \otimes v_1 + u_2 \otimes v_1.$$

Then $p + q = r$, and

$$A = \mathrm{span}\{u_1\}, \qquad B = U, \qquad C = U,$$
$$X = V, \qquad Y = V, \qquad Z = \mathrm{span}\{v_1\}.$$

Now let

$$a_1 = u_1,\ b_1 = u_2,\ c_1 = u_1 + u_2, \qquad x_1 = y_1 = z_1 = v_1,$$
$$a_2 = u_1,\ b_2 = -u_1,\ c_2 = 0, \qquad x_2 = y_2 = v_2,\ z_2 = v_1.$$

Then for each $i \in [2]$, $a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i$, and

$$p = a_1 \otimes x_1 + a_2 \otimes x_2,$$
$$q = b_1 \otimes y_1 + b_2 \otimes y_2,$$
$$r = c_1 \otimes z_1 + c_2 \otimes z_2.$$

Thus this is an LIU decomposition of $p + q = r$.

5.3. **Existence of a LIU decomposition of the equation $p + q = r$.**

Convention 5.9. For the remainder of this subsection, whenever we have a triple $p, q, r \in U \otimes V$ satisfying $p + q = r$, the spaces $A, B, C \leq U$ and $X, Y, Z \leq V$ will be subspaces as defined in Definition 5.7. It is always the case that $A + B \leq U$ and $X + Y \leq V$. We also assume without loss of generality that

$$U := A + B, \qquad V := X + Y,$$

because it is always the case that

$$p \in A \otimes X \leq (A + B) \otimes (X + Y) \leq U \otimes V$$
$$q \in B \otimes Y \leq (A + B) \otimes (X + Y) \leq U \otimes V$$
$$r \in \mathrm{Im}_L(r) \otimes \mathrm{Im}_R(r).$$

Furthermore, $\mathrm{Im}_L(r), \mathrm{Im}_R(r)$ satisfies

$$\mathrm{Im}_L(r) = \mathrm{Im}_L(p + q) \leq \mathrm{Im}_L(p) + \mathrm{Im}_L(q) = A + B,$$

$$\mathrm{Im}_R(r) = \mathrm{Im}_R(p+q) \le \mathrm{Im}_R(p) + \mathrm{Im}_R(q) = X + Y.$$

So we redefine U to just the space $A+B$, and redefine V to just the space $X+Y$.

Lemma 5.10. Let U be a $\mathbb{F}$-vector space, and let $A, B, C \le U$ with $U = A+B$. Define $U^{(1)} \le U$ by

$$U^{(1)} := A \cap B \cap C.$$

Define $U^{(2)} \le A \cap B$, $U^{(3)} \le A \cap C$, and $U^{(4)} \le B \cap C$ by choosing complements

$$A \cap B = U^{(1)} \oplus U^{(2)}, \qquad A \cap C = U^{(1)} \oplus U^{(3)}, \qquad B \cap C = U^{(1)} \oplus U^{(4)}.$$

Define $U^{(5)} \le A$ and $U^{(6)} \le B$ by choosing complements

$$A = (U^{(1)} + U^{(2)} + U^{(3)}) \oplus U^{(5)}, \qquad B = (U^{(1)} + U^{(2)} + U^{(4)}) \oplus U^{(6)}.$$

Then

$$U = \bigoplus_{i=1}^{6} U^{(i)}, \qquad A = \bigoplus_{i \in \{1,2,3,5\}} U^{(i)}, \qquad B = \bigoplus_{i \in \{1,2,4,6\}} U^{(i)}.$$

Proof. Let $d_i := \dim U^{(i)}$ for $1 \le i \le 6$. By construction, $\dim(A \cap B) = d_1 + d_2$, $\dim(A \cap C) = d_1 + d_3$. Using the fact

$$\dim(X+Y) = \dim X + \dim Y - \dim(X \cap Y),$$

with $X = A \cap B$, $Y = A \cap C$, we get

$$\dim\big((A \cap B) + (A \cap C)\big) = (d_1 + d_2) + (d_1 + d_3) - d_1 = d_1 + d_2 + d_3.$$

Since $A = ((A \cap B) + (A \cap C)) \oplus U^{(5)}$, this gives

$$\dim A = d_1 + d_2 + d_3 + d_5.$$

Similarly, from $B = ((A \cap B) + (B \cap C)) \oplus U^{(6)}$,

$$\dim B = d_1 + d_2 + d_4 + d_6.$$

Finally, as $U = A + B$,

$$\dim U = \dim(A+B) = \dim A + \dim B - \dim(A \cap B) = \sum_{i=1}^{6} d_i.$$

Recall

$$U = A + B = \sum_{i=1}^{6} U^{(i)}, \qquad A = \sum_{i \in \{1,2,3,5\}} U^{(i)}, \qquad B = \sum_{i \in \{1,2,4,6\}} U^{(i)},$$

and the fact that for any finite family of subspaces $\{W_i\}$, $\dim(\sum_i W_i) \le \sum_i \dim W_i$, with equality if and only if the sum is direct. Applying this to the spaces U, A, B concludes

$$U = \bigoplus_{i=1}^{6} U^{(i)}, \qquad A = \bigoplus_{i \in \{1,2,3,5\}} U^{(i)}, \qquad B = \bigoplus_{i \in \{1,2,4,6\}} U^{(i)}.$$

□

The above lemma is to prepare for below, which is the key one in proving the existence of a LIU decomposition for the equation $p + q = r$.

Lemma 5.11. Let $p, q, r \in U \otimes V$ satisfy $p + q = r$, and

$$A := \mathrm{Im}_L(p), \quad B := \mathrm{Im}_L(q), \quad C := \mathrm{Im}_L(r),$$
$$X := \mathrm{Im}_R(p), \quad Y := \mathrm{Im}_R(q), \quad Z := \mathrm{Im}_R(r).$$

Recall via Convention 5.9 that $U = A + B$, $V = X + Y$. With respect to the decomposition in Lemma 5.10, p, q, r satisfy:

$$p \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(3)} \otimes V^{(3)},$$
$$q \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(4)} \otimes V^{(4)},$$

$$r \in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(3)} \otimes V^{(3)} \oplus U^{(4)} \otimes V^{(4)},$$

where $U^{(\bullet)} := U^{(2)} \oplus \cdots \oplus U^{(6)}$ and $V^{(\bullet)} := V^{(2)} \oplus \cdots \oplus V^{(6)}$.

Proof. Apply Lemma 5.10 to the spaces $A, B, C \leq U$ and to the spaces $X, Y, Z \leq V$.

From the definitions, $A = U^{(1)} \oplus U^{(2)} \oplus U^{(3)} \oplus U^{(5)}$ and $B = U^{(1)} \oplus U^{(2)} \oplus U^{(4)} \oplus U^{(6)}$, and similarly $X = V^{(1)} \oplus V^{(2)} \oplus V^{(3)} \oplus V^{(5)}$ and $Y = V^{(1)} \oplus V^{(2)} \oplus V^{(4)} \oplus V^{(6)}$. Let $\pi_{ab} : U \otimes V \to U^{(a)} \otimes V^{(b)}$ be the block projections and write

$$\pi_{a\bullet} := \sum_{b=1}^{6} \pi_{ab}, \qquad \pi_{\bullet b} := \sum_{a=1}^{6} \pi_{ab}.$$

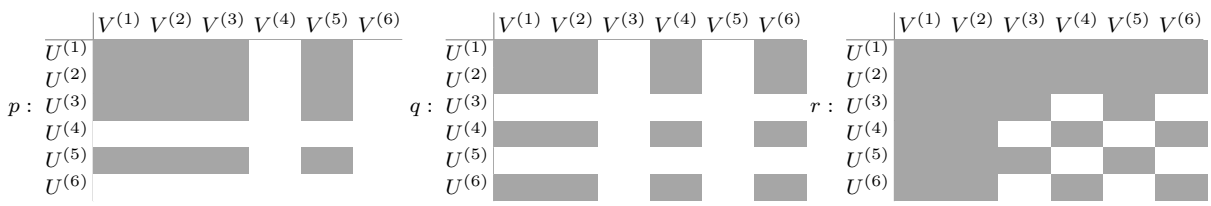
Hence $(\pi_{4\bullet} + \pi_{6\bullet})(p) = 0$ and $(\pi_{3\bullet} + \pi_{5\bullet})(q) = 0$, while $(\pi_{\bullet 4} + \pi_{\bullet 6})(p) = 0$ and $(\pi_{\bullet 3} + \pi_{\bullet 5})(q) = 0$. For $1 \leq a, b \leq 6$, set

$$p_{ab} := \pi_{ab}(p), \quad q_{ab} := \pi_{ab}(q), \quad r_{ab} := \pi_{ab}(r).$$

Applying π_{ab} to $p + q = r$ gives

$$p_{ab} + q_{ab} = r_{ab} \qquad (1 \leq a, b \leq 6).$$

This is a blockwise $p + q = r$ equality: if two entries in a block are zero, then the third is zero. Taking stock we visualize these relations below, where grey means *potentially nonzero*, white means *guaranteed to be zero*.



Row projections from known zero rows. Since $\pi_{3\bullet}(q) = 0$, define

$$s_3 := \pi_{3\bullet}(p) = \pi_{3\bullet}(r).$$

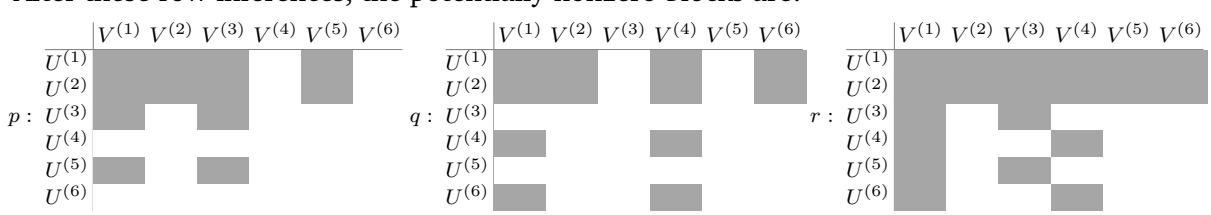
Then $s_3 \in U^{(3)} \otimes X$ and $s_3 \in U^{(3)} \otimes Z$, hence $s_3 \in U^{(3)} \otimes (X \cap Z) = U^{(3)} \otimes \big(V^{(1)} \oplus V^{(3)}\big)$. So row $U^{(3)}$ of p, r has support only in columns $V^{(1)}, V^{(3)}$. The same argument with $U^{(5)}$ gives row $U^{(5)}$ of p, r also in $V^{(1)} \oplus V^{(3)}$.

Since $\pi_{4\bullet}(p) = 0$, define

$$t_4 := \pi_{4\bullet}(q) = \pi_{4\bullet}(r).$$

Then $t_4 \in U^{(4)} \otimes Y$ and $t_4 \in U^{(4)} \otimes Z$, hence $t_4 \in U^{(4)} \otimes (Y \cap Z) = U^{(4)} \otimes \big(V^{(1)} \oplus V^{(4)}\big)$. So row $U^{(4)}$ of q, r has support only in columns $V^{(1)}, V^{(4)}$. The same argument with $U^{(6)}$ gives row $U^{(6)}$ of q, r also in $V^{(1)} \oplus V^{(4)}$.

After these row inferences, the potentially nonzero blocks are:



Column projections from known zero columns. Since $\pi_{\bullet 3}(q) = 0$, define

$$u_3 := \pi_{\bullet 3}(p) = \pi_{\bullet 3}(r).$$

Then $u_3 \in A \otimes V^{(3)}$ and $u_3 \in C \otimes V^{(3)}$, hence $u_3 \in (A \cap C) \otimes V^{(3)} = \big(U^{(1)} \oplus U^{(3)}\big) \otimes V^{(3)}$. So column $V^{(3)}$ of p, r has support only in rows $U^{(1)}, U^{(3)}$.

Since $\pi_{\bullet 4}(p) = 0$, define

$$u_4 := \pi_{\bullet 4}(q) = \pi_{\bullet 4}(r).$$

Then $u_4 \in B \otimes V^{(4)}$ and $u_4 \in C \otimes V^{(4)}$, hence $u_4 \in (B \cap C) \otimes V^{(4)} = \big(U^{(1)} \oplus U^{(4)}\big) \otimes V^{(4)}$. So column $V^{(4)}$ of q, r has support only in rows $U^{(1)}, U^{(4)}$.

Since $\pi_{\bullet 5}(q) = 0$, define

$$u_5 := \pi_{\bullet 5}(p) = \pi_{\bullet 5}(r).$$

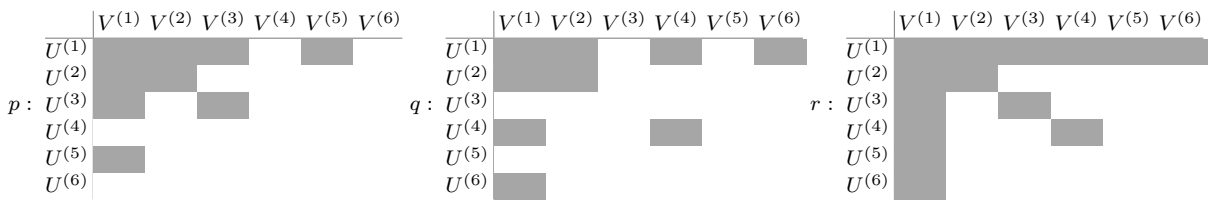
Then $u_5 \in (A \cap C) \otimes V^{(5)} = \big(U^{(1)} \oplus U^{(3)}\big) \otimes V^{(5)}$. Combining this with the row-$U^{(3)}$ and row-$U^{(5)}$ restrictions above forces support in column $V^{(5)}$ to lie only in row $U^{(1)}$.

Since $\pi_{\bullet 6}(p) = 0$, define

$$u_6 := \pi_{\bullet 6}(q) = \pi_{\bullet 6}(r).$$

Then $u_6 \in (B \cap C) \otimes V^{(6)} = \big(U^{(1)} \oplus U^{(4)}\big) \otimes V^{(6)}$. Combining this with the row-$U^{(4)}$ and row-$U^{(6)}$ restrictions above forces support in column $V^{(6)}$ to lie only in row $U^{(1)}$.

After these column inferences, the potentially nonzero blocks are:



The $(2,2)$ block relation. Set $v_2 := \pi_{2\bullet}(r)$. By the previous row/column inferences, row $U^{(2)}$ of both p and q is supported in $V^{(1)} \oplus V^{(2)} = X \cap Y$, hence

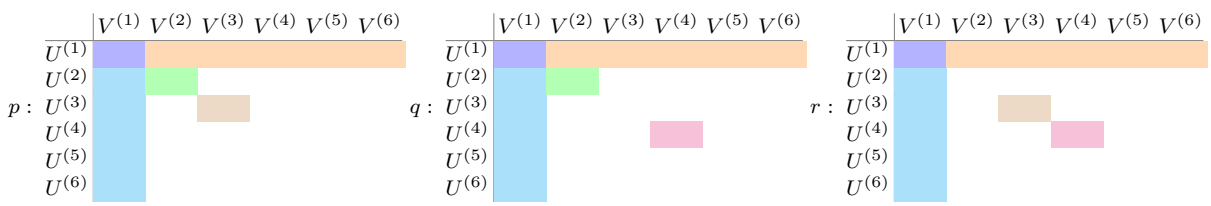
$$v_2 \in U^{(2)} \otimes (X \cap Y).$$

Also $r \in C \otimes Z$, so $v_2 \in U^{(2)} \otimes Z$. Therefore

$$v_2 \in U^{(2)} \otimes \big((X \cap Y) \cap Z\big) = U^{(2)} \otimes \big(X \cap Y \cap Z\big) = U^{(2)} \otimes V^{(1)}.$$

Thus $r_{22} = 0$, and the block equation gives $p_{22} + q_{22} = 0$.

What remains are the following, where blue is $U^{(1)} \otimes V^{(1)}$, orange is $U^{(1)} \otimes V^{(\bullet)}$, cyan is $U^{(\bullet)} \otimes V^{(1)}$, green is $U^{(2)} \otimes V^{(2)}$, brown is $U^{(3)} \otimes V^{(3)}$, and pink is $U^{(4)} \otimes V^{(4)}$, and blank entries are zero.



This is the claimed support pattern

$$\begin{aligned}
p &\in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(3)} \otimes V^{(3)}, \\
q &\in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(2)} \otimes V^{(2)} \oplus U^{(4)} \otimes V^{(4)}, \\
r &\in U^{(1)} \otimes V^{(1)} \oplus U^{(1)} \otimes V^{(\bullet)} \oplus U^{(\bullet)} \otimes V^{(1)} \oplus U^{(3)} \otimes V^{(3)} \oplus U^{(4)} \otimes V^{(4)}.
\end{aligned}$$

□

Example 5.12. Let $\mathbb{F} = \mathbb{F}_{997}$, let $U = V = \mathbb{F}^6$ with standard bases, and identify $U \otimes V$ with $\mathbb{M}_6(\mathbb{F})$ via $u \otimes v \mapsto [u][v]^T$ where $[u]$ is the column vector of coordinates.

Let $p, q, r \in U \otimes V$ correspond to the 6×6 matrices below.

$$p = \begin{bmatrix} 373 & 831 & 66 & 724 & 159 & 68 \\ 772 & 812 & 372 & 663 & 579 & 209 \\ 155 & 549 & 499 & 757 & 500 & 64 \\ 218 & 733 & 547 & 132 & 342 & 122 \\ 859 & 708 & 25 & 767 & 78 & 648 \\ 992 & 398 & 894 & 611 & 169 & 953 \end{bmatrix} \quad q = \begin{bmatrix} 176 & 355 & 631 & 671 & 759 & 955 \\ 252 & 645 & 740 & 978 & 500 & 638 \\ 308 & 485 & 963 & 867 & 212 & 39 \\ 146 & 830 & 155 & 854 & 264 & 14 \\ 30 & 375 & 193 & 887 & 38 & 692 \\ 107 & 281 & 218 & 640 & 676 & 450 \end{bmatrix} \quad r = \begin{bmatrix} 549 & 189 & 697 & 398 & 918 & 26 \\ 27 & 460 & 115 & 644 & 82 & 847 \\ 463 & 37 & 465 & 627 & 712 & 103 \\ 364 & 566 & 702 & 986 & 606 & 136 \\ 889 & 86 & 218 & 657 & 116 & 343 \\ 102 & 679 & 115 & 254 & 845 & 406 \end{bmatrix}$$

so that $p + q = r$ in $\mathbb{F}^{6\times 6}$. Choose change-of-basis matrices M, N so that the new coordinates are adapted to the decomposition $U = \bigoplus_{i=1}^{6} U^{(i)}$ and $V = \bigoplus_{i=1}^{6} V^{(i)}$:

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 310 & 34 & 931 & 898 & 971 & 773 \\ 104 & 130 & 918 & 780 & 84 & 647 \\ 795 & 834 & 145 & 316 & 939 & 574 \\ 386 & 64 & 11 & 618 & 494 & 754 \\ 732 & 237 & 949 & 667 & 810 & 127 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 655 & 49 & 786 & 153 & 994 \\ 0 & 757 & 318 & 920 & 875 & 310 \\ 0 & 304 & 386 & 307 & 423 & 363 \\ 0 & 988 & 354 & 652 & 88 & 712 \\ 0 & 780 & 227 & 987 & 917 & 291 \\ 0 & 59 & 338 & 600 & 869 & 429 \end{bmatrix},$$

the transformed matrices $p' = MpN$, $q' = MqN$, $r' = MrN$ have the support patterns as described by Lemma 5.11.

$p' =$

	$V^{(1)}$	$V^{(2)}$	$V^{(3)}$	$V^{(4)}$	$V^{(5)}$	$V^{(6)}$
$U^{(1)}$	373	16	259	0	968	0
$U^{(2)}$	119	905	0	0	0	0
$U^{(3)}$	966	0	123	0	0	0
$U^{(4)}$	0	0	0	0	0	0
$U^{(5)}$	647	0	0	0	0	0
$U^{(6)}$	0	0	0	0	0	0

$q' =$

	$V^{(1)}$	$V^{(2)}$	$V^{(3)}$	$V^{(4)}$	$V^{(5)}$	$V^{(6)}$
$U^{(1)}$	176	828	0	553	0	244
$U^{(2)}$	607	92	0	0	0	0
$U^{(3)}$	0	0	0	0	0	0
$U^{(4)}$	68	0	0	815	0	0
$U^{(5)}$	0	0	0	0	0	0
$U^{(6)}$	970	0	0	0	0	0

$r' =$

	$V^{(1)}$	$V^{(2)}$	$V^{(3)}$	$V^{(4)}$	$V^{(5)}$	$V^{(6)}$
$U^{(1)}$	549	844	259	553	968	244
$U^{(2)}$	726	0	0	0	0	0
$U^{(3)}$	966	0	123	0	0	0
$U^{(4)}$	68	0	0	815	0	0
$U^{(5)}$	647	0	0	0	0	0
$U^{(6)}$	970	0	0	0	0	0

Lemma 5.13. Let $p, q, r \in U \otimes V$ satisfy $p + q = r$. Then (p, q, r) admits an LIU decomposition.

Proof. Recall from Convention 5.9 the spaces $A, B, C \leq U$, and $X, Y, Z \leq V$ where $U = A + B$ and $V = X + Y$. Decompose U and V by Lemma 5.10

$$U \otimes V = \bigoplus_{a,b=1}^{6} U^{(a)} \otimes V^{(b)}.$$

Let $\pi_{ab} : U \otimes V \to U^{(a)} \otimes V^{(b)}$ be the corresponding projections. Define

$$p_{11} := \pi_{11}(p), \quad p_{1\bullet} := \sum_{b=2}^{6} \pi_{1b}(p), \quad p_{\bullet 1} := \sum_{a=2}^{6} \pi_{a1}(p), \quad p_{22} := \pi_{22}(p), \quad p_{33} := \pi_{33}(p),$$

and analogously

$$q_{11},\ q_{1\bullet},\ q_{\bullet 1},\ q_{22},\ q_{44},$$
$$r_{11},\ r_{1\bullet},\ r_{\bullet 1},\ r_{33},\ r_{44}.$$

By Lemma 5.11, all other blocks vanish, so

$$p = p_{11} + p_{1\bullet} + p_{\bullet 1} + p_{22} + p_{33},$$
$$q = q_{11} + q_{1\bullet} + q_{\bullet 1} + q_{22} + q_{44},$$
$$r = r_{11} + r_{1\bullet} + r_{\bullet 1} + r_{33} + r_{44}.$$

Applying the relevant projections to $p + q = r$ gives equations

$$p_{11} + q_{11} = r_{11}, \quad p_{1\bullet} + q_{1\bullet} = r_{1\bullet}, \quad p_{\bullet 1} + q_{\bullet 1} = r_{\bullet 1},$$
$$p_{22} + q_{22} = 0, \qquad p_{33} = r_{33}, \qquad q_{44} = r_{44}.$$

We build an LIU decomposition for each block. After, summing the decompositions gives the desired result.

In the blue block $U^{(1)} \otimes V^{(1)}$, choose bases of $U^{(1)}$ and $V^{(1)}$. Writing p_{11}, q_{11}, r_{11} in that basis, gives coefficientwise local equalities. The green ($U^{(2)} \otimes V^{(2)}$), brown ($U^{(3)} \otimes V^{(3)}$), and pink ($U^{(4)} \otimes V^{(4)}$) blocks are special cases of the blue-block argument.

In the orange block $U^{(1)} \otimes V^{(\bullet)}$, choose a basis $\{u_m\}_{m=1}^{d}$ of $U^{(1)}$ and write

$$p_{1\bullet} = \sum_{m=1}^{d} u_m \otimes f_m, \qquad q_{1\bullet} = \sum_{m=1}^{d} u_m \otimes g_m, \qquad r_{1\bullet} = \sum_{m=1}^{d} u_m \otimes h_m.$$

Since $p_{1\bullet} + q_{1\bullet} = r_{1\bullet}$, we get

$$\forall m \in [d], \qquad u_m \otimes f_m + u_m \otimes g_m = u_m \otimes h_m.$$

This gives local equalities with a common left factor in $A \cap B \cap C$. The cyan block ($U^{(\bullet)} \otimes V^{(1)}$) case is analogous. $\square$

Example 5.14. We demonstrate this splitting for Example 5.12. Recall all coefficients are in $\mathbb{F}_{997}$.

Let u_i and v_j be basis vectors with $U^{(i)} = \langle u_i \rangle$ and $V^{(j)} = \langle v_j \rangle$.

For the blue block,

$$p_{11} = 373\, u_1 \otimes v_1, \qquad q_{11} = 176\, u_1 \otimes v_1, \qquad r_{11} = 549\, u_1 \otimes v_1,$$

so $p_{11} + q_{11} = (373 + 176)\, u_1 \otimes v_1 = 549\, u_1 \otimes v_1 = r_{11}$.

For the orange block,

$$p_{1\bullet} = u_1 \otimes (16v_2 + 259v_3 + 968v_5), \quad q_{1\bullet} = u_1 \otimes (828v_2 + 553v_4 + 244v_6),$$

so $p_{1\bullet} + q_{1\bullet} = u_1 \otimes (844v_2 + 259v_3 + 553v_4 + 968v_5 + 244v_6) = r_{1\bullet}$.

Likewise,

$$p_{\bullet 1} = (119u_2 + 966u_3 + 647u_5) \otimes v_1, \quad q_{\bullet 1} = (607u_2 + 68u_4 + 970u_6) \otimes v_1,$$

hence $p_{\bullet 1} + q_{\bullet 1} = (726u_2 + 966u_3 + 68u_4 + 647u_5 + 970u_6) \otimes v_1 = r_{\bullet 1}$.

5.4. **Proof of Theorem 5.1.** We apply the LIU decomposition to axis-action images and match each block to one summand in Equation (5.1). First, we need to describe the partitioning of the index sets of a LIU decomposition.

Remark 5.15. Given an LIU decomposition of $p + q = r$, with

$$p = \sum_{i=1}^{n} a_i \otimes x_i, \qquad q = \sum_{i=1}^{n} b_i \otimes y_i, \qquad r = \sum_{i=1}^{n} c_i \otimes z_i,$$

and local equalities

$$\forall i \in [n], \qquad a_i \otimes x_i + b_i \otimes y_i = c_i \otimes z_i.$$

Define index sets $I_{11},\ I_{1\bullet},\ I_{\bullet 1},\ I_{22},\ I_{33},\ I_{44}$, where I_{ab} consists of those indices i for which the three local terms $a_i \otimes x_i$, $b_i \otimes y_i$, $c_i \otimes z_i$ lie in the block $U^{(a)} \otimes V^{(b)}$. Then these sets give a partition

$$[n] = I_{11} \sqcup I_{1\bullet} \sqcup I_{\bullet 1} \sqcup I_{22} \sqcup I_{33} \sqcup I_{44}.$$

Proof. ($\subseteq$) We prove the inclusion

$$\begin{aligned}\mathrm{Der}(s \otimes t) \subseteq\ & \mathrm{Der}(s) \boxtimes \mathcal{C}(t) + \mathcal{C}(s) \boxtimes \mathrm{Der}(t) \\ & + \iota_{20}(\mathcal{L}(s) \boxtimes \mathcal{L}(t)) + \iota_{10}(\mathcal{R}(s) \boxtimes \mathcal{R}(t)) + \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t)).\end{aligned}$$

Let $(\delta^{(2)}, \delta^{(1)}, \delta^{(0)}) \in \mathrm{Der}(s \otimes t)$. Define

$$p := \Xi((s \otimes t) \circ_2 \delta^{(2)}), \qquad q := \Xi((s \otimes t) \circ_1 \delta^{(1)}), \qquad r := \Xi((s \otimes t) \circ_0 \delta^{(0)}).$$

Then $p + q = r$ in $U \otimes V$, where $U = \mathrm{Bil}(U_2, U_1; U_0), V = \mathrm{Bil}(V_2, V_1; V_0)$.

The strategy is to apply Lemma 5.13 to $p+q = r$, obtain an LIU decomposition, then lift the local equalities to relations on the axis operators. With

$$\begin{aligned} A &:= \mathrm{Im}_L(p), \quad B := \mathrm{Im}_L(q), \quad C := \mathrm{Im}_L(r), \\ X &:= \mathrm{Im}_R(p), \quad Y := \mathrm{Im}_R(q), \quad Z := \mathrm{Im}_R(r). \end{aligned}$$

By Proposition 4.5, $p \in A \otimes X$, $q \in B \otimes Y$, and $r \in C \otimes Z$. Let $(a_i, b_i, c_i, x_i, y_i, z_i)_{i=1}^n$ be the LIU decomposition by applying Lemma 5.13. By Remark 5.15, the LIU decomposition index set $[n]$ is partitioned into

$$I_{11} \sqcup I_{1\bullet} \sqcup I_{\bullet 1} \sqcup I_{22} \sqcup I_{33} \sqcup I_{44},$$

where each subset indexes local terms supported in the correspondingly colored block ($U^{(1)} \otimes V^{(1)}$, $U^{(1)} \otimes V^{(\bullet)}$, $U^{(\bullet)} \otimes V^{(1)}$, $U^{(2)} \otimes V^{(2)}$, $U^{(3)} \otimes V^{(3)}$, $U^{(4)} \otimes V^{(4)}$). We show each part gives an operator triple that's in one of the summands on the right-hand side of Equation (5.1).

- $i \in I_{1\bullet}$ ($U^{(1)} \otimes V^{(\bullet)}$): Each local equality satisfies
$$a_i = b_i = c_i, \qquad x_i + y_i = z_i,$$
with $a_i, b_i, c_i \in A \cap B \cap C$, $x_i \in X$, $y_i \in Y$, $z_i \in Z$. Recall from Lemma 4.13 that
$$A \leq \mathrm{Im}(\Lambda_2^s), \quad B \leq \mathrm{Im}(\Lambda_1^s), \quad C \leq \mathrm{Im}(\Lambda_0^s),$$
$$X \leq \mathrm{Im}(\Lambda_2^t), \quad Y \leq \mathrm{Im}(\Lambda_1^t), \quad Z \leq \mathrm{Im}(\Lambda_0^t).$$
So each of $a_i, b_i, c_i, x_i, y_i, z_i$ has a preimage under the corresponding axis action maps. Choose
$$\alpha_i \in \mathrm{End}(U_2),\ \beta_i \in \mathrm{End}(U_1),\ \gamma_i \in \mathrm{End}(U_0)$$
$$f_i \in \mathrm{End}(V_2),\ g_i \in \mathrm{End}(V_1),\ h_i \in \mathrm{End}(V_0)$$
such that $s \circ_2 \alpha_i = a_i,\ s \circ_1 \beta_i = b_i,\ s \circ_0 \gamma_i = c_i$, and $t \circ_2 f_i = x_i,\ t \circ_1 g_i = y_i,\ t \circ_0 h_i = z_i$.
Then $(\alpha_i, \beta_i, \gamma_i) \in \mathcal{C}(s)$ and $(f_i, g_i, h_i) \in \mathrm{Der}(t)$, so
$$(\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i), \kappa_0(\gamma_i \otimes h_i)) \in \mathcal{C}(s) \boxtimes \mathrm{Der}(t).$$
- $i \in I_{\bullet 1}$ ($U^{(\bullet)} \otimes V^{(1)}$): Here each local equality satisfies $x_i = y_i = z_i$ and $a_i + b_i = c_i$. Choose
$$\alpha_i \in \mathrm{End}(U_2),\ \beta_i \in \mathrm{End}(U_1),\ \gamma_i \in \mathrm{End}(U_0)$$
$$f_i \in \mathrm{End}(V_2),\ g_i \in \mathrm{End}(V_1),\ h_i \in \mathrm{End}(V_0)$$
such that
$$s \circ_2 \alpha_i = a_i,\ s \circ_1 \beta_i = b_i,\ s \circ_0 \gamma_i = c_i, \qquad t \circ_2 f_i = x_i,\ t \circ_1 g_i = y_i,\ t \circ_0 h_i = z_i.$$
Then $(\alpha_i, \beta_i, \gamma_i) \in \mathrm{Der}(s)$ and $(f_i, g_i, h_i) \in \mathcal{C}(t)$, hence
$$(\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i), \kappa_0(\gamma_i \otimes h_i)) \in \mathrm{Der}(s) \boxtimes \mathcal{C}(t).$$
- $i \in I_{22}, I_{33}, I_{44}$: these are the nucleus cases with one side equal to 0. We show the I_{22} case, which has $c_i \otimes z_i = 0$. The local equality is $a_i \otimes x_i + b_i \otimes y_i = 0$, with $a_i, b_i \in A \cap B$ and $x_i, y_i \in X \cap Y$. Choose operators
$$\alpha_i \in \mathrm{End}(U_2),\ \beta_i \in \mathrm{End}(U_1), \qquad f_i \in \mathrm{End}(V_2),\ g_i \in \mathrm{End}(V_1)$$
satisfying
$$s \circ_2 \alpha_i = a_i,\ s \circ_1 \beta_i = b_i, \qquad t \circ_2 f_i = x_i,\ t \circ_1 g_i = y_i.$$
Then for all i,
$$(s \otimes t) \circ_2 \kappa_2(\alpha_i \otimes f_i) = a_i \otimes x_i = -\, b_i \otimes y_i = -\,(s \otimes t) \circ_1 \kappa_1(\beta_i \otimes g_i).$$
Hence
$$\big(-\kappa_2(\alpha_i \otimes f_i),\ \kappa_1(\beta_i \otimes g_i)\big) \in \mathcal{M}(s \otimes t),$$
and therefore
$$(\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i), 0) = \iota_{21}\big(-\kappa_2(\alpha_i \otimes f_i), \kappa_1(\beta_i \otimes g_i)\big) \in \iota_{21}(\mathcal{M}(s \otimes t)).$$
Applying Theorem 4.1, $\iota_{21}(\mathcal{M}(s \otimes t)) = \iota_{21}(\mathcal{M}(s) \boxtimes \mathcal{M}(t))$, so the operator triple lies in one of the summands on the right-hand side of Equation (5.1).
The I_{33} and I_{44} cases are analogous.
- $i \in I_{11}$ ($U^{(1)} \otimes V^{(1)}$): This is the same as the $I_{1\bullet}$ case, now with $x_i, y_i, z_i \in X \cap Y \cap Z$ as well.

For each i, let $(\delta_i^{(2)}, \delta_i^{(1)}, \delta_i^{(0)}) \in \mathrm{End}(U_2 \otimes V_2) \times \mathrm{End}(U_1 \otimes V_1) \times \mathrm{End}(U_0 \otimes V_0)$ be the resulting operator triple constructed above. By construction, each triple lies in one of the summands on the right-hand side. Summing over i,
$$\Xi\left((s \otimes t) \circ_2 \sum_{i=1}^{n} \delta_i^{(2)}\right) = p = \Xi((s \otimes t) \circ_2 \delta^{(2)}),$$
$$\Xi\left((s \otimes t) \circ_1 \sum_{i=1}^{n} \delta_i^{(1)}\right) = q = \Xi((s \otimes t) \circ_1 \delta^{(1)}),$$

$$\Xi\left((s\otimes t)\circ_0 \sum_{i=1}^{n}\delta_i^{(0)}\right) = r = \Xi((s\otimes t)\circ_0 \delta^{(0)}).$$

Since Ξ is an isomorphism and $\Lambda_2^{s\otimes t}, \Lambda_1^{s\otimes t}, \Lambda_0^{s\otimes t}$ are injective, we get

$$\sum_{i=1}^{n}\delta_i^{(2)} = \delta^{(2)}, \qquad \sum_{i=1}^{n}\delta_i^{(1)} = \delta^{(1)}, \qquad \sum_{i=1}^{n}\delta_i^{(0)} = \delta^{(0)}.$$

Therefore $(\delta^{(2)},\delta^{(1)},\delta^{(0)}) = \sum_{i=1}^{n}(\delta_i^{(2)},\delta_i^{(1)},\delta_i^{(0)})$. Since each triple $(\delta_i^{(2)},\delta_i^{(1)},\delta_i^{(0)})$ lies in one summand on the right-hand side, their sum lies in that right-hand side subspace sum as well.

($\supseteq$) We prove $\mathrm{Der}(s\otimes t) \supseteq \mathrm{Der}(s)\boxtimes\mathcal{C}(t) + \mathcal{C}(s)\boxtimes\mathrm{Der}(t) + \iota_{20}(\mathcal{L}(s)\boxtimes\mathcal{L}(t)) + \iota_{10}(\mathcal{R}(s)\boxtimes\mathcal{R}(t)) + \iota_{21}(\mathcal{M}(s)\boxtimes\mathcal{M}(t))$ by proving each summand on the right belongs to $\mathrm{Der}(s\otimes t)$.

For $\mathrm{Der}(s)\boxtimes\mathcal{C}(t)$, take

$$(\alpha,\beta,\gamma)\in\mathrm{Der}(s), \qquad (f,g,h)\in\mathcal{C}(t).$$

Then $(\alpha,\beta,\gamma)\boxtimes(f,g,h) = \big(\kappa_2(\alpha\otimes f),\kappa_1(\beta\otimes g),\kappa_0(\gamma\otimes h)\big)$.

As $t\circ_2 f = t\circ_1 g = t\circ_0 h$ and $s\circ_2\alpha + s\circ_1\beta = s\circ_0\gamma$,

$$\begin{aligned}(s\otimes t)\circ_2\kappa_2(\alpha\otimes f) + (s\otimes t)\circ_1\kappa_1(\beta\otimes g) &= s\circ_2\alpha\otimes t\circ_2 f + s\circ_1\beta\otimes t\circ_1 g\\ &= (s\circ_2\alpha + s\circ_1\beta)\otimes t\circ_0 h\\ &= s\circ_0\gamma\otimes t\circ_0 h\\ &= (s\otimes t)\circ_0\kappa_0(\gamma\otimes h).\end{aligned}$$

Taking span, $\mathrm{Der}(s)\boxtimes\mathcal{C}(t)$ lies in $\mathrm{Der}(s\otimes t)$. Hence $\mathrm{Der}(s)\boxtimes\mathcal{C}(t)\subseteq\mathrm{Der}(s\otimes t)$ by linearity.

The argument for $\mathcal{C}(s)\boxtimes\mathrm{Der}(t)\subseteq\mathrm{Der}(s\otimes t)$ is identical, and the argument for nucleus terms is also analogous after setting one of the three terms to 0. □

A worked out derivation decomposition example is the Appendix (Example D.2).

5.5. **Non-existence in higher valence.** In this subsection, we show that the LIU decomposition strategy for understanding the derivation of tensor product bimaps does not extend to higher valence in general. For trilinear maps, the corresponding four term relation is $p+q+r=s$. We formalize the obstruction and give an explicit counterexample.

Let $U := \mathrm{Tri}(U_3,U_2,U_1;U_0)$, and $V := \mathrm{Tri}(V_3,V_2,V_1;V_0)$. Fix $t_1\in U$ and $t_2\in V$, and let $(\delta^{(3)},\delta^{(2)},\delta^{(1)},\delta^{(0)})\in\mathrm{Der}(t_1\otimes t_2)$. This gives tensors $p,q,r,s\in U\otimes V$ with $p+q+r=s$. We begin with the four-term definition of the LIU decomposition.

Definition 5.16. Let $p,q,r,s\in U\otimes V$ satisfy $p+q+r=s$. Let

$$\begin{aligned}A &:= \mathrm{Im}_L(p), & B &:= \mathrm{Im}_L(q), & C &:= \mathrm{Im}_L(r), & D &:= \mathrm{Im}_L(s),\\ X &:= \mathrm{Im}_R(p), & Y &:= \mathrm{Im}_R(q), & Z &:= \mathrm{Im}_R(r), & W &:= \mathrm{Im}_R(s).\end{aligned}$$

By Proposition 4.5, $p\in A\otimes X$, $q\in B\otimes Y$, $r\in C\otimes Z$, and $s\in D\otimes W$.

A (p,q,r,s) *LIU decomposition* consists of a natural number n, and vectors

$$\forall i\in[n],\quad a_i\in A,\ b_i\in B,\ c_i\in C,\ d_i\in D,\quad x_i\in X,\ y_i\in Y,\ z_i\in Z,\ w_i\in W$$

satisfying the following:

- **Decomposition equality.**

$$p=\sum_{i=1}^{n}a_i\otimes x_i,\quad q=\sum_{i=1}^{n}b_i\otimes y_i,\quad r=\sum_{i=1}^{n}c_i\otimes z_i,\quad s=\sum_{i=1}^{n}d_i\otimes w_i.$$

- **Local equality.**

$$\forall i\in[n],\quad a_i\otimes x_i + b_i\otimes y_i + c_i\otimes z_i = d_i\otimes w_i.$$

We shall show when $p+q+r=s$, there may not exist a LIU decomposition of the quadruple. This rules out the possibility to decompose $\mathrm{Der}(s \otimes t)$ in terms of algebraic invariants of s and t.

A natural question is whether one can reduce the $p+q+r=s$ case to the three-term case by setting $\tilde{p}=p+q$, and applying Lemma 5.13 to both $\tilde{p}+r=s$ and $p+q=\tilde{p}$.

However, separate LIU decompositions of $(\tilde{p}, r, s)$ and $(p, q, \tilde{p})$ does not guarantee a LIU decomposition of (p, q, r, s). Applying Lemma 5.13 to $(\tilde{p}, r, s)$ gives

$$\tilde{p}=\sum_i \tilde{a}_i \otimes \tilde{x}_i, \quad \tilde{a}_i \otimes \tilde{x}_i \in \mathrm{Im}_L(\tilde{p}) \otimes \mathrm{Im}_R(\tilde{p}).$$

The equation $p+q=\tilde{p}$ means $\mathrm{Im}_L(\tilde{p}) \leq \mathrm{Im}_L(p)+\mathrm{Im}_L(q)$ and $\mathrm{Im}_R(\tilde{p}) \leq \mathrm{Im}_R(p)+\mathrm{Im}_R(q)$. To recover a (p, q, r, s) LIU decomposition, we need the following equation to be satisfied for each i:

$$\tilde{a}_i \otimes \tilde{x}_i=a_i \otimes x_i+b_i \otimes y_i \quad (a_i \in \mathrm{Im}_L(p),\ x_i \in \mathrm{Im}_R(p),\ b_i \in \mathrm{Im}_L(q),\ y_i \in \mathrm{Im}_R(q)),$$

but this is not guaranteed. Rank-one terms in $(A+B) \otimes(X+Y)$ can contain cross blocks terms in $A \otimes Y$ and $B \otimes X$. For example, with $A=\langle e_1\rangle$, $B=\langle e_2\rangle$, $X=\langle f_1\rangle$, $Y=\langle f_2\rangle$,

$$(e_1+e_2) \otimes(f_1+f_2)=e_1 \otimes f_1+e_1 \otimes f_2+e_2 \otimes f_1+e_2 \otimes f_2,$$

so this rank-one tensor has nonzero $A \otimes Y$ and $B \otimes X$ parts and cannot be of the form $a \otimes x+b \otimes y$ with $a \in A$, $x \in X$, $b \in B$, $y \in Y$.

Lemma 5.17. Assume $s \neq 0$. If a LIU decomposition exists, then the intersection space

$$H \coloneqq(A \otimes X+B \otimes Y+C \otimes Z) \cap(D \otimes W)$$

contains a nonzero rank-one tensor.

Proof. Each local equality has right-hand side $d_i \otimes w_i \in D \otimes W$ and left-hand side in $A \otimes X+B \otimes Y+C \otimes Z$, hence $d_i \otimes w_i \in H$ for every i. Since

$$s=\sum_{i=1}^n d_i \otimes w_i \neq 0,$$

at least one $d_i \otimes w_i$ is nonzero, giving a nonzero rank-one tensor in H. □

We now show that LIU decompositions may not exist.

Theorem 5.18. Let $\mathbb{F}$ be a field such that $\mathrm{char}(\mathbb{F}) \neq 2$, and let $J \in \mathbb{M}_2(\mathbb{F})$ be a matrix with an irreducible minimal polynomial. Let $U=V=\mathbb{F}^4$ be vector spaces. Then there exist subspaces $A, B, C, D \leq U$ and $X, Y, Z, W \leq V$, and tensors $p \in A \otimes X$, $q \in B \otimes Y$, $r \in C \otimes Z$, $s \in D \otimes W$ satisfying $p+q+r=s$ with no LIU decomposition.

Proof. We start by defining the spaces $A, B, C, D \leq U$ and $X, Y, Z, W \leq V$, and the intersection space

$$H \coloneqq(A \otimes X+B \otimes Y+C \otimes Z) \cap(D \otimes W).$$

Then we construct a nonzero element $s \in H$ and show that every nonzero element of H has rank 2. Hence H contains no nonzero rank-one tensor, so Lemma 5.17 implies that a corresponding quadruple admits no LIU decomposition.

Take $U=V=\mathbb{F}^4$ with standard bases $\{e_i\}$ and $\{f_i\}$, and set

$$D=\mathrm{span}\{e_1, e_2\}, \quad D'=\mathrm{span}\{e_3, e_4\}, \qquad W=\mathrm{span}\{f_1, f_2\}, \quad W'=\mathrm{span}\{f_3, f_4\}.$$

Define isomorphisms

$$\iota_D: D \to D',\ e_1 \mapsto e_3,\ e_2 \mapsto e_4, \qquad \iota_W: W \to W',\ f_1 \mapsto f_3,\ f_2 \mapsto f_4.$$

For $J=(j_{ab}) \in \mathbb{M}_2(\mathbb{F})$, use the fixed bases (e_1, e_2) of D, (e_3, e_4) of D', (f_1, f_2) of W, and (f_3, f_4) of W' to define linear maps

$$J_D: D \to D', \qquad J_W: W \to W'.$$

Write $Jd \coloneqq J_D(d)$ and $wJ \coloneqq J_W(w)$. With this notation, define subspaces

$$A=\{(d, \iota_D d): d \in D\}, \quad B=\{(d,-\iota_D d): d \in D\}, \quad C=\{(d, Jd): d \in D\},$$

$$X = \{(w, \iota_W w) : w \in W\}, \quad Y = \{(w, -\iota_W w) : w \in W\}, \quad Z = \{(w, wJ) : w \in W\}.$$

Let $H \coloneqq (A \otimes X + B \otimes Y + C \otimes Z) \cap (D \otimes W)$. For

$$p \in A \otimes X, \quad q \in B \otimes Y, \quad r \in C \otimes Z, \quad s \in D \otimes W,$$

there exist $P, Q, R, S \in \mathbb{M}_2(\mathbb{F})$ with

$$p = \begin{bmatrix} P & P \\ P & P \end{bmatrix}, \qquad q = \begin{bmatrix} Q & -Q \\ -Q & Q \end{bmatrix}, \qquad r = \begin{bmatrix} R & RJ \\ JR & JRJ \end{bmatrix}, \qquad s = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix}.$$

with data given in 2×2 blocks relative to $U = D \oplus D'$ and $V = W \oplus W'$.

Hence $p + q + r = s$ is equivalent to

$$\begin{bmatrix} P + Q + R & P - Q + RJ \\ P - Q + JR & P + Q + JRJ \end{bmatrix} = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix},$$

meaning

$$P - Q = -RJ, \qquad P + Q = -RJ^2, \qquad S = R - RJ^2.$$

For any 2×2 matrix S, we can define P, Q, R in terms of S and the fixed J via

$$P = -\frac{1}{2}(RJ + RJ^2), \qquad Q = \frac{1}{2}(RJ - RJ^2), \qquad R = S(I_2 - J^2)^{-1}.$$

Here, invertibility of $I_2 - J^2$ is due to it being a nonzero element of $\mathbb{F}[J]$ which is a field. Fixing $S = I_2$ gives the existence part of the theorem.

To conclude, let $0 \neq h \in H$. It has block form $h = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix}$ for some $S \in \mathbb{M}_2(\mathbb{F})$. From the off-diagonal equations we get $P - Q + RJ = 0$ and $P - Q + JR = 0$, hence $RJ = JR$, so $R \in \mathbb{F}[J]$. So $S = R - RJ^2$ is in $\mathbb{F}[J]$. Since $h \neq 0$, we have $S \neq 0$. Since J has irreducible minimal polynomial, $\mathbb{F}[J]$ is a field, so S is invertible. Therefore h has rank 2. □

Example 5.19. Over $\mathbb{F} = \mathbb{F}_5$, take

$$J = \begin{bmatrix} 0 & 4 \\ 1 & 4 \end{bmatrix}, \qquad S = I_2.$$

Compute P, Q, R in terms of S, getting

$$R = (I_2 - J^2)^{-1} = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix}, \qquad P = \begin{bmatrix} 1 & 1 \\ 4 & 2 \end{bmatrix}, \qquad Q = \begin{bmatrix} 3 & 2 \\ 3 & 0 \end{bmatrix},$$

Recall $U = D \oplus D'$ and $V = W \oplus W'$, so $(u, u') \in D \oplus D'$ and $(v, v') \in W \oplus W'$ denote direct-sum coordinate elements of U and V. For $i \in [2]$, set

$$\begin{aligned} a_i &\coloneqq (e_i, \iota_D e_i), & x_i &\coloneqq (f_i, \iota_W f_i), \\ b_i &\coloneqq (e_i, -\iota_D e_i), & y_i &\coloneqq (f_i, -\iota_W f_i), \\ c_i &\coloneqq (e_i, Je_i), & z_i &\coloneqq (f_i, f_i J), \\ d_i &\coloneqq (e_i, 0), & w_i &\coloneqq (f_i, 0), \end{aligned}$$

With $a_i, b_i, c_i, d_i \in U$ and $x_i, y_i, z_i, w_i \in V$ spanning the subspaces $A, B, C, D \leq U$ and $X, Y, Z, W \leq V$ respectively, define

$$\begin{aligned} p &\coloneqq \sum_{i,j=1}^{2} P_{ij} a_i \otimes x_j \in A \otimes X, & q &\coloneqq \sum_{i,j=1}^{2} Q_{ij} b_i \otimes y_j \in B \otimes Y, \\ r &\coloneqq \sum_{i,j=1}^{2} R_{ij} c_i \otimes z_j \in C \otimes Z, & s &\coloneqq \sum_{i,j=1}^{2} S_{ij} d_i \otimes w_j \in D \otimes W. \end{aligned}$$

In the standard coordinates, they are

$$p = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 4 & 2 & 4 & 2 \\ 1 & 1 & 1 & 1 \\ 4 & 2 & 4 & 2 \end{bmatrix}, \quad q = \begin{bmatrix} 3 & 2 & 2 & 3 \\ 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & 2 \\ 2 & 0 & 3 & 0 \end{bmatrix},$$

$$r = \begin{bmatrix} 2 & 2 & 2 & 1 \\ 3 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 \\ 4 & 3 & 3 & 3 \end{bmatrix}, \quad s = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

One checks directly that $p + q + r = s$. Since this example arises from the construction in the proof of Theorem 5.18, every nonzero element of the associated space $H = ((A \otimes X) + (B \otimes Y) + (C \otimes Z)) \cap (D \otimes W)$ has rank 2. Hence the quadruple (p, q, r, s) does not admit a LIU decomposition.

6. Conclusion

Write last: a summary of what was done.

6.1. Future Directions.

(1) Depending on what gets written for the quick der chapter, randomized algorithms analysis
(2) Quick der/quicksylver for higher valence tensors
(3) X-ray idea with decomposing tensors.
(4) Derivation closure and porting the result on the derivation algebra to one on tensor spaces

Appendix A. Quicksylver and Quickder implementation performance

TODO: Add some performance numbers/charts from Quicksylver/derivation calculations from the algorithm implementation in Julia

Appendix B. QuickSylver/QuickDer examples

B.1. QuickSylver running example. Back to main text.

Example B.1. Let $\mathbb{F} = \mathbb{Q}$, with $a = b = r = s = 2$ and $c = 4$. Define

$$R[:,:,1] = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}, \quad R[:,:,2] = \begin{bmatrix} 3 & 1 \\ 4 & 3 \end{bmatrix}, \quad R[:,:,3] = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, \quad R[:,:,4] = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix},$$

$$S[:,:,1] = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}, \quad S[:,:,2] = \begin{bmatrix} 3 & 2 \\ 5 & 2 \end{bmatrix}, \quad S[:,:,3] = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}, \quad S[:,:,4] = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix},$$

$$T[:,:,1] = \begin{bmatrix} 11 & 7 \\ 24 & 25 \end{bmatrix}, \quad T[:,:,2] = \begin{bmatrix} 19 & 10 \\ 44 & 23 \end{bmatrix}, \quad T[:,:,3] = \begin{bmatrix} 3 & -1 \\ 20 & 15 \end{bmatrix}, \quad T[:,:,4] = \begin{bmatrix} 20 & 7 \\ 20 & 11 \end{bmatrix}.$$

The full system has the unique solution

$$X = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}, \qquad Y = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}.$$

Now run the solve-and-lift algorithm with $I = J = \{1\}$.

Double restricted. Set

$$X_I = \begin{bmatrix} x_1 & x_2 \end{bmatrix}, \qquad Y_J = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

The restricted system is

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ 3 & 4 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 11 \\ 19 \\ 3 \\ 20 \end{bmatrix}.$$

It has the unique solution

$$X_I = \begin{bmatrix} -1 & 1 \end{bmatrix}, \qquad Y_J = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Hence $\mathcal{S}_{\mathrm{DR}} = \{(X_I, Y_J)\}$.

Row restricted. With $\widehat{J} = \{2\}$,

$$M_R = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 0 \\ 4 & 1 \end{bmatrix}, \qquad N_R(X_I, Y_J) = \begin{bmatrix} 5 \\ 8 \\ 2 \\ 9 \end{bmatrix}.$$

Solving $M_R \widehat{Y} = N_R(X_I, Y_J)$ gives

$$\widehat{Y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad Y = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}.$$

So

$$\mathcal{S}_{\mathrm{R}} = \left\{ \left(\begin{bmatrix} -1 & 1 \end{bmatrix}, \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \right) \right\}.$$

Column restricted. With $\widehat{I} = \{2\}$,

$$M_C = \begin{bmatrix} 1 & 1 \\ 3 & 4 \\ 3 & 2 \\ 0 & 1 \end{bmatrix}, \qquad N_C(X_I, Y_J) = \begin{bmatrix} 5 \\ 18 \\ 12 \\ 3 \end{bmatrix}.$$

Solving $M_C \widehat{X} = N_C(X_I, Y_J)$ gives

$$\widehat{X} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad X = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}.$$

So

$$\mathcal{S}_{\mathrm{C}} = \left\{ \left(\begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right) \right\}.$$

Lift to full. The pullback is

$$P = \{(z_R, z_C) \in \mathcal{S}_{\mathrm{R}} \times \mathcal{S}_{\mathrm{C}} : \pi_J(z_R) = \pi_I(z_C)\} = \{p_0\},$$

where

$$p_0 := \big((X_I, Y), (X, Y_J)\big), \qquad X_I = X[I, :], \;\; Y_J = Y[:, J].$$

Then

$$\mathrm{Err}(\Psi(p_0)) = \big(XR[:,:,k] + S[:,:,k]Y - T[:,:,k]\big)_{k \in [4]} = 0_{2\times 2\times 4}.$$

Therefore LiftRestricted returns the unique solution (X, Y) above.

B.2. **QuickDer running example.** **Back to main text.**

Example B.2. We illustrate the solution strategy on a $2 \times 2 \times 2$ instance. Let $a = b = c = r = s = t = 2$, and $I = J = K = \{1\}$. Define the slices

$$R[:,:,1] = S[:,:,1] = T[:,:,1] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad R[:,:,2] = S[:,:,2] = T[:,:,2] = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Write the unknowns as

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}, \qquad Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}, \qquad Z = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}.$$

We solve the derivation equations by brute force first. They are

$$X\, R[:,:,k] + S[:,:,k]\, Y = (T^Z)[:,:,k] \qquad (k \in [2])$$

For $k = 1$, we have

$$X \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} Y = z_{11} T[:,:,1] + z_{21} T[:,:,2].$$

This gives the relations

$$\begin{bmatrix} x_{11} + y_{11} & y_{12} \\ x_{21} & 0 \end{bmatrix} = \begin{bmatrix} z_{11} & 0 \\ 0 & z_{21} \end{bmatrix},$$

so $y_{12} = x_{21} = z_{21} = 0$, and $x_{11} + y_{11} = z_{11}$.

And for $k = 2$,

$$X \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} Y = z_{12} T[:,:,1] + z_{22} T[:,:,2],$$

gives the relations

$$\begin{bmatrix} 0 & x_{12} \\ y_{21} & x_{22} + y_{22} \end{bmatrix} = \begin{bmatrix} z_{12} & 0 \\ 0 & z_{22} \end{bmatrix},$$

so $z_{12} = x_{12} = y_{21} = 0$, and $x_{22} + y_{22} = z_{22}$. Combined, the solution space is

$$X = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}, \qquad Y = \begin{bmatrix} \gamma & 0 \\ 0 & \delta \end{bmatrix}, \qquad Z = \begin{bmatrix} \alpha + \gamma & 0 \\ 0 & \beta + \delta \end{bmatrix}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{F},$$

We now construct the solution space using the steps described in this section.

Step 1: Triple-restricted solve. With $I = J = K = \{1\}$, the unknown blocks are

$$X_I = \begin{bmatrix} x_{11} & x_{12} \end{bmatrix}, \qquad Y_J = \begin{bmatrix} y_{11} \\ y_{21} \end{bmatrix}, \qquad Z_K = \begin{bmatrix} z_{11} \\ z_{21} \end{bmatrix},$$

and the triple-restricted equation has left hand and right hand sides

$$X_I R[:, J, 1] + S[I, :, 1] Y_J = \begin{bmatrix} x_{11} & x_{12} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_{11} \\ y_{21} \end{bmatrix} = x_{11} + y_{11},$$

$$(T^{Z_K})[I, J, 1] = \big(z_{11} T[:,:,1] + z_{21} T[:,:,2]\big)[1, 1] = z_{11}.$$

Hence the triple-restricted constraint is $x_{11} + y_{11} = z_{11}$.

Step 2: Column-depth lift. Recall $\widehat{I} = [a] \setminus I = \{2\}$ and $\widehat{X} = X[\widehat{I}, :]^T = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}$. Using the notation from above:

$$M_C = \big[(R[:, J, 1])^T\big] = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad N_C(X_I, Y_J, Z_K) = \big[(T^{Z_K})[\widehat{I}, J, 1] - S[\widehat{I}, :, 1] Y_J\big]^T = \begin{bmatrix} 0 \end{bmatrix}.$$

So the equation $M_C \widehat{X} = N_C(X_I, Y_J, Z_K)$ corresponds to the system

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix},$$

concluding $x_{21} = 0$.

Step 3: Row-depth lift. Recall $\widehat{J} = [b] \setminus J = \{2\}$ and $\widehat{Y} = Y[:, \widehat{J}] = \begin{bmatrix} y_{12} \\ y_{22} \end{bmatrix}$.

$$M_R = \big[S[I, :, 1]\big] = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad N_R(X_I, Y_J, Z_K) = \big[(T^{Z_K})[I, \widehat{J}, 1] - X_I R[:, \widehat{J}, 1]\big] = \begin{bmatrix} 0 \end{bmatrix}.$$

So the equation $M_R \widehat{Y} = N_R(X_I, Y_J, Z_K)$ corresponds to the system

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_{12} \\ y_{22} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix},$$

concluding $y_{12} = 0$.

Step 4: Row-col lift. Recall $\widehat{K} = [c] \setminus K = \{2\}$ and $\widehat{Z} = Z[:, \widehat{K}] = \begin{bmatrix} z_{12} \\ z_{22} \end{bmatrix}$. The row-col lift uses

$$M_D = \big[\mathrm{vec}(T[I, J, 1]) \quad \mathrm{vec}(T[I, J, 2])\big] = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

$$N_D(X_I, Y_J, Z_K) = \mathrm{vec}\big(X_I R[:, J, 2] + S[I, :, 2] Y_J\big) = \begin{bmatrix} 0 \end{bmatrix}.$$

So the equation $M_D\widehat{Z} = N_D(X_I, Y_J, Z_K)$ corresponds to the system

$$\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}z_{12}\\ z_{22}\end{bmatrix} = \begin{bmatrix}0\end{bmatrix},$$

concluding $z_{12} = 0$.

Step 5: Pullback and final residual solve. Using the three lifts and the pullback conditions, the compatible candidates are

$$X = \begin{bmatrix}x_{11} & x_{12}\\ 0 & x_{22}\end{bmatrix}, \qquad Y = \begin{bmatrix}y_{11} & 0\\ y_{21} & y_{22}\end{bmatrix}, \qquad Z = \begin{bmatrix}x_{11}+y_{11} & 0\\ z_{21} & z_{22}\end{bmatrix}.$$

Now apply the final lift condition $\mathrm{Err}(X, Y, Z) = 0$, where $\mathrm{Err}(X, Y, Z)[:, :, k] = X\,R[:, :, k] + S[:, :, k]Y - (T^Z)[:, :, k]$. For the candidate family above,

$$(T^Z)[:,:,1] = \begin{bmatrix}x_{11}+y_{11} & 0\\ 0 & z_{21}\end{bmatrix}, \qquad \mathrm{Err}(X,Y,Z)[:,:,1] = \begin{bmatrix}0 & 0\\ 0 & -z_{21}\end{bmatrix},$$

$$(T^Z)[:,:,2] = \begin{bmatrix}0 & 0\\ 0 & z_{22}\end{bmatrix}, \qquad \mathrm{Err}(X,Y,Z)[:,:,2] = \begin{bmatrix}0 & x_{12}\\ y_{21} & x_{22}+y_{22}-z_{22}\end{bmatrix}.$$

So $\mathrm{Err}(X, Y, Z) = 0$ requires

$$z_{21} = 0, \qquad x_{12} = 0, \qquad y_{21} = 0, \qquad z_{22} = x_{22} + y_{22}.$$

Hence the full derivation space matches what we computed initially as

$$X = \begin{bmatrix}\alpha & 0\\ 0 & \beta\end{bmatrix}, \qquad Y = \begin{bmatrix}\gamma & 0\\ 0 & \delta\end{bmatrix}, \qquad Z = \begin{bmatrix}\alpha+\gamma & 0\\ 0 & \beta+\delta\end{bmatrix}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{F},$$

Appendix C. Proof of Proposition 5.3

The strategy is to apply Theorem 5.1 to the multiplication tensors of algebras R and S, and then restrict to diagonal triples to recover the algebra derivations. We first record the kernel-intersection splitting lemma that will be needed later.

Lemma C.1. Let U, V be vector spaces, let $\epsilon : U \to V$ be a linear map, and let $X, Y, Z \leq U$ be subspaces. Suppose we have

$$\begin{aligned}\epsilon(X) \cap \big(\epsilon(Y) + \epsilon(Z)\big) &= \epsilon(X \cap Z),\\ \epsilon(Y) \cap \big(\epsilon(X) + \epsilon(Z)\big) &= \epsilon(Y \cap Z).\end{aligned} \tag{C.1}$$

Then $(X + Y + Z) \cap \ker(\epsilon) = (X \cap \ker(\epsilon)) + (Y \cap \ker(\epsilon)) + (Z \cap \ker(\epsilon))$.

Proof. The inclusion

$$(X \cap \ker(\epsilon)) + (Y \cap \ker(\epsilon)) + (Z \cap \ker(\epsilon)) \subseteq (X + Y + Z) \cap \ker(\epsilon)$$

is immediate.

For the reverse inclusion, let $u \in (X + Y + Z) \cap \ker(\epsilon)$. Write

$$u = x + y + z$$

with $x \in X$, $y \in Y$, and $z \in Z$. Since $u \in \ker(\epsilon)$, $0 = \epsilon(u) = \epsilon(x) + \epsilon(y) + \epsilon(z)$. Hence

$$\epsilon(x) = -\epsilon(y) - \epsilon(z) \in \epsilon(Y) + \epsilon(Z).$$

Since also $\epsilon(x) \in \epsilon(X)$, by Equation (C.1) we have

$$\epsilon(x) \in \epsilon(X) \cap \big(\epsilon(Y) + \epsilon(Z)\big) = \epsilon(X \cap Z).$$

Thus there exists $z_X \in X \cap Z$ such that

$$\epsilon(z_X) = \epsilon(x), \quad \text{and} \quad x - z_X \in X \cap \ker(\epsilon).$$

Similarly, there exists $z_Y \in Y \cap Z$ such that

$$\epsilon(z_Y) = \epsilon(y), \quad \text{and} \quad y - z_Y \in Y \cap \ker(\epsilon).$$

Finally, $z + z_X + z_Y \in Z$, and

$$\epsilon(z + z_X + z_Y) = \epsilon(z) + \epsilon(z_X) + \epsilon(z_Y) = \epsilon(z) + \epsilon(x) + \epsilon(y) = 0,$$

so $z + z_X + z_Y \in Z \cap \ker(\epsilon)$. Since

$$u = (x - z_X) + (y - z_Y) + (z + z_X + z_Y),$$

we get the desired conclusion

$$u \in (X \cap \ker(\epsilon)) + (Y \cap \ker(\epsilon)) + (Z \cap \ker(\epsilon)). \qquad \square$$

For Proposition 5.3, we now start by building a dictionary between algebra definitions and tensor definitions.

Lemma C.2. If A is a unital algebra and $\mu : A \times A \rightarrowtail A$ is its multiplication tensor, then μ is fully nondegenerate.

Proof. $\mu(a, 1) = a$ and $\mu(1, a) = a$ force both the left and right radical to be 0, and $\operatorname{span}(\mu(A, A)) = A$ since every $a \in A$ equals $\mu(a, 1)$. $\square$

The map $d \mapsto (d, d, d)$ identifies the derivation algebra of A, $\mathcal{D}_{\text{alg}}(A)$, with the diagonal of $\operatorname{Der}(\mu_A)$. For $x \in A$, write $\lambda_x(a) := xa$ and $\rho_x(a) := ax$ for the left and right multiplication operators. Recall the definitions for a unital algebra A: **TODO: cite non-associative algebra book for definitions**

$$\begin{aligned}
&\text{(left nucleus)} && \operatorname{Nuc}_\ell(A) := \{x \in A : \forall a, b \in A,\ (xa)b = x(ab)\},\\
&\text{(middle nucleus)} && \operatorname{Nuc}_m(A) := \{y \in A : \forall a, b \in A,\ (ay)b = a(yb)\},\\
&\text{(right nucleus)} && \operatorname{Nuc}_r(A) := \{z \in A : \forall a, b \in A,\ a(bz) = (ab)z\},\\
&\text{(nucleus)} && \operatorname{Nuc}(A) := \operatorname{Nuc}_\ell(A) \cap \operatorname{Nuc}_m(A) \cap \operatorname{Nuc}_r(A),\\
&\text{(center)} && Z(A) := \{c \in \operatorname{Nuc}(A) : \forall a \in A,\ ca = ac\}.
\end{aligned}$$

We do not define the centroid of an algebra A as it corresponds to the center in the unital case. Now we relate them to tensor definitions.

Proposition C.3. Let A be a unital algebra with multiplication tensor μ. Then

$$\begin{aligned}
\mathcal{M}(\mu) &= \{(\rho_y, \lambda_y) : y \in \operatorname{Nuc}_m(A)\}, & \mathcal{L}(\mu) &= \{(\lambda_x, \lambda_x) : x \in \operatorname{Nuc}_\ell(A)\},\\
\mathcal{R}(\mu) &= \{(\rho_z, \rho_z) : z \in \operatorname{Nuc}_r(A)\}, & \mathcal{C}(\mu) &= \{(\lambda_c, \lambda_c, \lambda_c) : c \in Z(A)\}.
\end{aligned}$$

Proof. We prove $\mathcal{M}(\mu) = \{(\rho_y, \lambda_y) : y \in \operatorname{Nuc}_m(A)\}$.

For $\subseteq$, let $(\alpha, \beta) \in \mathcal{M}(\mu)$. Then

$$(\forall a, b \in A) \quad (\alpha(a))b = a(\beta(b)).$$

Setting $b = 1$ gives $\alpha(a) = a\,\beta(1)$, so $\alpha = \rho_y$ with $y = \beta(1)$. Setting $a = 1$ gives $\beta(b) = \alpha(1)\,b$, so $\beta = \lambda_y$ with $y = \alpha(1)$. Hence $y = \alpha(1) = \beta(1)$ and $(ay)b = a(yb)$, so $y \in \operatorname{Nuc}_m(A)$. Therefore $(\alpha, \beta) = (\rho_y, \lambda_y)$ is in the right-hand side.

For $\supseteq$, let $y \in \operatorname{Nuc}_m(A)$. Then for all $a, b \in A$,

$$(\rho_y(a))b = (ay)b = a(yb) = a(\lambda_y(b)),$$

so $(\rho_y, \lambda_y) \in \mathcal{M}(\mu)$.

The equations for $\mathcal{L}(\mu)$ and $\mathcal{R}(\mu)$ are proved identically. Finally,

$$\begin{aligned}
\mathcal{C}(\mu) &= \{(\alpha, \beta, \gamma) : \mu \circ_2 \alpha = \mu \circ_1 \beta = \mu \circ_0 \gamma\}\\
&= \{(\alpha, \beta, \gamma) : (\alpha, \beta) \in \mathcal{M}(\mu),\ (\alpha, \gamma) \in \mathcal{L}(\mu),\ (\beta, \gamma) \in \mathcal{R}(\mu)\}.
\end{aligned}$$

Using the previous three parts, let $(\alpha, \beta, \gamma) \in \mathcal{C}(\mu)$. Then

$$(\alpha, \gamma) = (\lambda_x, \lambda_x), \quad (\beta, \gamma) = (\rho_z, \rho_z), \quad (\alpha, \beta) = (\rho_y, \lambda_y)$$

for some $x \in \operatorname{Nuc}_\ell(A)$, $z \in \operatorname{Nuc}_r(A)$, $y \in \operatorname{Nuc}_m(A)$. Hence $\lambda_x = \rho_z$, $\lambda_x = \rho_y$, and $\rho_z = \lambda_y$, so $x = y = z =: c$ and $\lambda_c = \rho_c$. Thus $c \in \operatorname{Nuc}_\ell(A) \cap \operatorname{Nuc}_m(A) \cap \operatorname{Nuc}_r(A)$ and $ca = ac$ for all a. This means $c \in Z(A)$.

Conversely, if $c \in Z(A)$ then $(\lambda_c, \lambda_c, \lambda_c) \in \mathcal{C}(\mu)$. $\square$

For any algebra A and $u \in \mathrm{Nuc}(A)$, let $\mathrm{ad}(u) := \lambda_u - \rho_u$. A calculation shows $\mathrm{ad}(u) \in \mathcal{D}_{\mathrm{alg}}(A)$. Next, we set up some definitions necessary to relate derivations of tensors to the derivations of algebras.

Definition C.4. Let V be a vector space. For a subspace $W \leq \mathrm{End}(V)^3$, define

$$W^{\mathrm{diag}} := \{d \in \mathrm{End}(V) : (d, d, d) \in W\}.$$

The below proposition is key to identifying pieces of a unital algebra derivation with pieces of tensor derivations.

Proposition C.5. Let A be a unital algebra with multiplication tensor μ, and let $N_A := \iota_{20}(\mathcal{L}(\mu)) + \iota_{10}(\mathcal{R}(\mu)) + \iota_{21}(\mathcal{M}(\mu))$. Then

$$N_A^{\mathrm{diag}} = \mathrm{ad}(\mathrm{Nuc}(A)).$$

Proof. For $\supseteq$, let $u \in \mathrm{Nuc}(A)$. By Proposition C.3,

$$(\lambda_u, \lambda_u) \in \mathcal{L}(\mu), \qquad (\rho_u, \rho_u) \in \mathcal{R}(\mu), \qquad (\rho_u, \lambda_u) \in \mathcal{M}(\mu).$$

Hence $\iota_{20}(\lambda_u, \lambda_u) - \iota_{10}(\rho_u, \rho_u) + \iota_{21}(\rho_u, \lambda_u) = (\mathrm{ad}(u), \mathrm{ad}(u), \mathrm{ad}(u)) \in N_A$, so $\mathrm{ad}(u) \in N_A^{\mathrm{diag}}$.

For $\subseteq$, let $d \in N_A^{\mathrm{diag}}$. Then $(d, d, d) \in N_A$, so

$$(d, d, d) = \iota_{20}(\alpha, \gamma) + \iota_{10}(\beta, \gamma') + \iota_{21}(\alpha', \beta').$$

By Proposition C.3, there exist $a \in \mathrm{Nuc}_\ell(A)$, $b \in \mathrm{Nuc}_r(A)$, and $c \in \mathrm{Nuc}_m(A)$ with

$$(\alpha, \gamma) = (\lambda_a, \lambda_a), \quad (\beta, \gamma') = (\rho_b, \rho_b), \quad (\alpha', \beta') = (\rho_c, \lambda_c).$$

Therefore $(d, d, d) = (\lambda_a - \rho_c,\ \rho_b + \lambda_c,\ \lambda_a + \rho_b)$. Comparing components gives $\rho_b = -\rho_c$, and $\lambda_a = \lambda_c$. Since A is unital, left and right multiplication are injective, so $a = c$, and $b = -c$. Hence $c \in \mathrm{Nuc}_\ell(A) \cap \mathrm{Nuc}_m(A) \cap \mathrm{Nuc}_r(A) = \mathrm{Nuc}(A)$ and

$$(d, d, d) = (\mathrm{ad}(c), \mathrm{ad}(c), \mathrm{ad}(c)),$$

so $d \in \mathrm{ad}(\mathrm{Nuc}(A))$. $\square$

Now we set up some notation and preliminary lemmas.

$$\begin{aligned}
A &:= R \otimes S, \quad X := \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S), \quad Y := \mathcal{C}(\mu_R) \boxtimes \mathrm{Der}(\mu_S), \\
N &:= \iota_{20}(\mathcal{L}(\mu_R) \boxtimes \mathcal{L}(\mu_S)) + \iota_{10}(\mathcal{R}(\mu_R) \boxtimes \mathcal{R}(\mu_S)) + \iota_{21}(\mathcal{M}(\mu_R) \boxtimes \mathcal{M}(\mu_S)), \\
N_R &:= \iota_{20}(\mathcal{L}(\mu_R)) + \iota_{10}(\mathcal{R}(\mu_R)) + \iota_{21}(\mathcal{M}(\mu_R)), \\
N_S &:= \iota_{20}(\mathcal{L}(\mu_S)) + \iota_{10}(\mathcal{R}(\mu_S)) + \iota_{21}(\mathcal{M}(\mu_S)).
\end{aligned}$$

Lemma C.6. For R a unital algebra, define

$$\epsilon_R : \mathrm{Der}(\mu_R) \to R^2, \qquad \epsilon_R(\alpha, \beta, \gamma) := (\alpha(1_R), \beta(1_R)).$$

Then $\mathrm{Ker}(\epsilon_R) = \{(d, d, d) : d \in \mathcal{D}_{\mathrm{alg}}(R)\}$.

Proof. Let $(\alpha, \beta, \gamma) \in \mathrm{Der}(\mu_R)$ with $\alpha(1_R) = \beta(1_R) = 0$. Setting $y = 1_R$ in

$$\alpha(x)y + x\beta(y) = \gamma(xy)$$

gives $\alpha(x) = \gamma(x)$, and setting $x = 1_R$ gives $\beta(y) = \gamma(y)$. Hence letting $\alpha = \beta = \gamma = d$. Computing, $d(xy) = d(x)y + xd(y)$, so $d \in \mathcal{D}_{\mathrm{alg}}(R)$. The reverse inclusion is immediate. $\square$

For $(\alpha, \beta, \gamma) \in \mathrm{End}(R)^3$ and $(f, g, h) \in \mathrm{End}(S)^3$, evaluating at $(1_R \otimes 1_S)$ gives

$$\epsilon_A\big((\alpha, \beta, \gamma) \boxtimes (f, g, h)\big) = (\alpha(1_R) \otimes f(1_S),\ \beta(1_R) \otimes g(1_S)). \tag{C.2}$$

Lemma C.7. With the notation above,

$$X \cap N = N_R \boxtimes \mathcal{C}(\mu_S), \qquad Y \cap N = \mathcal{C}(\mu_R) \boxtimes N_S.$$

Proof. We first prove $X \cap N = N_R \boxtimes \mathcal{C}(\mu_S)$.

For the inclusion $\supseteq$, note that $N_R \leq \mathrm{Der}(\mu_R)$, so $N_R \boxtimes \mathcal{C}(\mu_S) \leq X$. Also, as every pure $\boxtimes$ term of $N_R \boxtimes \mathcal{C}(\mu_S)$ lies in N, so $N_R \boxtimes \mathcal{C}(\mu_S) \leq X \cap N$.

For the reverse inclusion, let $w \in X \cap N$. Since $w \in X = \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S)$, by Proposition C.3 we may choose a basis $z_1, \ldots, z_m$ of $Z(S)$ and elements $d_i = (d_{2,i}, d_{1,i}, d_{0,i}) \in \mathrm{Der}(\mu_R)$ such that

$$w = \Big(\sum_{i=1}^m d_{2,i} \otimes \lambda_{z_i},\ \sum_{i=1}^m d_{1,i} \otimes \lambda_{z_i},\ \sum_{i=1}^m d_{0,i} \otimes \lambda_{z_i}\Big). \tag{C.3}$$

Since $w \in N$, also write $w = L + R + M$, with

$$L \in \iota_{20}(\mathcal{L}(\mu_R) \boxtimes \mathcal{L}(\mu_S)), \quad R \in \iota_{10}(\mathcal{R}(\mu_R) \boxtimes \mathcal{R}(\mu_S)), \quad M \in \iota_{21}(\mathcal{M}(\mu_R) \boxtimes \mathcal{M}(\mu_S)).$$

Expanding L, M, R as sums of $\boxtimes$ terms, there exists elements

$$L_j \in \mathcal{L}(\mu_R),\ a_j \in \mathrm{Nuc}_\ell(S), \qquad R_j \in \mathcal{R}(\mu_R),\ b_j \in \mathrm{Nuc}_r(S), \qquad M_j \in \mathcal{M}(\mu_R),\ c_j \in \mathrm{Nuc}_m(S)$$

such that

$$L = \sum_j \iota_{20}\big(L_j \boxtimes (\lambda_{a_j}, \lambda_{a_j})\big), \quad R = \sum_j \iota_{10}\big(R_j \boxtimes (\rho_{b_j}, \rho_{b_j})\big), \quad M = \sum_j \iota_{21}\big(M_j \boxtimes (\rho_{c_j}, \lambda_{c_j})\big).$$

Let $W \leq S$ be the span of $\{z_1, \ldots, z_m,\ a_j,\ b_j,\ c_j\}$, and extend $z_1, \ldots, z_m$ to a basis

$$z_1, \ldots, z_m, s_{m+1}, \ldots, s_n$$

of W. Expanding each a_j, b_j, c_j in this basis and regrouping gives elements

$$L_i \in \mathcal{L}(\mu_R), \qquad R_i \in \mathcal{R}(\mu_R), \qquad M_i \in \mathcal{M}(\mu_R) \qquad (1 \leq i \leq n),$$

such that

$$w = \sum_{i=1}^n \Big(\iota_{20}\big(L_i \boxtimes (\lambda_{s_i}, \lambda_{s_i})\big) + \iota_{10}\big(R_i \boxtimes (\rho_{s_i}, \rho_{s_i})\big) + \iota_{21}\big(M_i \boxtimes (\rho_{s_i}, \lambda_{s_i})\big)\Big), \tag{C.4}$$

with $s_i = z_i$ for $1 \leq i \leq m$.

Now for every $r \in R$, apply the three components of w to $r \otimes 1_S$ using Equations (C.3) and (C.4). Since $\lambda_{s_i}(1_S) = \rho_{s_i}(1_S) = s_i$, this yields

$$\sum_{i=1}^m (d_{2,i}(r), d_{1,i}(r), d_{0,i}(r)) \otimes z_i = \sum_{i=1}^n \big((L_{i,2} + M_{i,2})(r),\ (R_{i,1} + M_{i,1})(r),\ (L_{i,0} + R_{i,0})(r)\big) \otimes s_i.$$

By linear independence of the basis $z_1, \ldots, z_m, s_{m+1}, \ldots, s_n$, for each $1 \leq i \leq m$ we have

$$(d_{2,i}, d_{1,i}, d_{0,i}) = \iota_{20}(L_i) + \iota_{10}(R_i) + \iota_{21}(M_i) \in N_R.$$

Since $z_i \in Z(S)$, we have $\lambda_{z_i} = \rho_{z_i}$, concluding

$$w = \sum_{i=1}^m \big(\iota_{20}(L_i) + \iota_{10}(R_i) + \iota_{21}(M_i)\big) \boxtimes (\lambda_{z_i}, \lambda_{z_i}, \lambda_{z_i}) \in N_R \boxtimes \mathcal{C}(\mu_S).$$

TODO: Shorten and sharpen this direction. □

Below is a key lemma allowing us to analyze the diagonal part of the right hand side of Equation (5.1) in Theorem 5.1 as the sum of the diagonal part of each summand. Each of these summands will correspond to a piece in Brešar's decomposition.

Lemma C.8. With the notation above,

$$(X + Y + N)^{\mathrm{diag}} = X^{\mathrm{diag}} + Y^{\mathrm{diag}} + N^{\mathrm{diag}}.$$

Proof. We will apply Lemma C.1 to

$$U = \mathrm{Der}(\mu_A), \qquad V = A^2, \qquad \epsilon = \epsilon_A$$

and the subspaces $X, Y, N \leq U$. Recall

$$X = \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S), \quad Y = \mathcal{C}(\mu_R) \boxtimes \mathrm{Der}(\mu_S), \quad N = \iota_{20}(\mathcal{L}(\mu_R \otimes \mu_S)) + \iota_{10}(\mathcal{R}(\mu_R \otimes \mu_S)) + \iota_{21}(\mathcal{M}(\mu_R \otimes \mu_S)).$$

To apply Lemma C.1, we must verify

$$\epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big) = \epsilon_A(X \cap N), \qquad \epsilon_A(Y) \cap \big(\epsilon_A(X) + \epsilon_A(N)\big) = \epsilon_A(Y \cap N).$$

To check these, let

$$E_R \coloneqq \epsilon_R(\mathrm{Der}(\mu_R)), \quad F_R \coloneqq \epsilon_R(N_R), \quad E_S \coloneqq \epsilon_S(\mathrm{Der}(\mu_S)), \quad F_S \coloneqq \epsilon_S(N_S).$$

By Equation (C.2),

$$\epsilon_A(X) = E_R \otimes Z(S), \qquad \epsilon_A(Y) = Z(R) \otimes E_S. \tag{C.5}$$

Because $Y = \mathcal{C}(\mu_R) \boxtimes \mathrm{Der}(\mu_S)$, the first two R-components of $\epsilon_A(Y)$ lie in $Z(R)^2 \leq F_R$, so $\epsilon_A(Y) \leq F_R \otimes S$. Also, N is built from N_R on the R-side, so $\epsilon_A(N) \leq F_R \otimes S$. This gives the first relation in the equation below, and the second is analogous.

$$\epsilon_A(Y) + \epsilon_A(N) \leq F_R \otimes S, \qquad \epsilon_A(X) + \epsilon_A(N) \leq R \otimes F_S. \tag{C.6}$$

By Lemma C.7 and Equation (C.2),

$$\epsilon_A(X \cap N) = F_R \otimes Z(S), \qquad \epsilon_A(Y \cap N) = Z(R) \otimes F_S. \tag{C.7}$$

Now we conclude

$$\begin{aligned}
\epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big) &\subseteq (E_R \otimes Z(S)) \cap (F_R \otimes S) && \text{by (C.5) and (C.6)} \\
&= F_R \otimes Z(S) && \text{by Lemma 2.2 and } F_R \leq E_R \\
&= \epsilon_A(X \cap N) && \text{by (C.7).}
\end{aligned}$$

Conversely, $X \cap N \leq X$ and $X \cap N \leq Y + N$, so

$$\epsilon_A(X \cap N) \subseteq \epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big).$$

Hence $\epsilon_A(X) \cap \big(\epsilon_A(Y) + \epsilon_A(N)\big) = \epsilon_A(X \cap N)$. Analogously, $\epsilon_A(Y) \cap \big(\epsilon_A(X) + \epsilon_A(N)\big) = \epsilon_A(Y \cap N)$.

Finally, applying Lemma C.1 concludes

$$(X + Y + N) \cap \ker(\epsilon_A) = (X \cap \ker(\epsilon_A)) + (Y \cap \ker(\epsilon_A)) + (N \cap \ker(\epsilon_A)),$$

giving

$$(X + Y + N)^{\mathrm{diag}} = X^{\mathrm{diag}} + Y^{\mathrm{diag}} + N^{\mathrm{diag}}.$$

□

Proof of Proposition 5.3. Recall

$$X = \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S), \quad Y = \mathcal{C}(\mu_R) \boxtimes \mathrm{Der}(\mu_S), \quad N = \iota_{20}(\mathcal{L}(\mu_R \otimes \mu_S)) + \iota_{10}(\mathcal{R}(\mu_R \otimes \mu_S)) + \iota_{21}(\mathcal{M}(\mu_R \otimes \mu_S)).$$

By Lemma C.2, μ_R, μ_S are fully nondegenerate, so applying Theorem 5.1 gives

$$\mathrm{Der}(\mu_R \otimes \mu_S) = X + Y + N.$$

By definition,

$$\mathcal{D}_{\mathrm{alg}}(R \otimes S) = \mathrm{Der}(\mu_R \otimes \mu_S)^{\mathrm{diag}},$$

and by Lemma C.8,

$$\mathrm{Der}(\mu_R \otimes \mu_S)^{\mathrm{diag}} = (X + Y + N)^{\mathrm{diag}} = X^{\mathrm{diag}} + Y^{\mathrm{diag}} + N^{\mathrm{diag}}.$$

It remains to analyze X^{diag},Y^{diag}, and N^{diag} and match them to the right hand side of Equation (5.2).

For X^{diag}, the inclusion $\mathcal{D}_{\mathrm{alg}}(R) \otimes Z(S) \subseteq X^{\mathrm{diag}}$ is immediate: if $d \in \mathcal{D}_{\mathrm{alg}}(R)$ and $c \in Z(S)$, then $(d, d, d) \in \mathrm{Der}(\mu_R)$, and $(\lambda_c, \lambda_c, \lambda_c) \in \mathcal{C}(\mu_S)$.

For the reverse inclusion, let $u \in X^{\mathrm{diag}}$, so $(u, u, u) \in X$. Recall that $X = \mathrm{Der}(\mu_R) \boxtimes \mathcal{C}(\mu_S)$.

From Proposition C.3, the map

$$Z(S) \to \mathcal{C}(\mu_S), \qquad z \mapsto (\lambda_z, \lambda_z, \lambda_z)$$

is an isomorphism. Thus a basis $z_1, \dots, z_m$ of $Z(S)$, can be used to expand $\mathcal{C}(\mu_S)$ in the corresponding basis $\{\lambda_{z_i}\}$. Hence there exists $d_i = (d_{2,i}, d_{1,i}, d_{0,i}) \in \mathrm{Der}(\mu_R)$ such that

$$(u, u, u) = \sum_{i=1}^{m} d_i \boxtimes (\lambda_{z_i}, \lambda_{z_i}, \lambda_{z_i}) = \Big(\sum_{i=1}^{m} d_{2,i} \otimes \lambda_{z_i}, \sum_{i=1}^{m} d_{1,i} \otimes \lambda_{z_i}, \sum_{i=1}^{m} d_{0,i} \otimes \lambda_{z_i} \Big).$$

Equality of the three components therefore implies

$$d_{2,i} = d_{1,i} = d_{0,i}$$

for each i. Hence each d_i is of the form $d_i = (\delta_i, \delta_i, \delta_i)$, and $\delta_i \in \mathcal{D}_{\text{alg}}(R)$. As

$$u = \sum_{i=1}^{m} \delta_i \otimes \lambda_{z_i} \in \mathcal{D}_{\text{alg}}(R) \otimes Z(S),$$

we conclude $\mathcal{D}_{\text{alg}}(R) \otimes Z(S) \supseteq X^{\text{diag}}$. The proof that $Y^{\text{diag}} = Z(R) \otimes \mathcal{D}_{\text{alg}}(S)$ is analogous.

For N^{diag}, apply Proposition C.5 to the algebra $A = R \otimes S$. We use Brešar's nucleus identity $\mathrm{Nuc}(R \otimes S) = \mathrm{Nuc}(R) \otimes \mathrm{Nuc}(S)$ [Bre17, p. 2] to conclude

$$N^{\text{diag}} = \mathrm{ad}(\mathrm{Nuc}(R \otimes S)) = \mathrm{ad}(\mathrm{Nuc}(R) \otimes \mathrm{Nuc}(S)). \qquad \square$$

Appendix D. Tensor product bimap decomposition examples

D.1. Nucleus decomposition example. Back to main text.

Example D.1. **TODO: Shorten or make clearer this concrete nucleus deecomposition example.** We decompose a term in $\mathcal{M}(s \otimes t)$ as sum of terms from $\mathcal{M}(s) \boxtimes \mathcal{M}(t)$. Let $\mathbb{F} = \mathbb{F}_5$, and let

$$s, t : \mathbb{F}^2 \times \mathbb{F}^2 \rightarrowtail \mathbb{F}$$

be bilinear maps given by

$$S = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad T = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

respectively. Given matrices

$$X = \begin{bmatrix} 0 & 4 & 2 & 0 \\ 2 & 1 & 1 & 3 \\ 1 & 3 & 3 & 1 \\ 2 & 1 & 2 & 1 \end{bmatrix}, \qquad Y = \begin{bmatrix} 4 & 4 & 0 & 3 \\ 1 & 3 & 0 & 1 \\ 1 & 3 & 2 & 3 \\ 4 & 1 & 3 & 1 \end{bmatrix}.$$

A direct check verifies $X(S \otimes T) = (S \otimes T)Y$. Let x, y correspond to the matrices X and Y. Then $(s \otimes t) \circ_2 x = (s \otimes t) \circ_1 y$, so $(x, y) \in \mathcal{M}(s \otimes t)$. We give a decomposition of (x, y).

Set $w := \Xi\big((s \otimes t) \circ_2 x\big) = \Xi\big((s \otimes t) \circ_1 y\big)$, and write a minimal rank decomposition $w = u_1 \otimes v_1 + u_2 \otimes v_2$. Apply Lemma 4.12. Then there are $\alpha_1, \alpha_2, \beta_1, \beta_2, f_1, f_2, g_1, g_2 \in \mathrm{End}(\mathbb{F}^2)$ with

$$s \circ_2 \alpha_1 = u_1 = s \circ_1 \beta_1, \qquad s \circ_2 \alpha_2 = u_2 = s \circ_1 \beta_2,$$

$$t \circ_2 f_1 = v_1 = t \circ_1 g_1, \qquad t \circ_2 f_2 = v_2 = t \circ_1 g_2,$$

and $x = \kappa_2(\alpha_1 \otimes f_1) + \kappa_2(\alpha_2 \otimes f_2)$, and $y = \kappa_1(\beta_1 \otimes g_1) + \kappa_1(\beta_2 \otimes g_2)$. For this example, one such choice is

$$\alpha_1 = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}, \quad f_1 = \begin{bmatrix} 1 & 4 \\ 2 & 2 \end{bmatrix}, \quad f_2 = \begin{bmatrix} 0 & 3 \\ 1 & 4 \end{bmatrix}.$$

and

$$\beta_1 = \begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}, \quad \beta_2 = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}, \quad g_1 = \begin{bmatrix} 4 & 2 \\ 1 & 4 \end{bmatrix}, \quad g_2 = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix}.$$

We verify

$$X(S\otimes T) = (s\circ_2 \alpha_1)\otimes(t\circ_2 f_1) + (s\circ_2 \alpha_2)\otimes(t\circ_2 f_2) = \begin{bmatrix}1&0&4&1\\0&0&0&2\\1&1&3&4\\0&2&0&3\end{bmatrix}, \qquad (S\otimes T)Y = (s\circ_1 \beta_1)\otimes(t\circ_1 g_1) + (s\circ_1 \beta_2)\otimes(t\circ_1 g_2) = \begin{bmatrix}1&0&4&1\\0&0&0&2\\1&1&3&4\\0&2&0&3\end{bmatrix}.$$

Apply Remark 4.10 to relate the sums $(s\circ_2 \alpha_1)\otimes(t\circ_2 f_1) + (s\circ_2 \alpha_2)\otimes(t\circ_2 f_2)$ with $(s\circ_1 \beta_1)\otimes(t\circ_1 g_1) + (s\circ_1 \beta_2)\otimes(t\circ_1 g_2)$, we obtain

$$M = \begin{bmatrix}1&1\\0&1\end{bmatrix}, \qquad M^{-1} = \begin{bmatrix}1&4\\0&1\end{bmatrix},$$

A calculation using M gives

$$s\circ_2 \alpha_1 + s\circ_2 \alpha_2 = s\circ_1 \beta_1, \qquad s\circ_2 \alpha_2 = s\circ_1 \beta_2,$$
$$t\circ_2 f_1 = t\circ_1 g_1, \qquad t\circ_2 4f_1 + t\circ_2 f_2 = t\circ_1 g_2.$$

Using linearity and that s and t are non-degenerate, we can conclude $(x,y) = (\alpha_1+\alpha_2,\beta_1)\boxtimes(f_1,g_1) + (\alpha_2,\beta_2)\boxtimes(4f_1+f_2,g_2) \in \mathcal{M}(s)\boxtimes\mathcal{M}(t)$.

D.2. **Derivation decomposition example.** **Back to main text.**

Example D.2. **TODO: Shorten or get rid of this explicit derivation of a tensor product bimap example!** We decompose $(\delta^{(2)},\delta^{(1)},\delta^{(0)}) \in \mathrm{Der}(s\otimes t)$ as sum of terms from $\mathrm{Der}(s)\boxtimes\mathcal{C}(t)$ and $\mathcal{M}(s)\boxtimes\mathcal{M}(t)$. Let $\mathbb{F}=\mathbb{F}_5$. Let $E_{ij}\in\mathbb{M}_2(\mathbb{F})$ denote matrix units. Let $s,t:\mathbb{F}^2\times\mathbb{F}^2 \rightarrowtail \mathbb{F}$ be given by matrices

$$S = I_2, \qquad T = E_{11}+E_{12}+E_{22}.$$

Consider $\Delta = (\Delta^{(2)},\Delta^{(1)},\Delta^{(0)})$, a derivation of $S\otimes T$.

$$\Delta^{(2)} = \begin{bmatrix}1&0&0&0\\0&1&0&0\\1&0&0&0\\0&0&0&0\end{bmatrix}, \qquad \Delta^{(1)} = \begin{bmatrix}0&0&0&0\\0&0&0&0\\4&4&1&0\\0&0&0&1\end{bmatrix}, \qquad \Delta^{(0)} = \begin{bmatrix}1\end{bmatrix}.$$

We verify

$$\Delta^{(2)}(S\otimes T) + (S\otimes T)\Delta^{(1)} = (S\otimes T)^{\Delta^{(0)}}.$$

Let $\delta = (\delta^{(2)},\delta^{(1)},\delta^{(0)})$ be the operators corresponding to Δ. Set

$$p := (s\otimes t)\circ_2 \delta^{(2)}, \qquad q := (s\otimes t)\circ_1 \delta^{(1)}, \qquad r := (s\otimes t)\circ_0 \delta^{(0)}.$$

As $p+q=r$, $(\delta^{(2)},\delta^{(1)},\delta^{(0)}) \in \mathrm{Der}(s\otimes t)$.

We apply Lemma 5.13 to $p+q=r$, getting the subspaces

$$U^{(1)} = \mathrm{span}\{E_{11},E_{22}\}, \quad U^{(2)} = \mathrm{span}\{E_{21}\}, \quad V^{(1)} = \mathrm{span}\{T\}, \quad V^{(2)} = \mathrm{span}\{E_{11}+E_{12}\}.$$

Then the nonzero blocks are

$$p_{11} = E_{11}\otimes T, \quad q_{11} = E_{22}\otimes T, \quad r_{11} = I_2\otimes T, \quad p_{11}+q_{11} = r_{11},$$
$$p_{22} = E_{21}\otimes(E_{11}+E_{12}), \quad q_{22} = -E_{21}\otimes(E_{11}+E_{12}), \quad r_{22} = 0, \quad p_{22}+q_{22} = 0.$$

Therefore $p+q=r$ has an LIU decomposition with two local equalities.

$$a_1 = E_{11}, \quad b_1 = E_{22}, \quad c_1 = I_2, \quad x_1 = y_1 = z_1 = T, \quad a_1\otimes x_1 + b_1\otimes y_1 = c_1\otimes z_1,$$
$$a_2 = E_{21}, \quad b_2 = -E_{21}, \quad c_2 = 0, \quad x_2 = y_2 = z_2 = E_{11}+E_{12}, \quad a_2\otimes x_2 + b_2\otimes y_2 = c_2\otimes z_2.$$

Let $\alpha_D = E_{11},\ \beta_D = E_{22},\ \gamma_D = 1$, and $f_D = g_D = I_2,\ h_D = 1$. Then

$$a_1 = s\circ_2 \alpha_D,\ b_1 = s\circ_1 \beta_D,\ c_1 = s\circ_0 \gamma_D, \qquad x_1 = t\circ_2 f_D,\ y_1 = t\circ_1 g_D,\ z_1 = t\circ_0 h_D,$$

with $(\alpha_D,\beta_D,\gamma_D)\in\mathrm{Der}(s)$ and $(f_D,g_D,h_D)\in\mathcal{C}(t)$.

Similarly, let $\alpha_N = E_{12},\ \beta_N = E_{21}$, and $f_N = E_{11},\ g_N = E_{11}+E_{12}$. Then $a_2 = s\circ_2 \alpha_N,\ b_2 = -s\circ_1 \beta_N$, and $x_2 = t\circ_2 f_N,\ y_2 = t\circ_1 g_N$, meaning $(\alpha_N,\beta_N)\in\mathcal{M}(s)$, $(f_N,g_N)\in\mathcal{M}(t)$.

Therefore

$$\delta^{(2)} = \kappa_2(\alpha_D \otimes f_D) + \kappa_2(\alpha_N \otimes f_N) \quad \delta^{(1)} = \kappa_1(\beta_D \otimes g_D) - \kappa_1(\beta_N \otimes g_N), \quad \delta^{(0)} = \kappa_0(\gamma_D \otimes h_D).$$

So $(\delta^{(2)}, \delta^{(1)}, \delta^{(0)})$ is a sum of a term from $\mathrm{Der}(s) \boxtimes \mathcal{C}(t)$ and a term from $\iota_{21}\big(\mathcal{M}(s) \boxtimes \mathcal{M}(t)\big)$.

Appendix E. Extras (will be deleted before dissertation)

Everything below the line are writing I had in the prelim or written up elsewhere that I think can potentially go somewhere in this doc, but I haven't had the chance to update their notation or figure out what background is needed for them to be coherent. Each block of writing is delineated by dashed lines.

- - - -

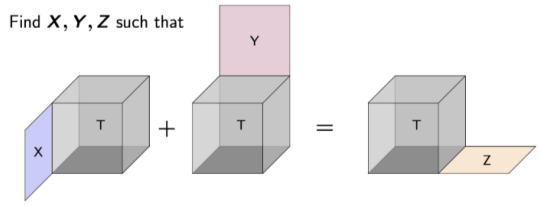
Then

$$M \otimes N = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \otimes \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} = \left(\begin{array}{cc|cc} 3 & 4 & \multicolumn{2}{c}{\multirow{2}{*}{$\mathbf{0}$}} \\ 5 & 6 & & \\ \hline \multicolumn{2}{c|}{\multirow{2}{*}{$\mathbf{0}$}} & 6 & 8 \\ & & 10 & 12 \end{array}\right)$$

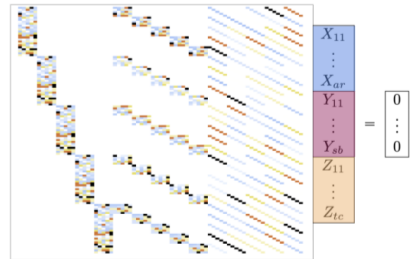
The 4×4 matrix $M \otimes N$ is given as an element of $\mathbb{M}_2(\mathbb{M}_2(K)) \cong \mathbb{M}_4(K)$. It has interpretation $\langle M \otimes N| : (K^2 \otimes K^2) \rightarrowtail (K^2 \otimes K^2)$ given by mapping basis element $e_i \otimes e_j$ (dual basis $\epsilon_i \otimes \epsilon_j$) to the $2(i-1)+j$th column (row) of the matrix. Specifically, $(\epsilon_k \otimes \epsilon_l)\langle M \otimes N|e_i \otimes e_j\rangle$ is row $2k-1+l$, column $2i-1+j$ of $M \otimes N$, given by the product of the (k,i)th entry of M with the (l,j)th entry of N.

- - - -

Computing the derivation algebra



appears to require solving equations of the form



- .

(In gradshow poster) Small but concrete example that may be useful somewhere

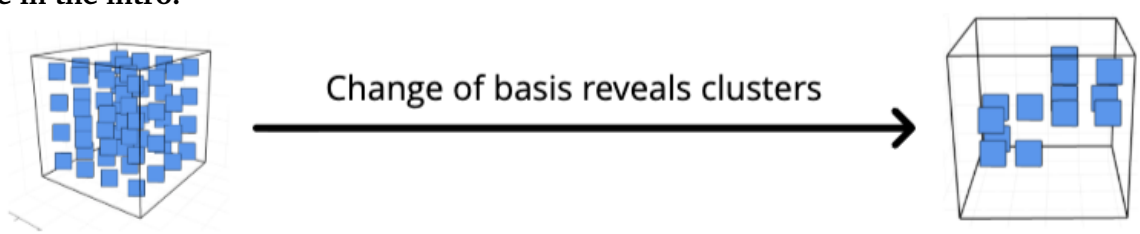
Example

$$t_1 = \begin{bmatrix} 10 & 5 \\ 5 & 6 \end{bmatrix} \quad t_2 = \begin{bmatrix} 9 & 3 \\ 3 & 7 \end{bmatrix}$$

$$T = [t_1, t_2]$$

$$\mathrm{Der}(T) = (X, Y, Z) = \mathrm{nullspace}\left(\begin{array}{cccc|cccc|cccc}
x_{11} & x_{12} & x_{21} & x_{22} & y_{11} & y_{21} & y_{12} & y_{22} & z_{11} & z_{12} & z_{21} & z_{22} \\
\hline
10 & 5 & & & 10 & 5 & & & -10 & & -9 & \\
9 & 3 & & & 9 & 3 & & & & -10 & & -9 \\
5 & 6 & & & & & 10 & 5 & -5 & & -3 & \\
3 & 7 & & & & & 9 & 3 & & -5 & & -3 \\
 & & 10 & 5 & 5 & 6 & & 0 & -5 & & -3 & \\
 & & 9 & 3 & 3 & 7 & & 0 & & -5 & & -3 \\
 & & 5 & 6 & & & 5 & 6 & -6 & & -7 & \\
 & & 3 & 7 & & & 3 & 7 & & -6 & & -7
\end{array}\right)$$

(In gradshow poster) Concrete change of basis clustering example, visual may help motivate in the intro.



Numerical example

$$T_1 = \begin{bmatrix} 5 & 6 & 6 & 8 \\ 5 & 6 & 6 & 8 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 4 & 4 \end{bmatrix} \quad T_2 = \begin{bmatrix} 9 & 12 & 14 & 14 \\ 5 & 10 & 10 & 14 \\ 4 & 10 & 12 & 8 \\ 5 & 14 & 14 & 16 \end{bmatrix}$$

$$T_3 = \begin{bmatrix} 10 & 4 & 4 & 12 \\ 10 & 4 & 4 & 12 \\ 0 & 0 & 0 & 0 \\ 10 & 12 & 12 & 16 \end{bmatrix} \quad T_4 = \begin{bmatrix} 16 & 17 & 20 & 20 \\ 14 & 14 & 16 & 18 \\ 6 & 9 & 12 & 6 \\ 11 & 16 & 18 & 16 \end{bmatrix}$$

$$R = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 0 \\ 2 & 2 & 2 & 0 \\ 2 & 0 & 2 & 2 \end{bmatrix}$$

$$\tilde{T}_1 = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{T}_2 = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\tilde{T}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 2 & 0 \end{bmatrix} \quad \tilde{T}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$